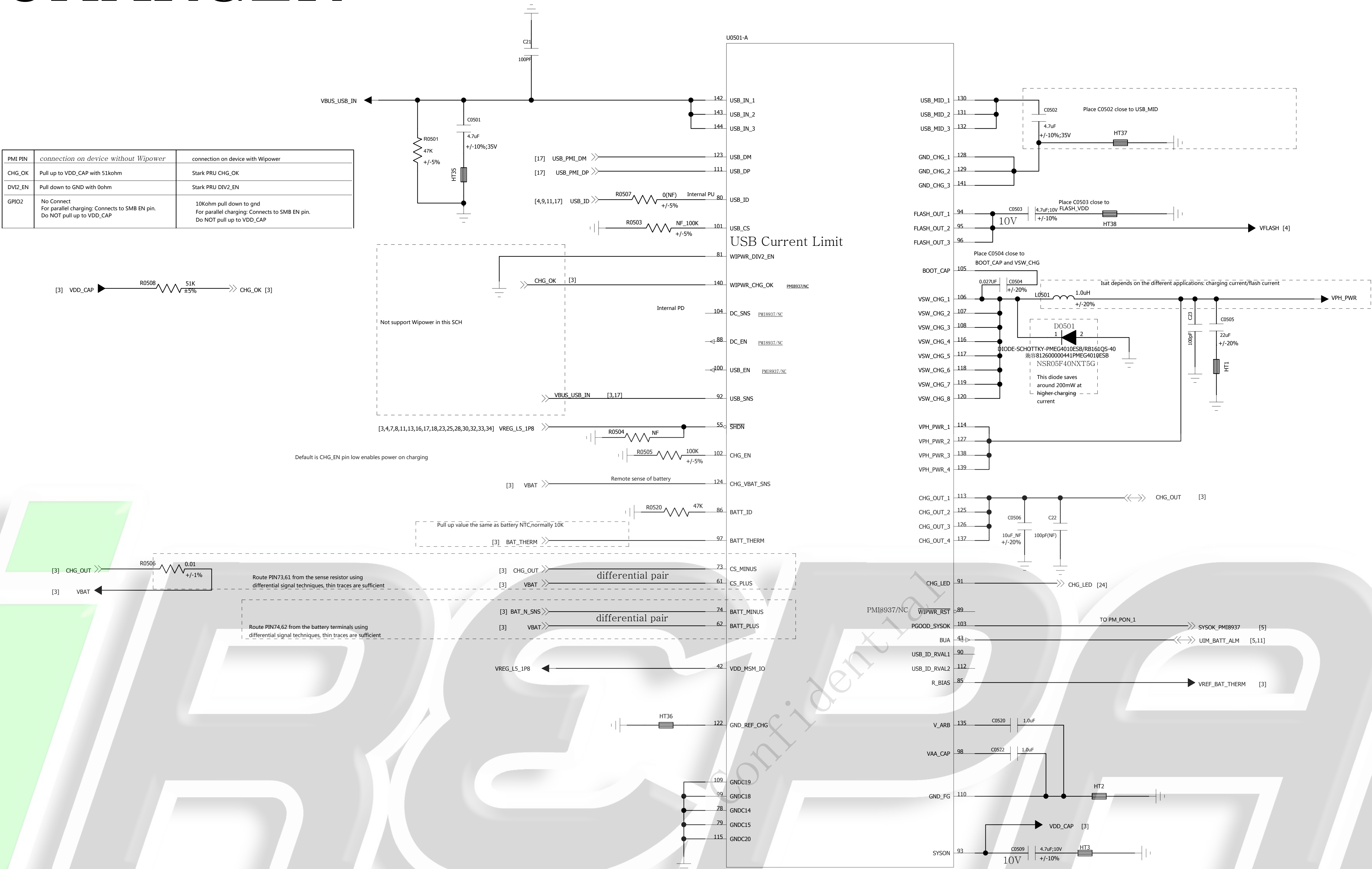


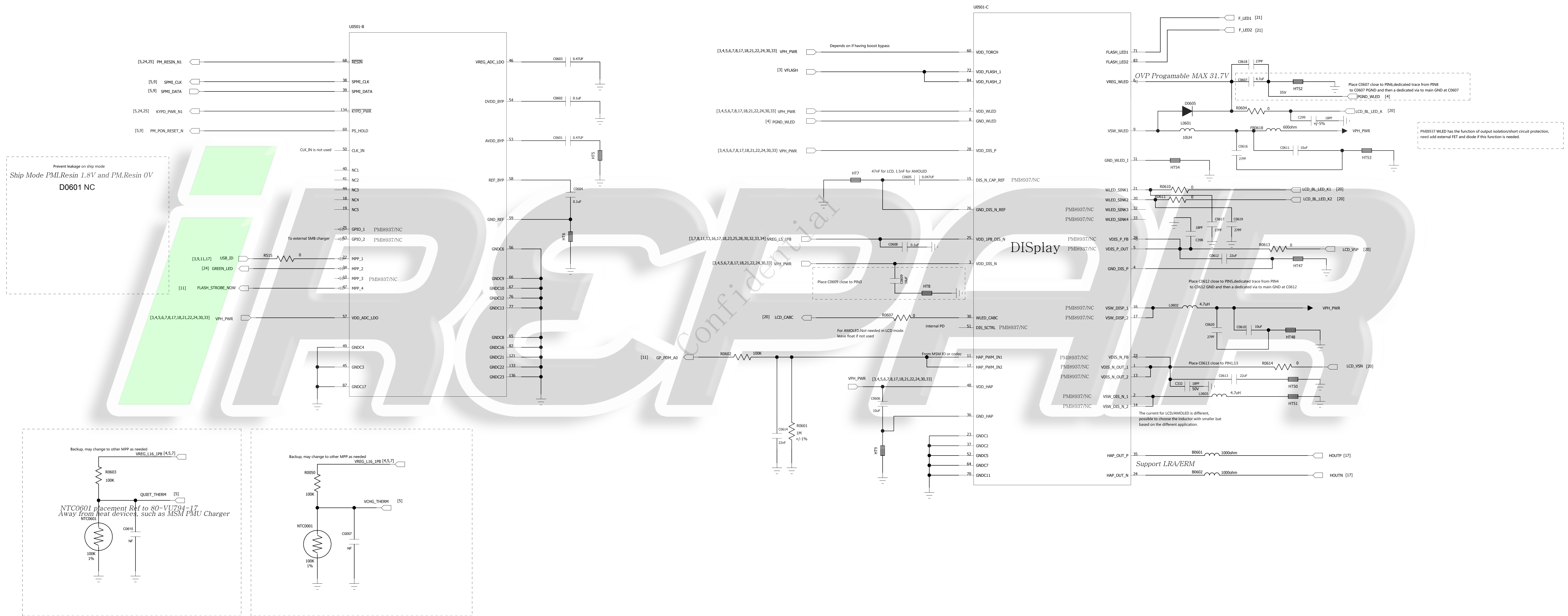
PMI8940 CHARGER



BATID Resistor Pull down to GND in BAT Module.
BAT_THERM NTC(68K B=4250) Pull down to GND in BAT Module.

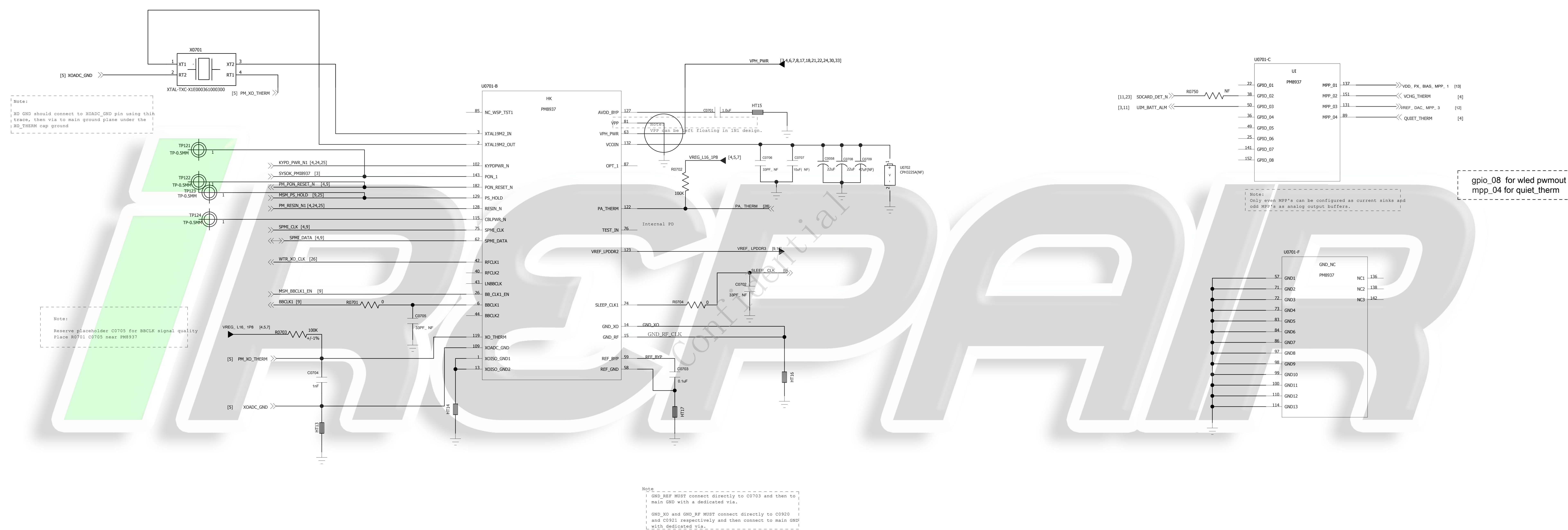
BATTERY CONNECTOR

PMI8940 Control/Interface



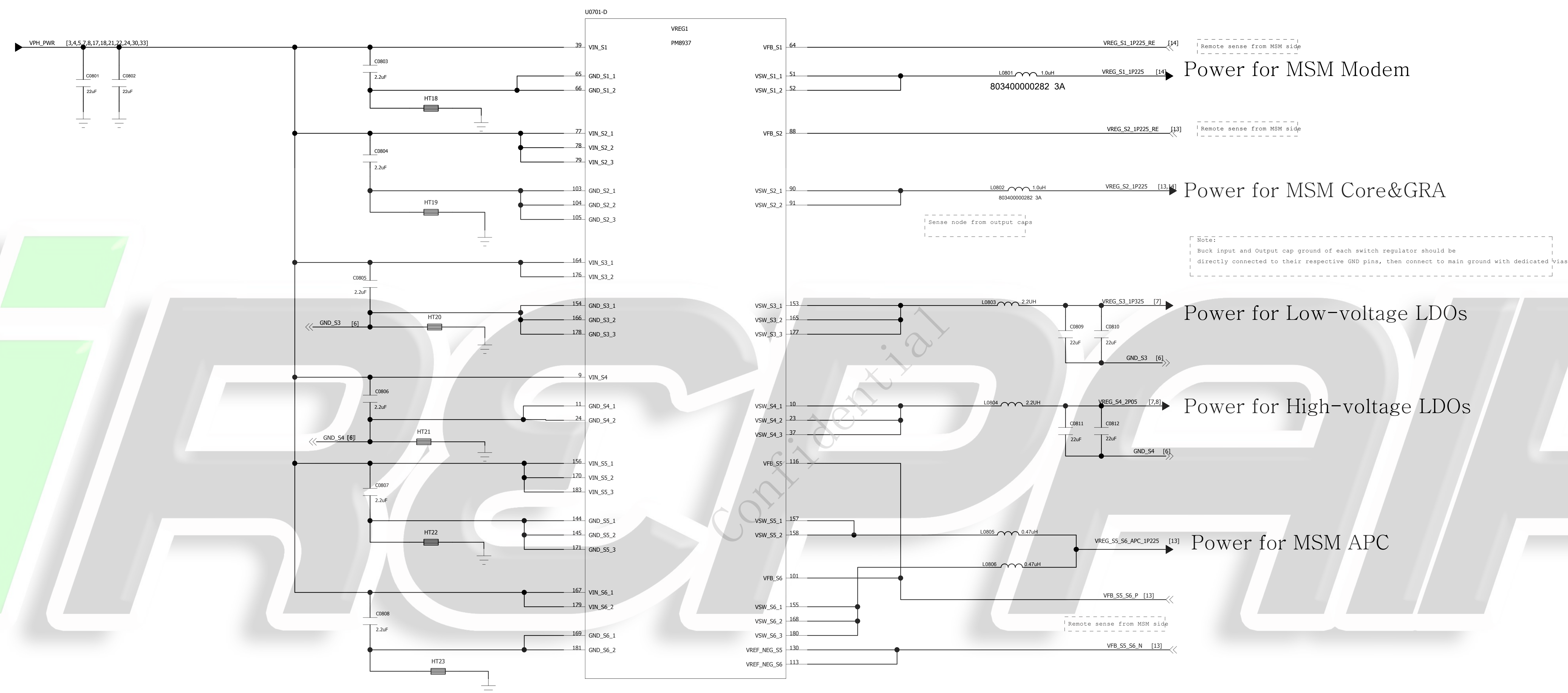
PROJECT:		REV:	<REV>
DOCUMENT NO.: 06_PM18940_CONTROL/INTERFACE		SIZED:	D
DEPARTMENT: Hardware DEPT.			
COMPANY: 			
DESIGNER:	<DESIGNER>	Last Saved Date:	2017/11/20
		SHEET:	4 OF 35

PM8940 Control and MPP/Clock



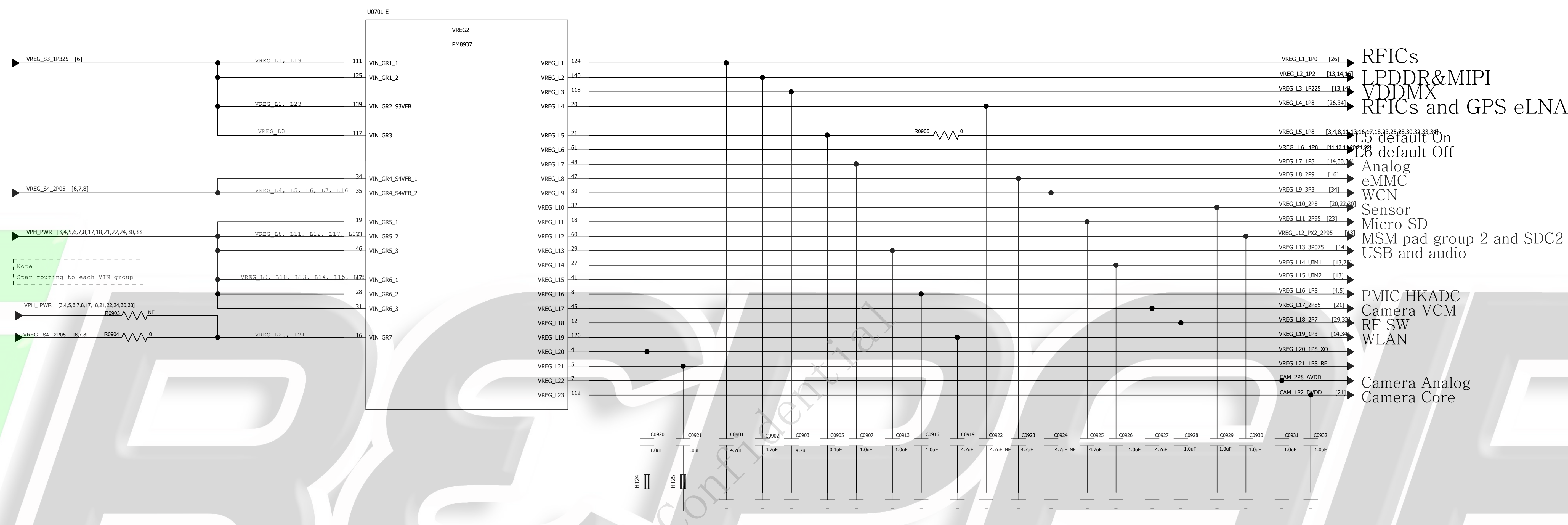
PROJECT:			REV:	<REV>
DOCUMENT NO.:	07_08940_CONTROL_AND_MPP/CLOCK		SIZED:	D
DEPARTMENT:	Hardware DEPT.			
COMPANY:				
DESIGNER:	<DESIGNER>	Last Saved Date:	2017/11/20	SHEET: 5 OF 35

PM8940 Buck converter



PROJECT:		REV:	<REV>
DOCUMENT NO.: 08_PM8940_BUCK_CONVERTER		SIZED:	D
DEPARTMENT: Hardware DEPT.			
COMPANY:			
DESIGNER:	<DESIGNER>	Last Saved Date:	2017/11/20
		SHEET:	6 OF 35

PM8940 LDO circuits

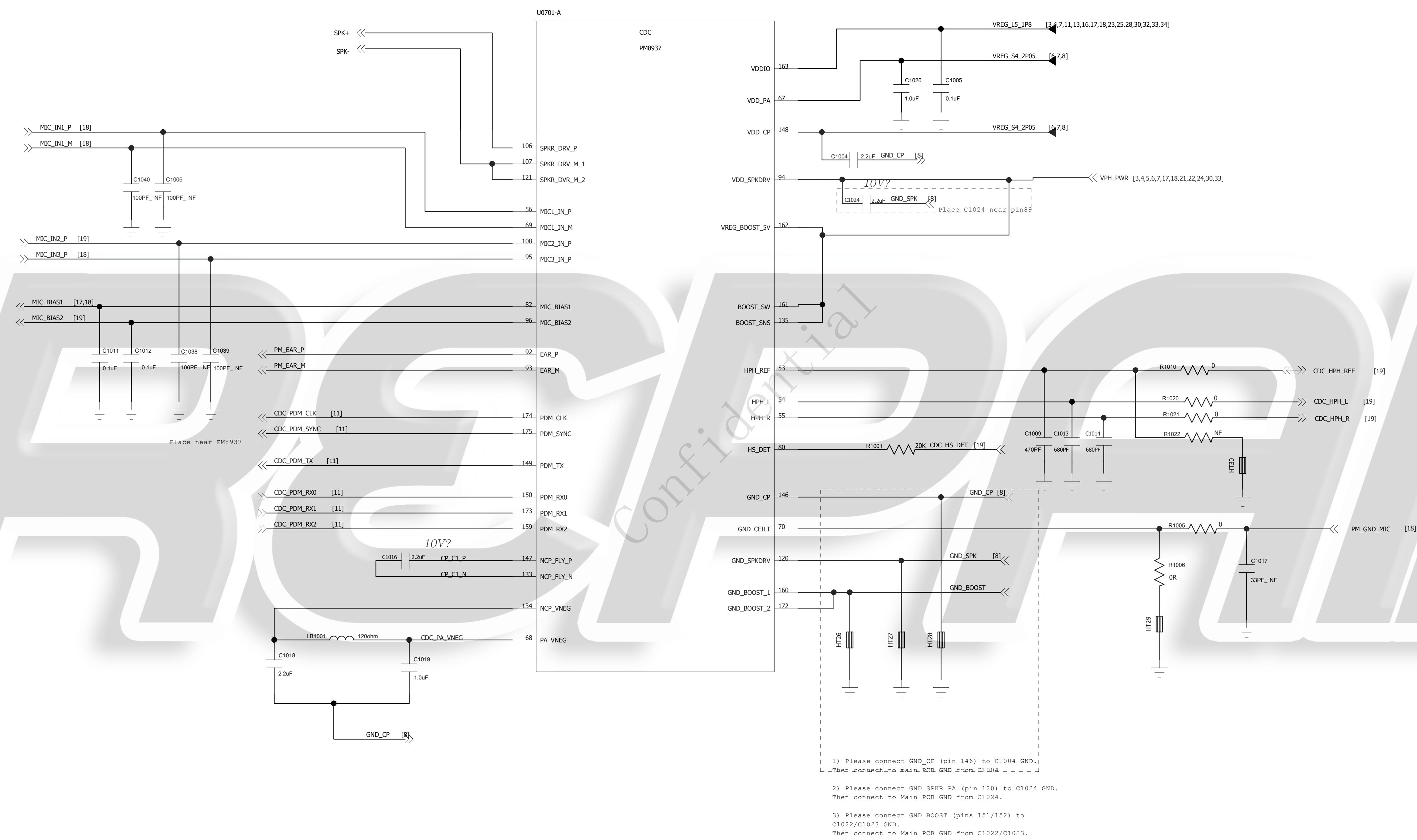


PSEUDO CAPLESS LDOs

L4*, L6, L8*, L9*, L10, L11*, L12, L14, L15, L17*, L18, L22, L23
L4*, L8*, L9*, L11*, L17*, still need validation as Pseudo-capless
L5, L7, L13, L16 have internal PMIC Loads and require local CAPS.

PROJECT:		REV:	<REV>
DOCUMENT NO.:	09_PM8940_LDO_CIRCUITS	SIZED:	D
DEPARTMENT:	Hardware DEPT.		
COMPANY:			
DESIGNER:	<DESIGNER>	Last Saved Date:	2017/11/20
SHEET:	7	OF	35

PM8940 Codec



PROJECT:			REV:	<REV>
DOCUMENT NO.:	10_PM8940_CODEC		SIZED:	D
DEPARTMENT:	Hardware DEPT.			
COMPANY:				
DESIGNER:	<DESIGNER>	Last Saved Date:	2017/11/20	SHEET: 8 OF 35

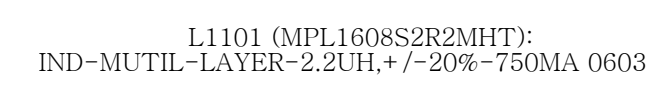
The diagram illustrates the connection of the MSM8660Q to the SDG10000. The MSM8660Q pinout is shown on the left, and the SDG10000 pinout is shown on the right. The physical layout of the board is shown in the center, with various components like resistors (R1102, R1104, R1103) and capacitors (C1103) placed near the pins. A large 'R' watermark is visible in the background.

MSM8660Q Pinout:

- PM_PON_RESET_N [4,5]
- RESIN_N
- RESOUT_N
- MODE_0
- MODE_1
- MSM_PS_HOLD [5,25]
- PS_HOLD
- ITAG_RESOUT_N [25]
- SIRST_N
- ITAG_TCK [25]
- TCK
- ITAG_TDI [25]
- TDI
- ITAG_TDO [25]
- TDO
- ITAG_TMS [25]
- TMS
- ITAG_TRST_N [25]
- TRST_N
- VREF_LP0DK3 [5,16]
- VREF_EB0_CH0_CA
- VSSK_0_1
- PULL_TEST_SE

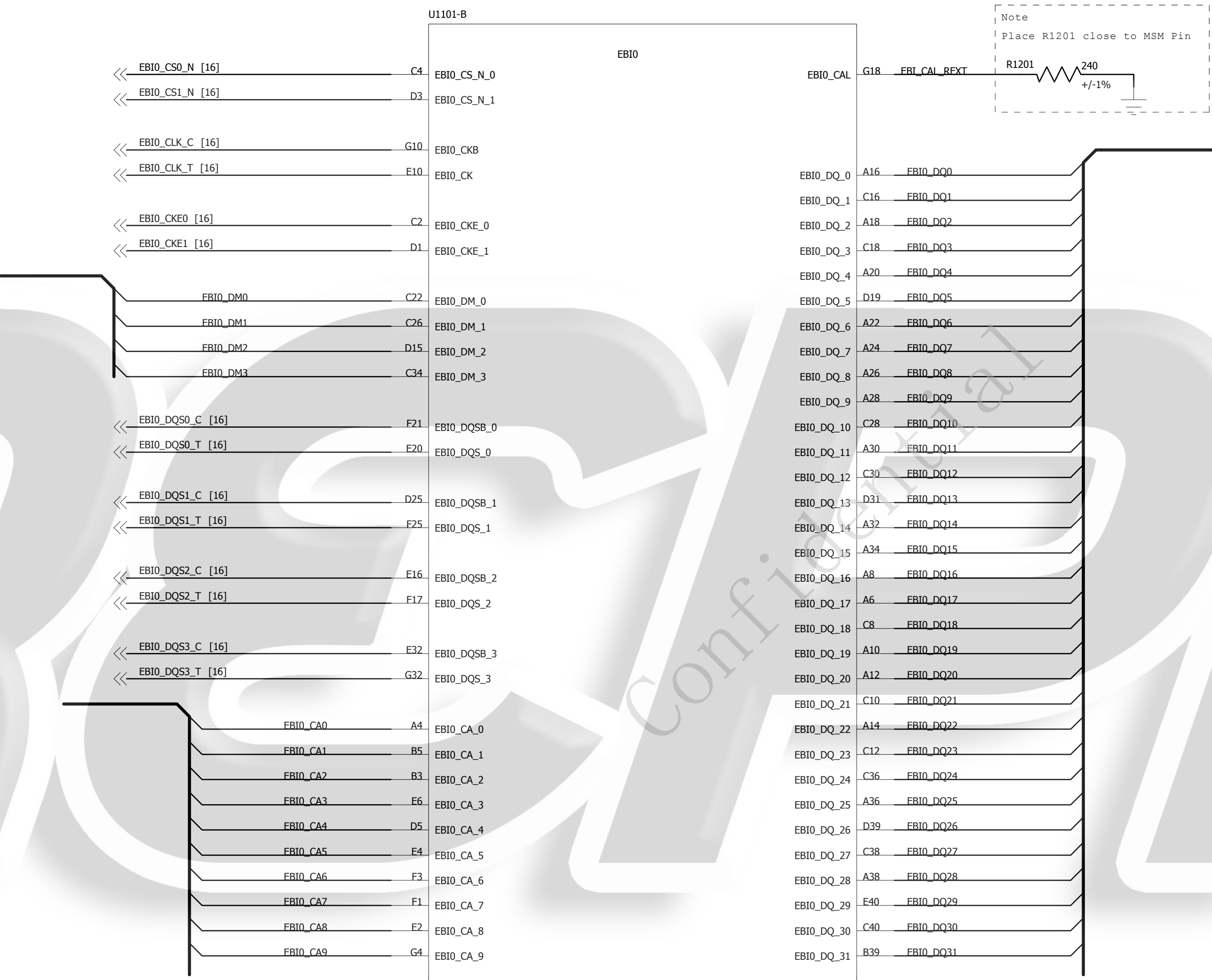
SDG10000 Pinout:

- SDC1_DATA_0 [16]
- SDC1_DATA_1 [16]
- SDC1_DATA_2 [16]
- SDC1_DATA_3 [16]
- SDC1_DATA_4 [16]
- SDC1_DATA_5 [16]
- SDC1_DATA_6 [16]
- SDC1_DATA_7 [16]
- SDC2_CLK_1 [23]
- SDC2_CLK_2 [23]
- SDC2_CLK_3 [23]
- SDC2_CLK_4 [23]
- SDC2_CLK_5 [23]
- SDC2_CLK_6 [23]
- SDC2_CLK_7 [23]
- SDC2_CLK_8 [23]
- SDC2_CLK_9 [23]
- SDC2_CLK_10 [23]
- SDC2_CLK_11 [23]
- SDC2_CLK_12 [23]
- SDC2_CLK_13 [23]
- SDC2_CLK_14 [23]
- SDC2_CLK_15 [23]
- SDC2_CLK_16 [23]
- SDC2_CLK_17 [23]
- SDC2_CLK_18 [23]
- SDC2_CLK_19 [23]
- SDC2_CLK_20 [23]
- SDC2_CLK_21 [23]
- SDC2_CLK_22 [23]
- SDC2_CLK_23 [23]
- SDC2_CLK_24 [23]
- SDC2_CLK_25 [23]
- SDC2_CLK_26 [23]
- SDC2_CLK_27 [23]
- SDC2_CLK_28 [23]
- SDC2_CLK_29 [23]
- SDC2_CLK_30 [23]
- SDC2_CLK_31 [23]
- SDC2_CLK_32 [23]
- SDC2_CLK_33 [23]
- SDC2_CLK_34 [23]
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- SDC2_CLK_203 [23]
- SDC2_CLK_204 [23]
- SDC2_CLK_205 [23]
- SDC2_CLK_206 [23]
- SDC2_CLK_207 [23]
- SDC2_CLK_208 [23]
- SDC2_CLK_20



Note
MIPI_DSI can be configured as DCDC or LDO mode.
When configured as LDO mode, pin AF9 should be left floating and L1101 should be DNI

MSM8920 EBI

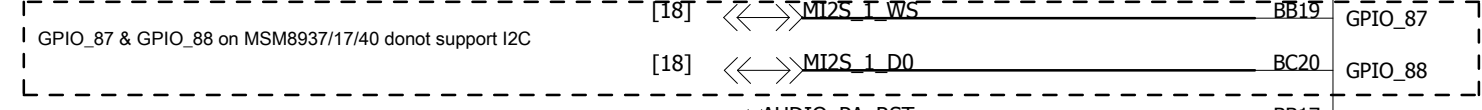
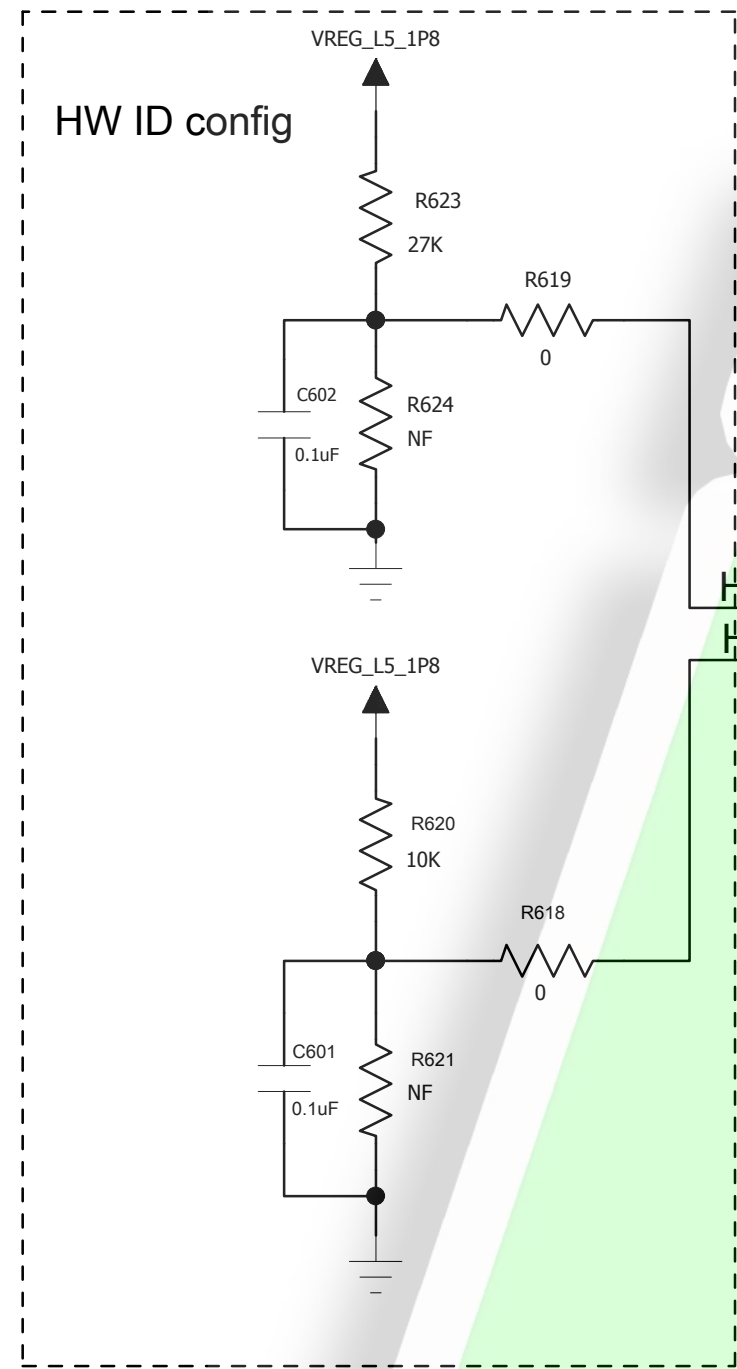


PROJECT:	REV: <REV>
DOCUMENT NO.: 12_MSM8920: EBI	SIZED: D
DEPARTMENT: Hardware DEPT.	
COMPANY: WINGTECH	
DESIGNER: <DESIGNER>	Last Saved Date: 2017/11/20
	SHEET: 10 OF 35

01	GP_PDM_AD [4]	→
13	SDCARD_DEF_N [5,23]	→
32		
18	CDC_PDM_CLK [8]	→
19	CDC_PDM_SYNC [8]	→
26	CDC_PDM_TX [8]	→
27	CDC_PDM_RXD [8]	→
28	CDC_PDM_RXI [8]	→

Note: Asterisks (*) indicate modem power management (MPM) wake-up pins

R1315	R1218	R1317	R1327	QPO12810	
Y8	NC	Y8	NC	NEL	S88739AA1
NC	Y8	Y8	NC	LAL	S88739BA1
Y8	NC	NC	Y8	HEH	S88740AA1
NC	Y8	NC	Y8	LEH	S88740BA1



Note: Do not have pull-ups on the following GPIOs unless intend for boot or secure- boot related configurations:
GPIO_91, 107, 109

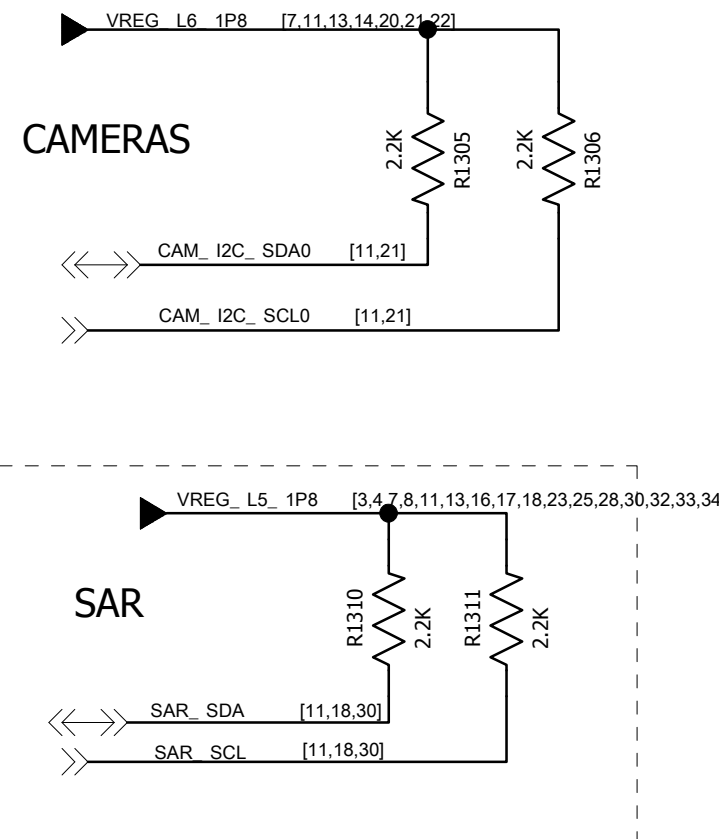
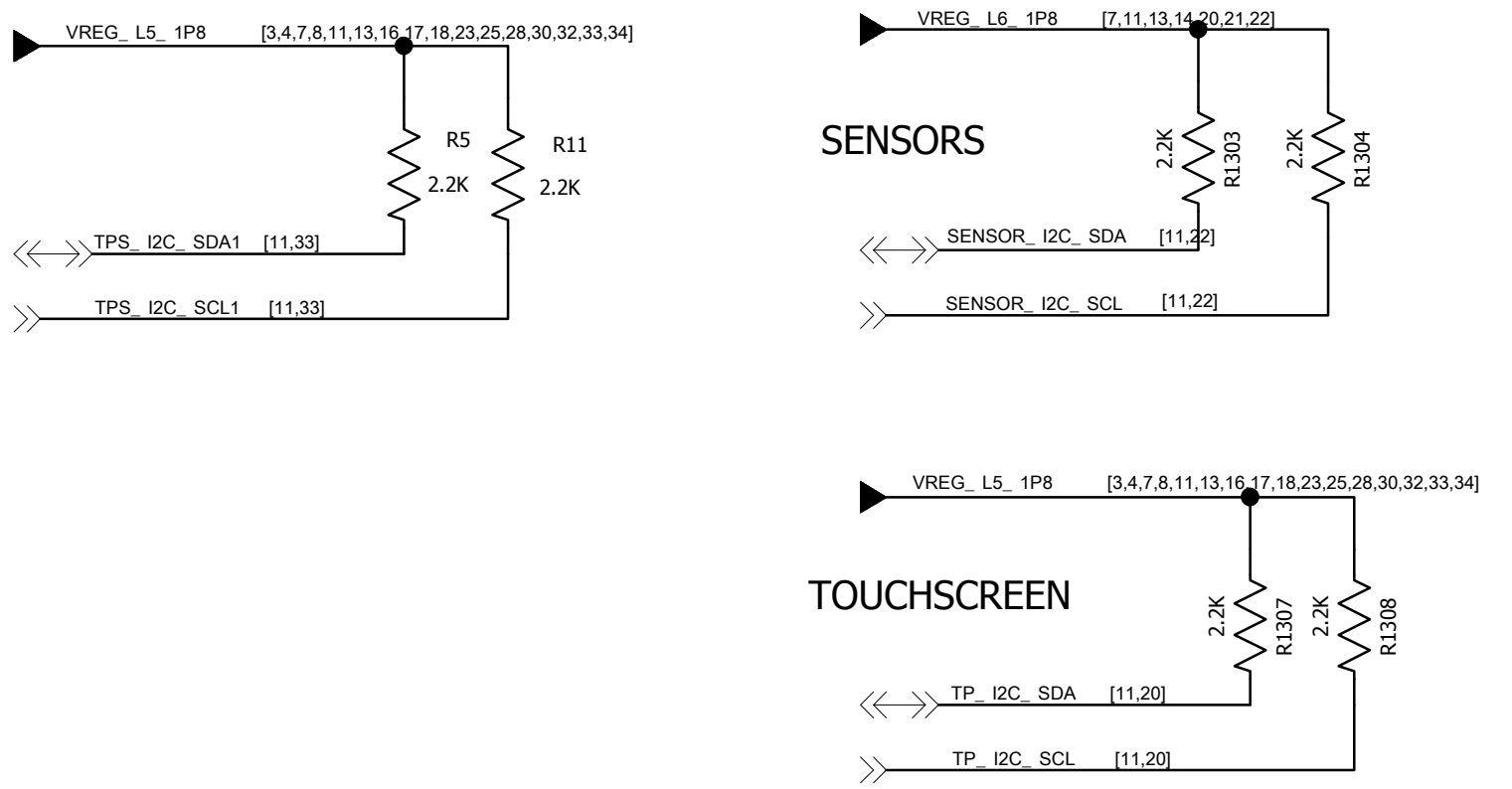
GPIO	BOOT_CONFIG
GPIO_3	BOOT_CONFIG[0]/WDG_DISABLE
GPIO_111	BOOT_CONFIG[1]
GPIO_112	BOOT_CONFIG[2]
GPIO_88	BOOT_CONFIG[3]

BOOT_CONFIG[3:1]	BOOT_CONFIG
0b000	SDC1 -> SDC2 -> USB2.0
0b010	SDC1
0b100	SDC2-> SDC1

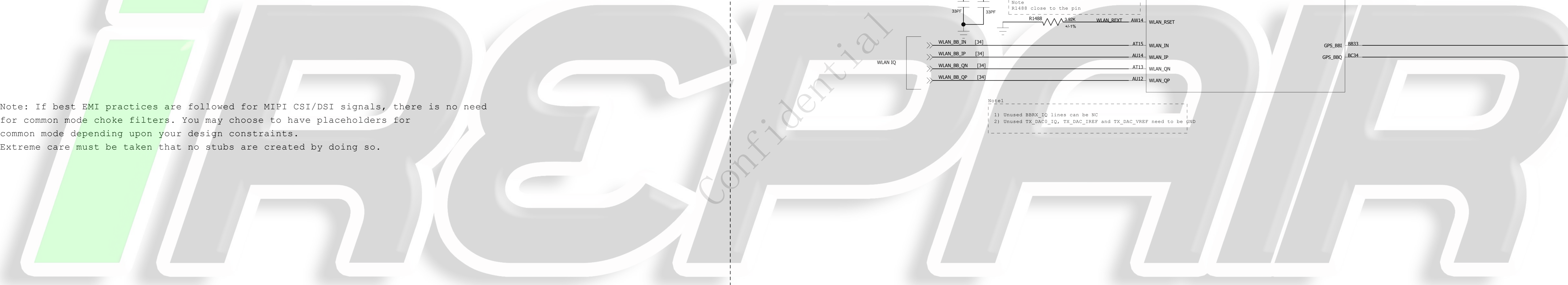
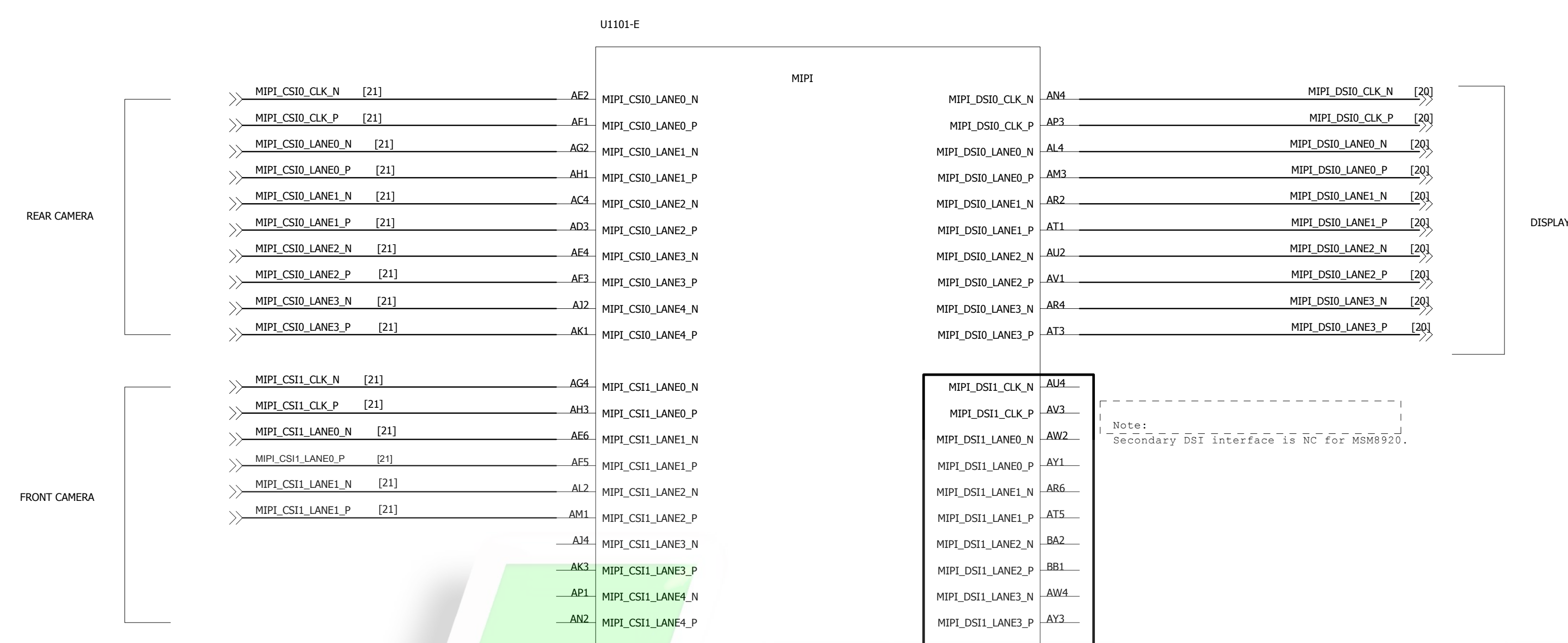
Default Boot Config (0b000) is SDC1(eMMC)

I2C PULL-UP RESISTORS

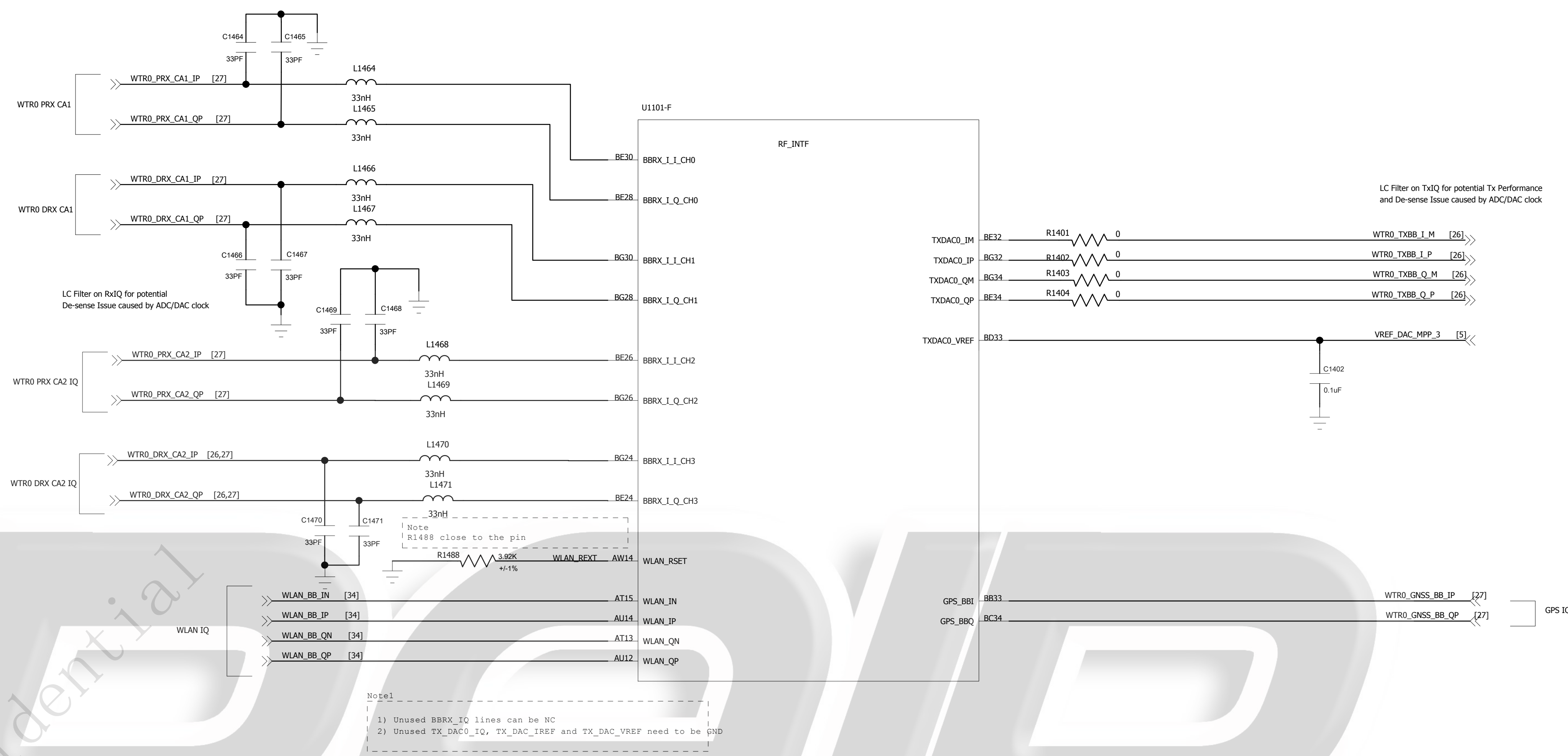
Note: Ensure SW sets these GPIOs (Sensor, CTP and Camera I2C bus) to inout pull down when the peripherals are powered off to eliminate leakage.



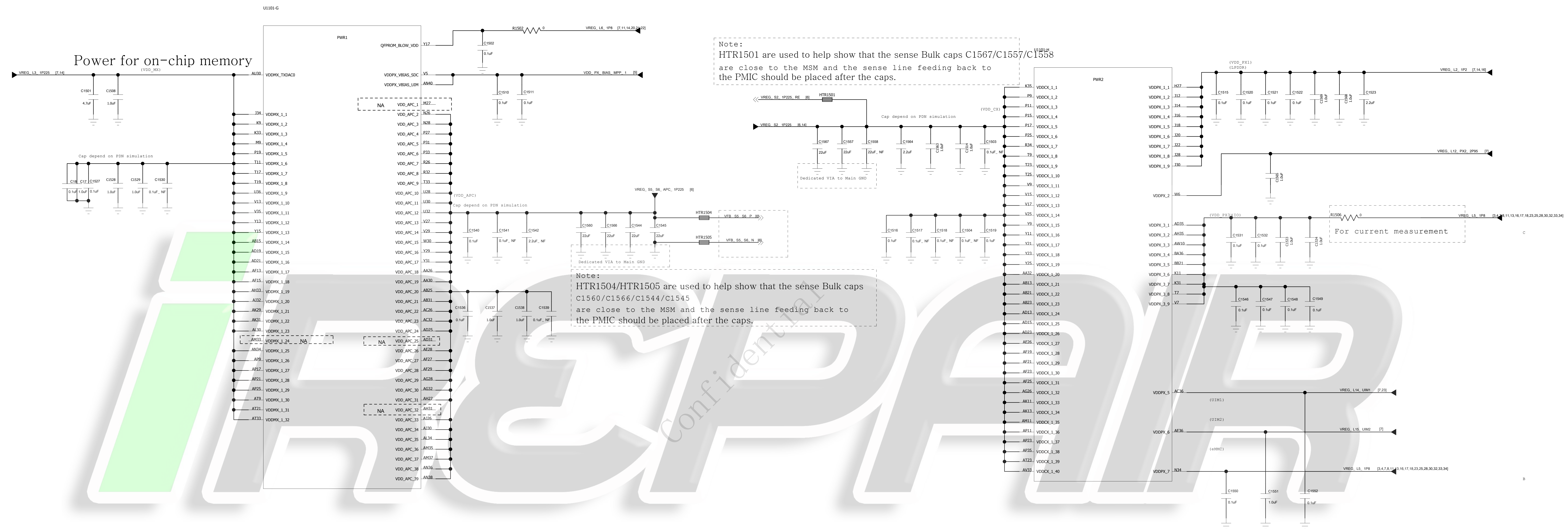
MSM8920 MIPI and RF Control



Note: If best EMI practices are followed for MIPI CSI/DSI signals, there is no need for common mode choke filters. You may choose to have placeholders for common mode depending upon your design constraints. Extreme care must be taken that no stubs are created by doing so.

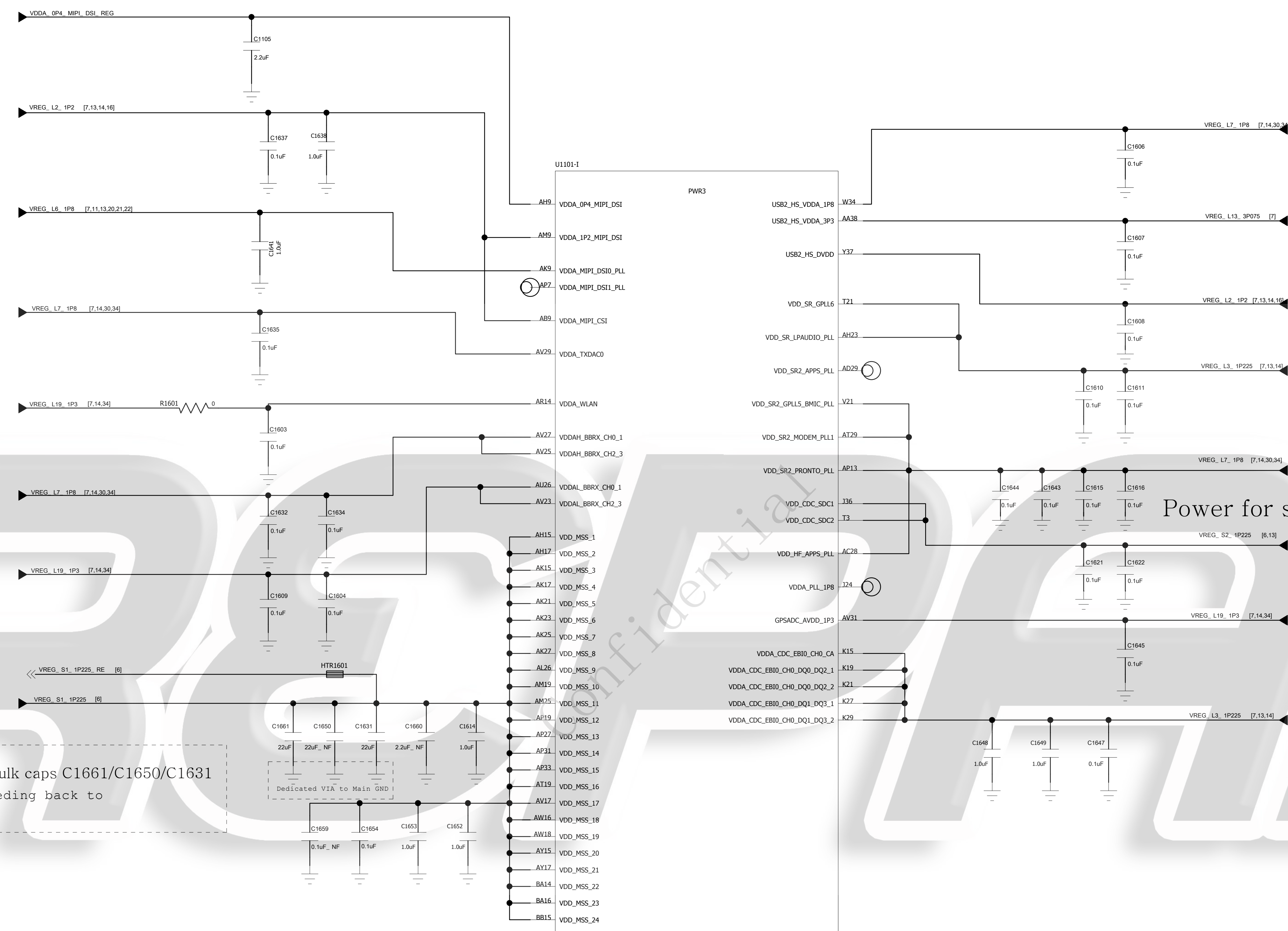


MSM8920 POWER1



PROJECT:			REV:	<REV>
DOCUMENT NO.:	_15_MSM8920_-POWER1		SIZED:	D
DEPARTMENT:	Hardware DEPT.			
COMPANY:				
DESIGNER:	<DESIGNER>	Limit Saved Date:	2017/11/20	SHEET: 13 OF 35

MSM8920 POWER2

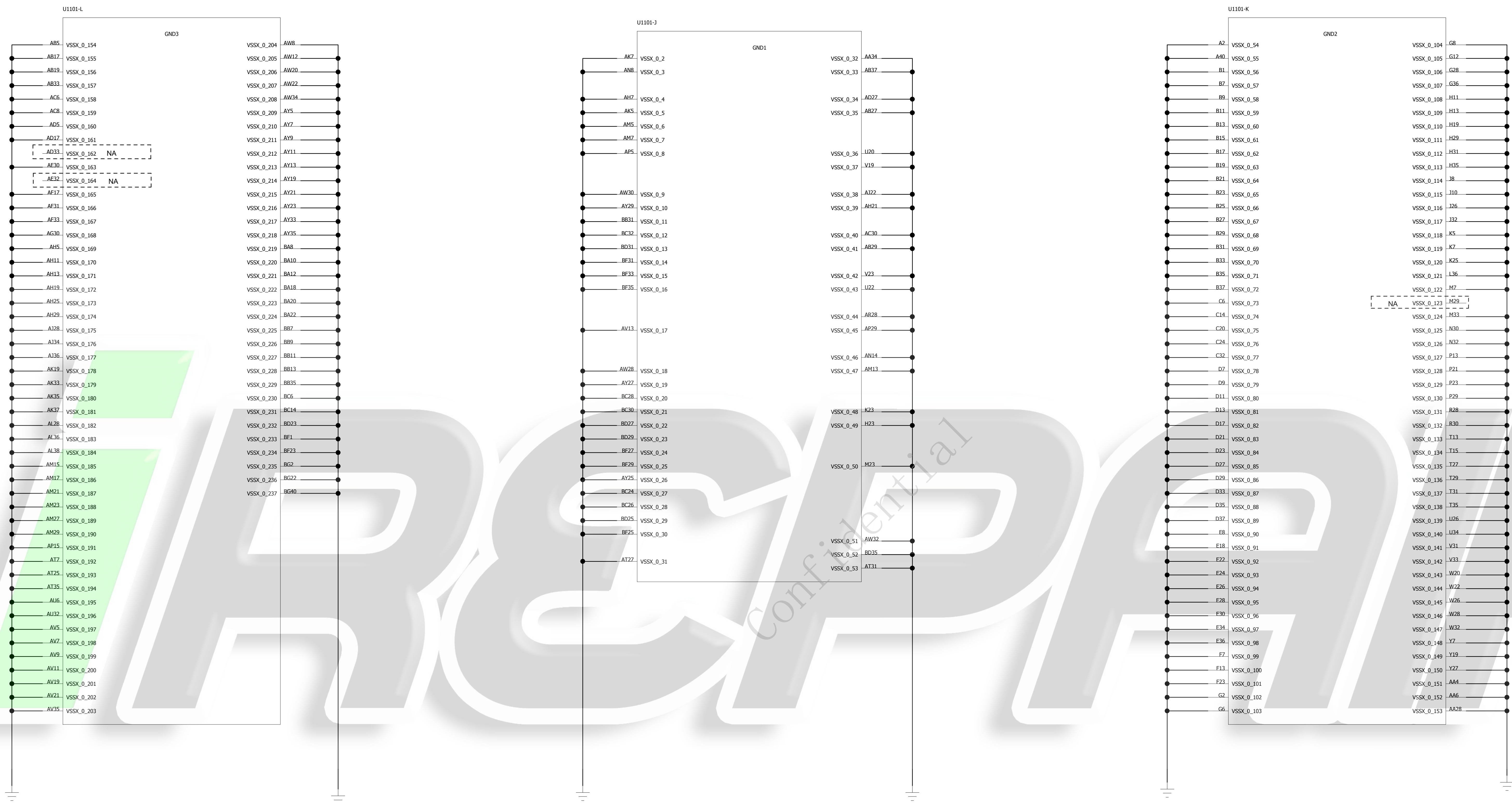


Note:
HTRI601 are used to help show that the sense Bulk caps C1661/C1650/C1631 are close to the MSM and the sense line feeding back to the PMIC should be placed after the caps.

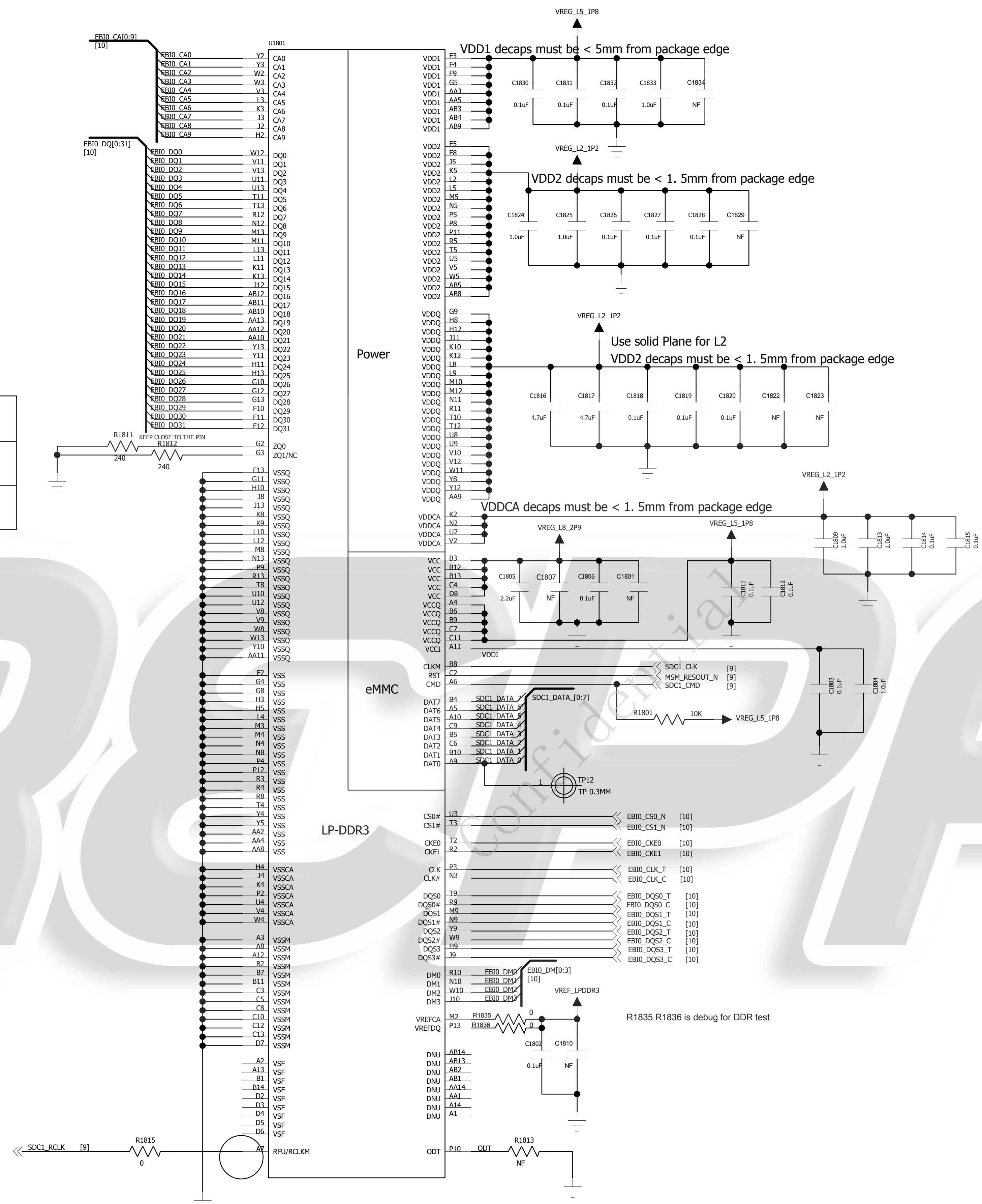
Power for secure digital calibration delay circuits

PROJECT:		REV:	<REV>
DOCUMENT NO.:	16_MSM8920_POWER2	SIZED:	D
DEPARTMENT:	Hardware DEPT.		
COMPANY:			
DESIGNER:	<DESIGNER>	Issue Saved Date:	2017/11/20
		SHEET:	14 OF 35

MSM8920 GND



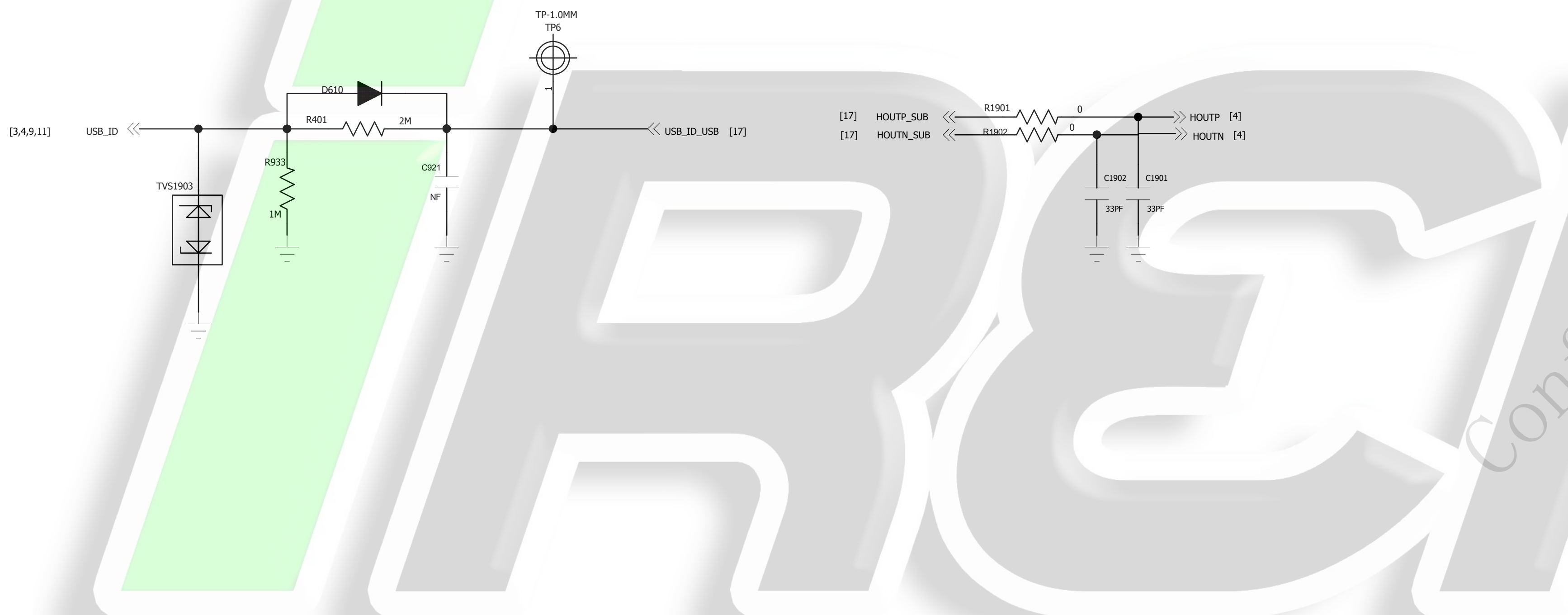
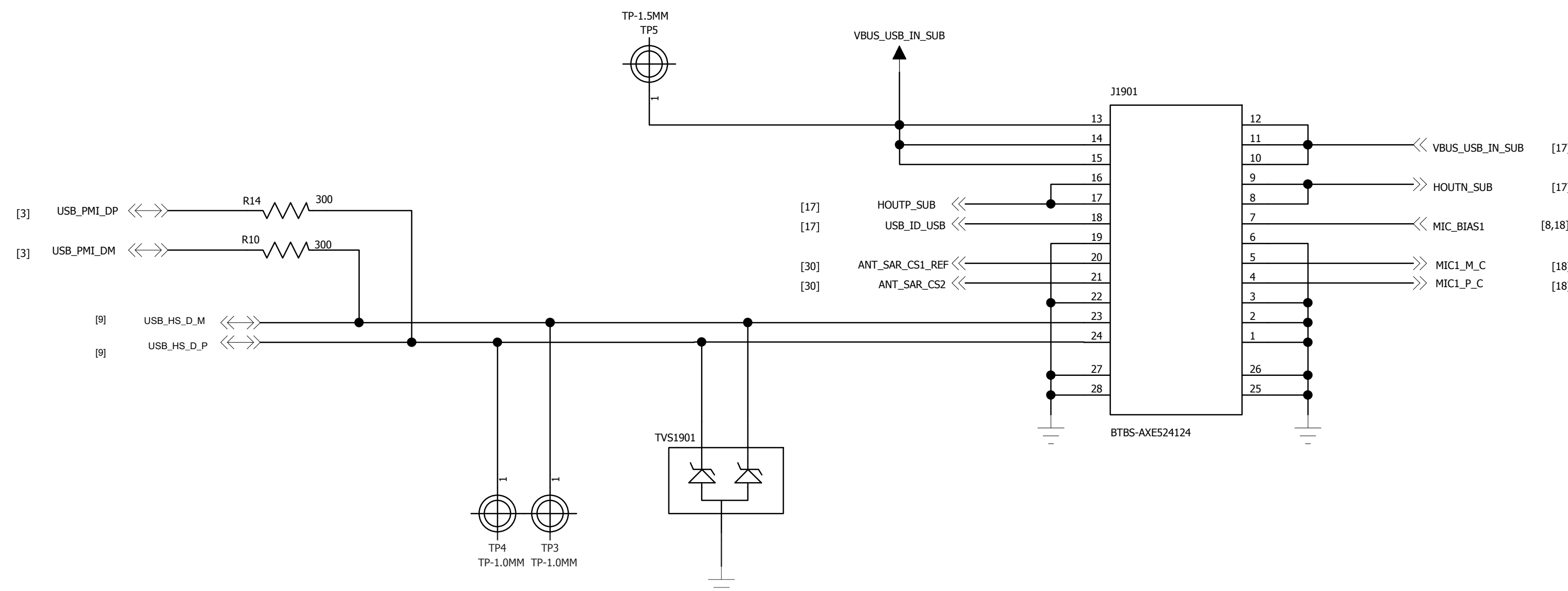
MEMORY:LPDDR3+ EMMC



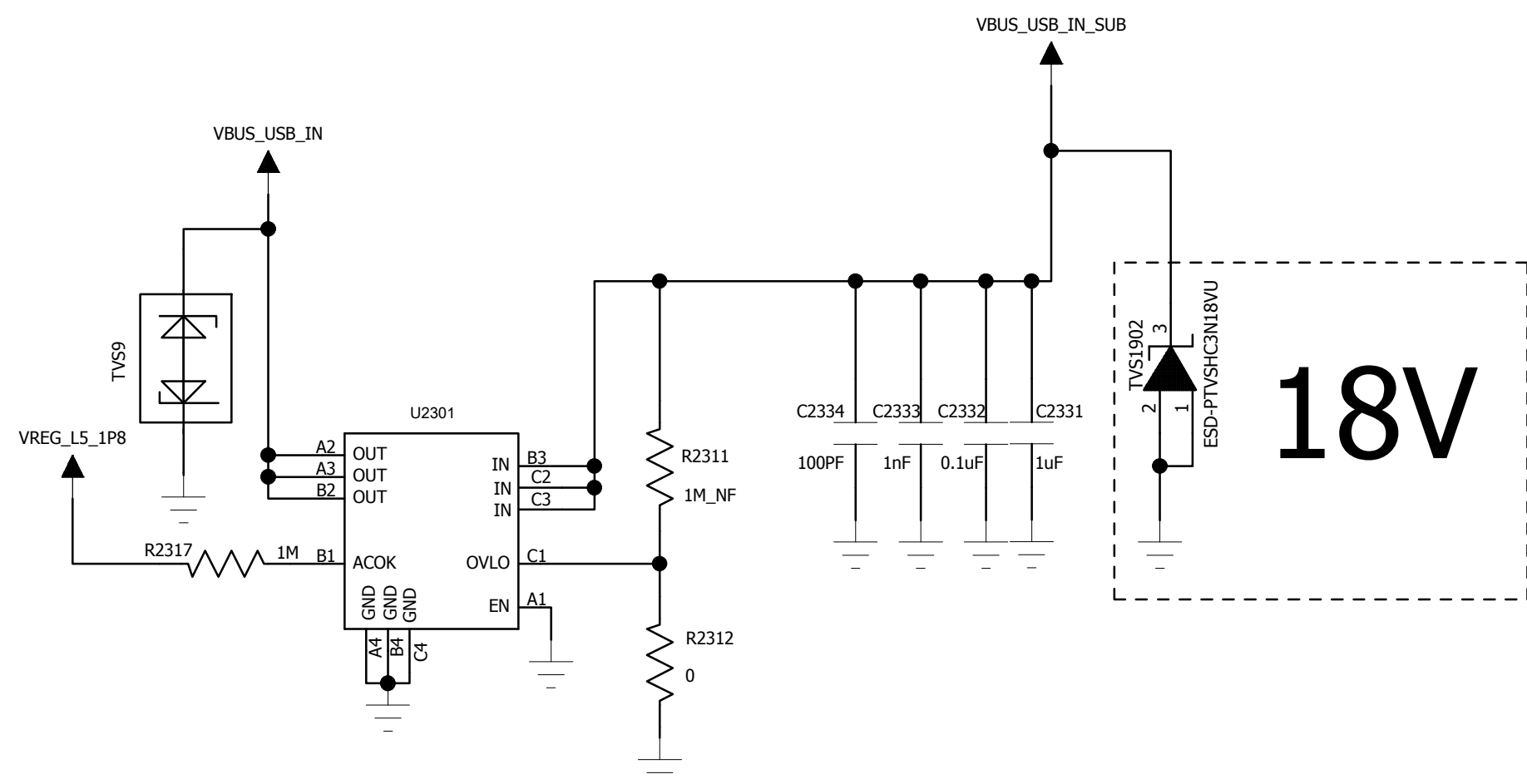
Note: Pull-up resistors on SDC1_DATA are PCB and eMMC vender dependent.

	MICRO/HYNIX	SAMSUNG
U1501.G3	ZQ1	NC
U1501.A7	RFU	RCLK

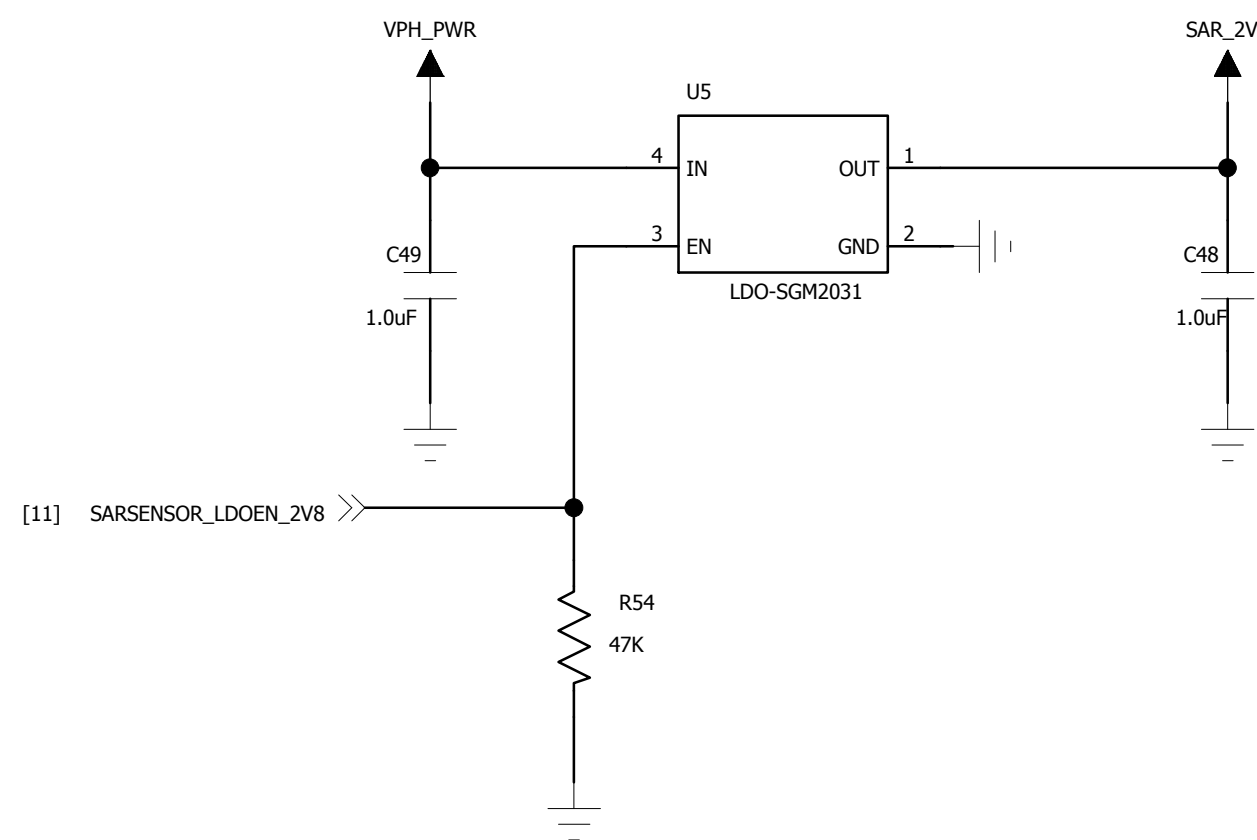
USB-MIC+Motor+RF



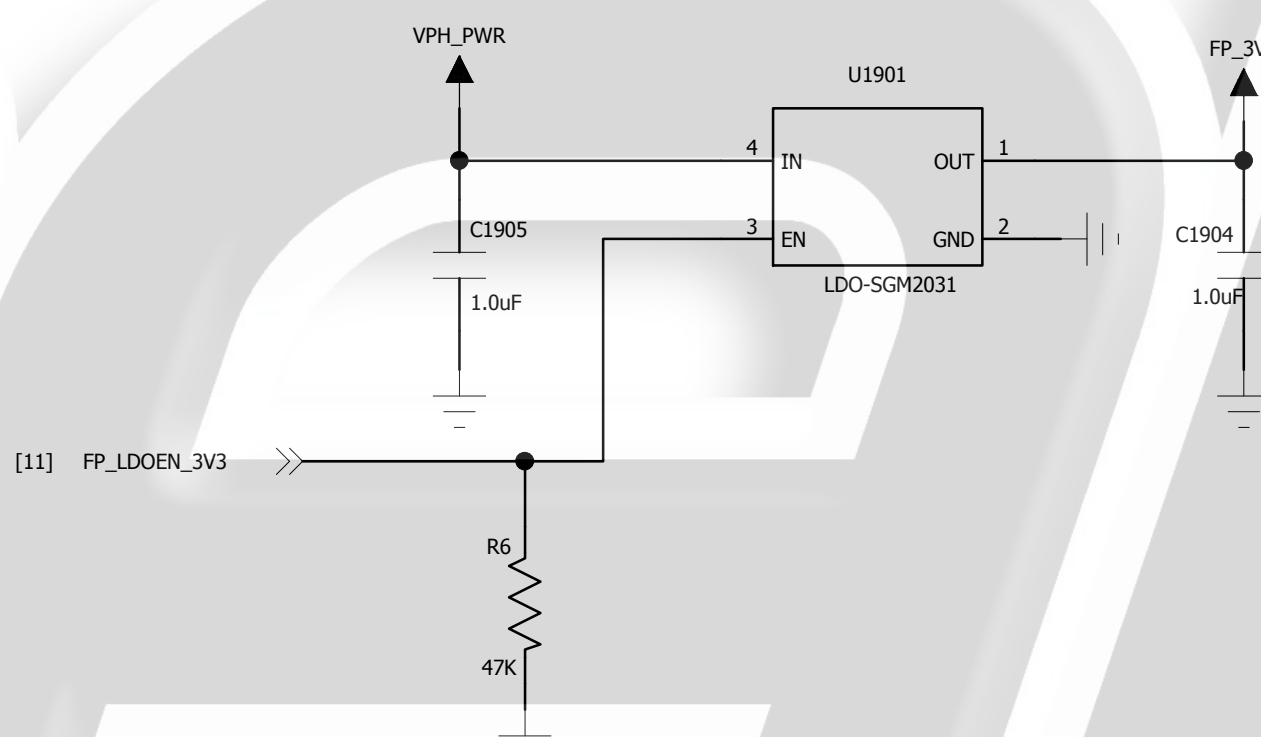
OVP



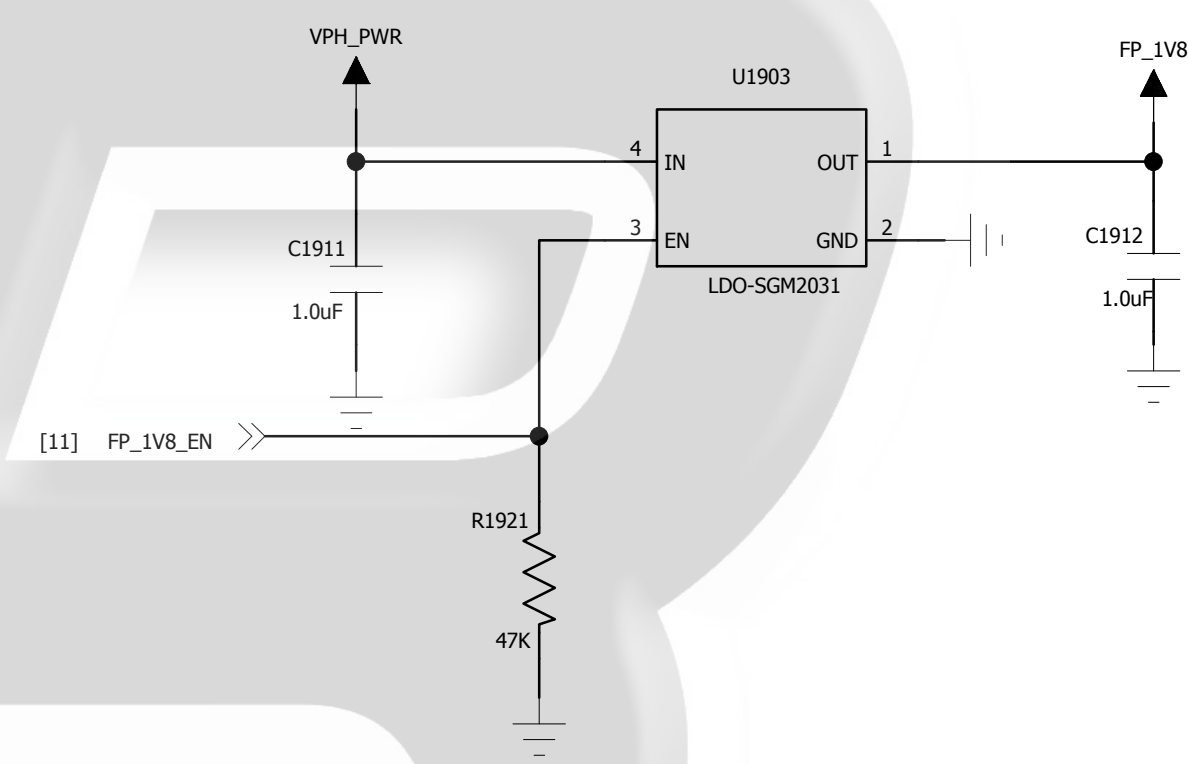
SAR SENSOR 2.8V LDO



3.3V LDO

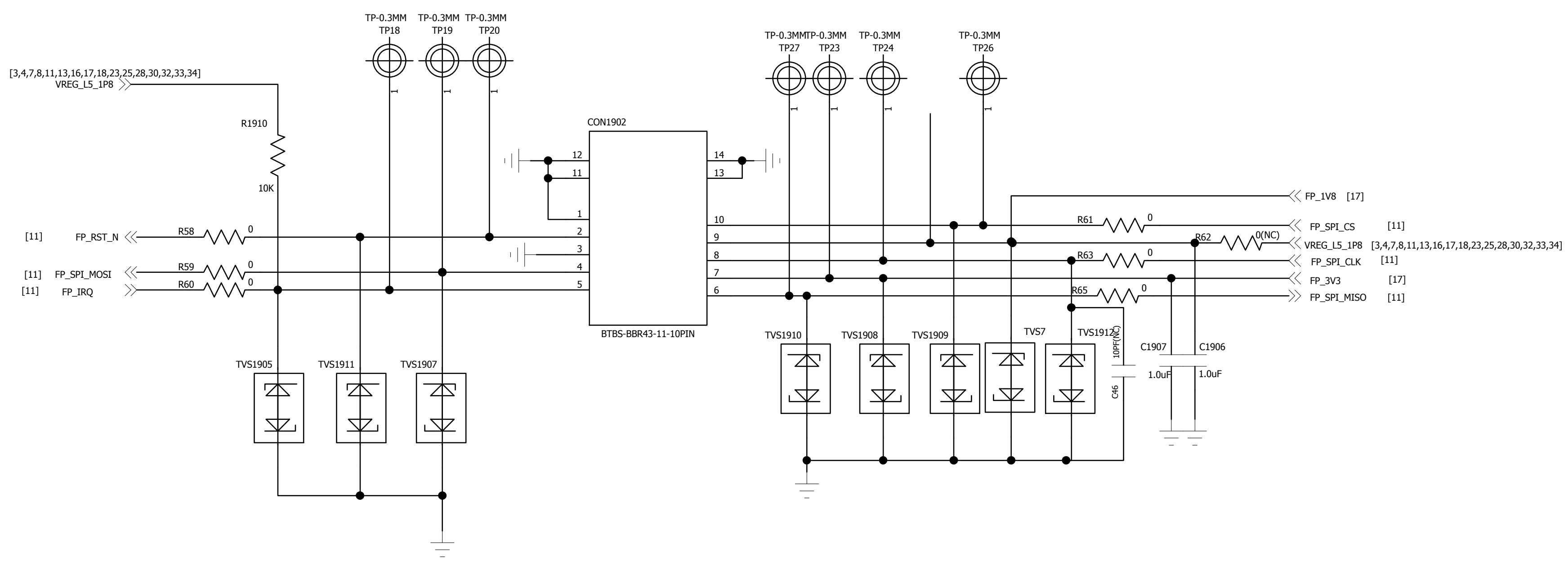


1.8V LDO



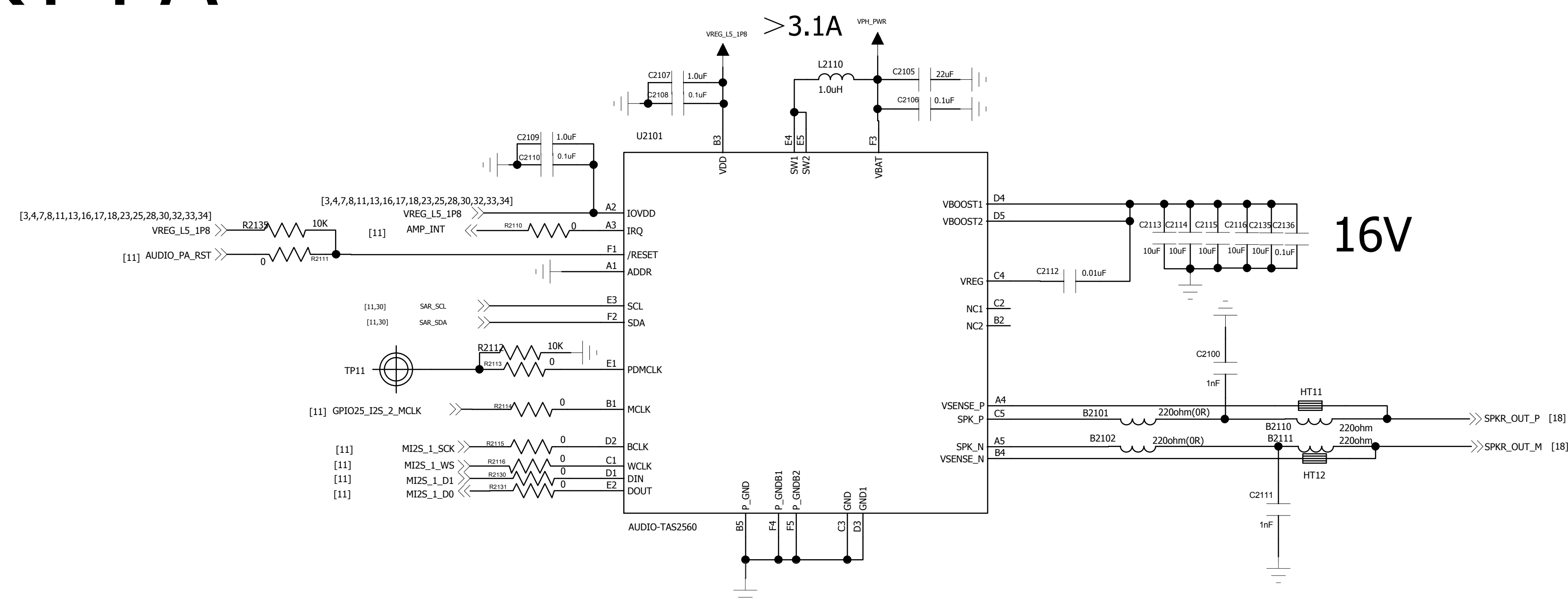
IC	
ET516B	

Finger Print

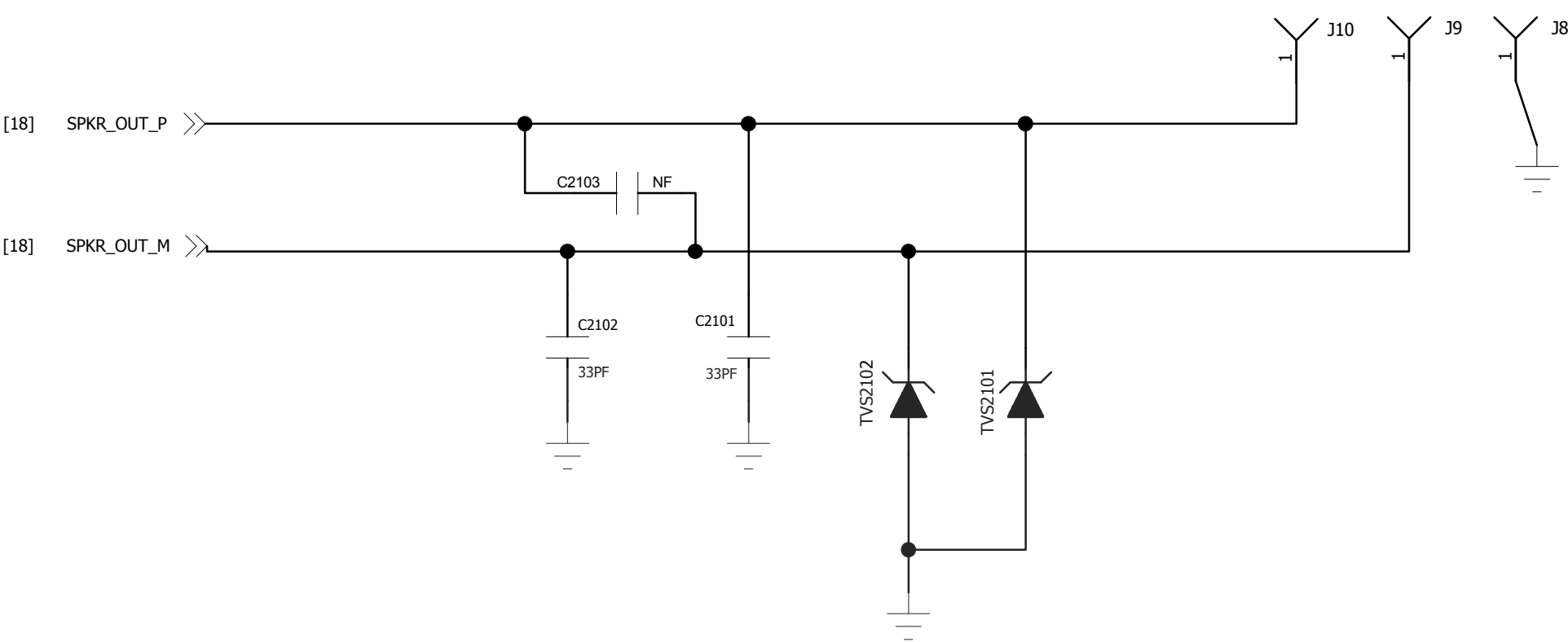


AUX MIC/RECEIVER/SMART PA

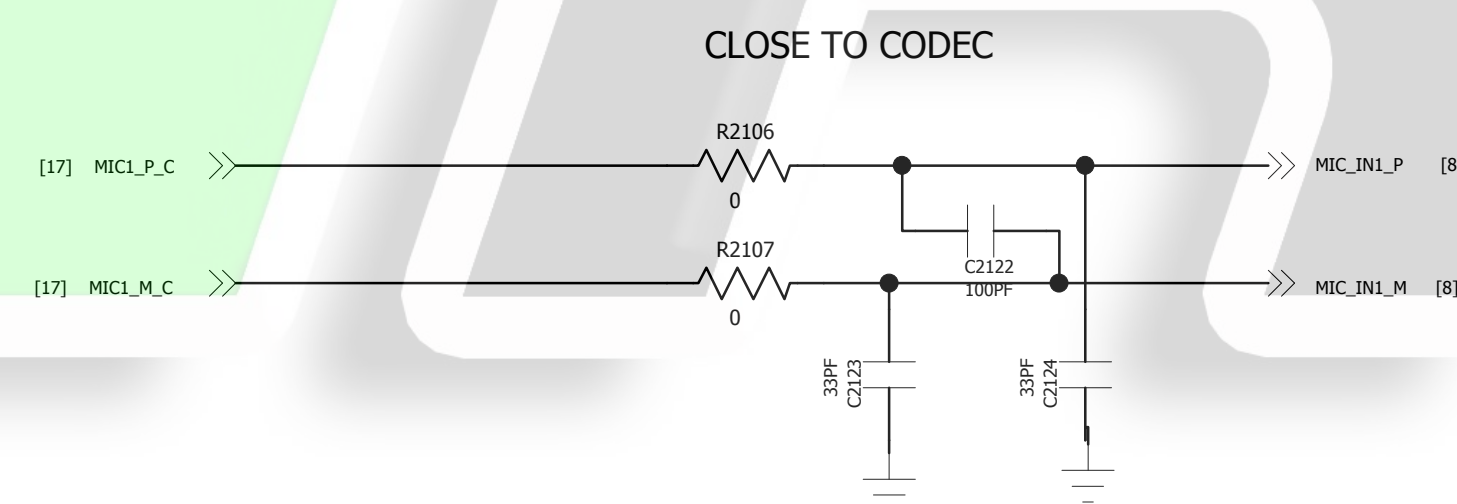
SMART PA



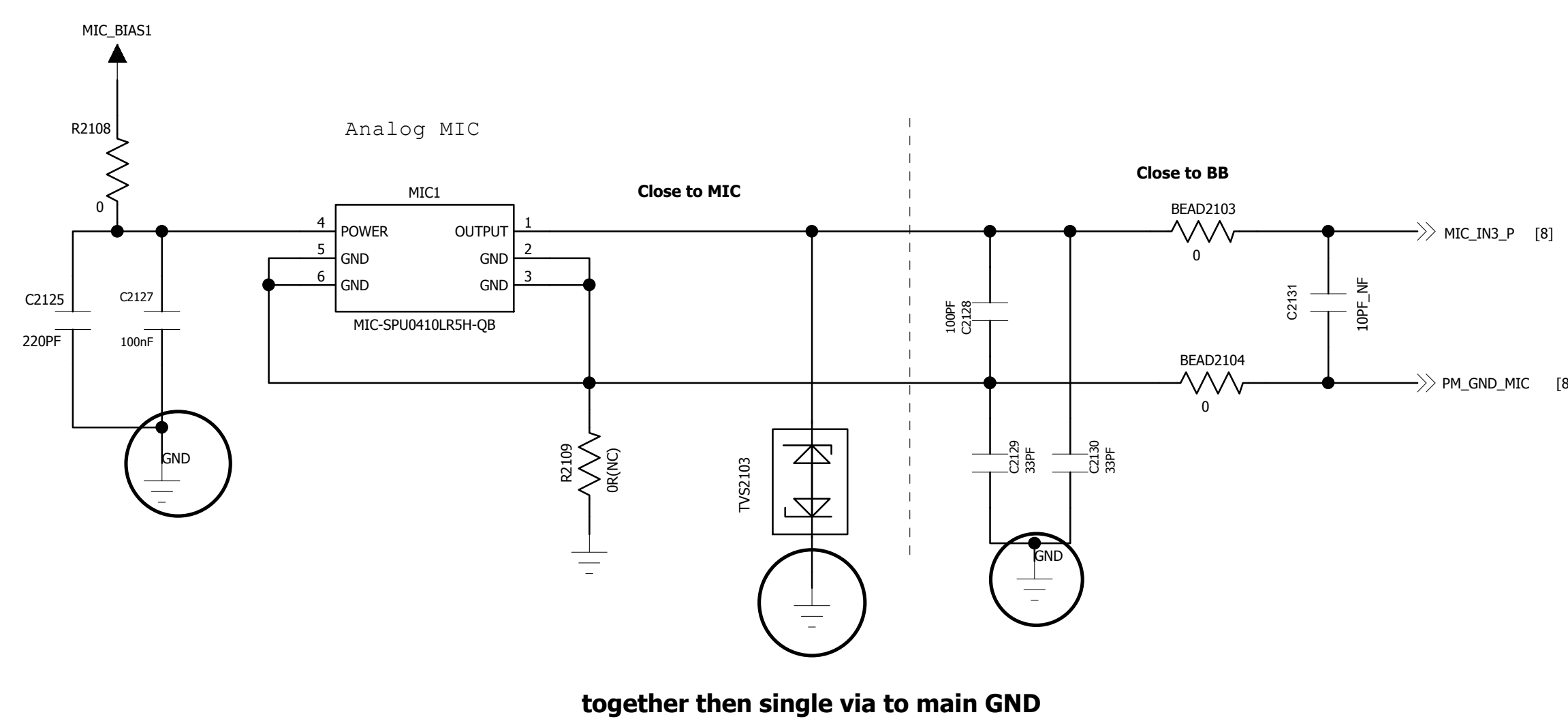
RECEIVER&SPK



PRIMARY MIC

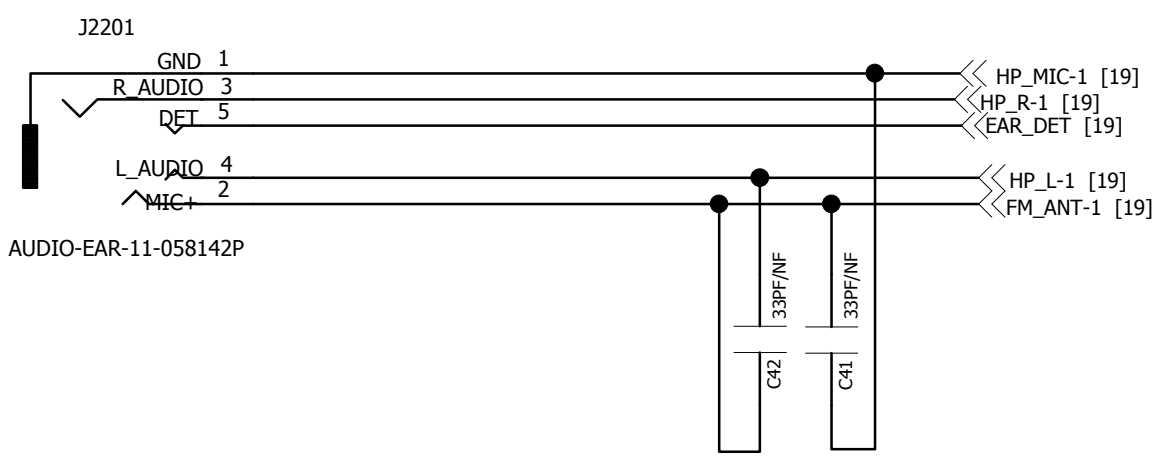


AUXMIC



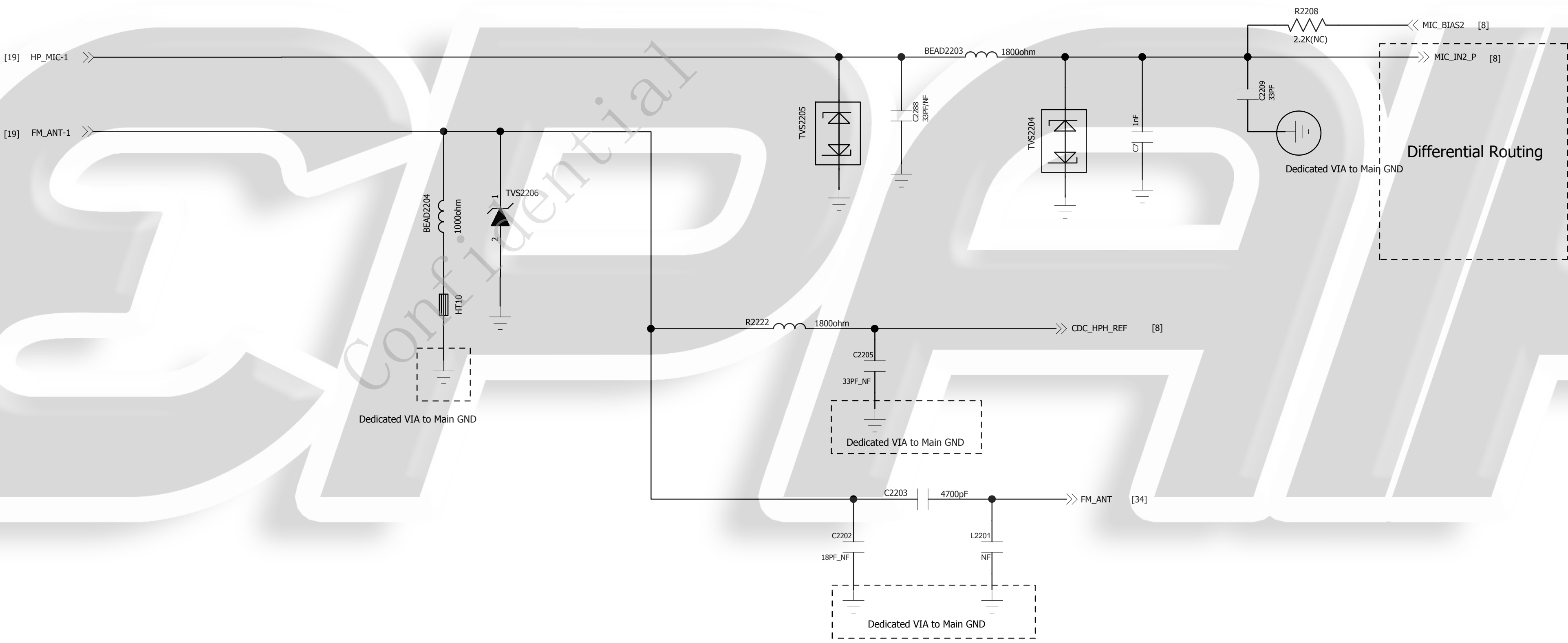
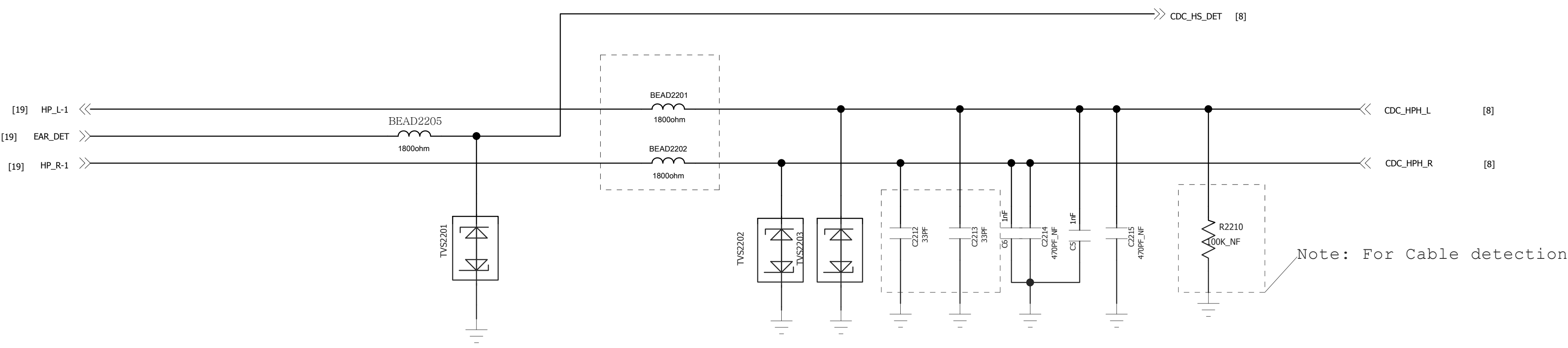
EARPHONE

AUDIO JACK



American Standard
HEADSET PLUG IN:HP_JACKDET "0"
HEADSET PLUG OUT:HP_JACKDET "1"

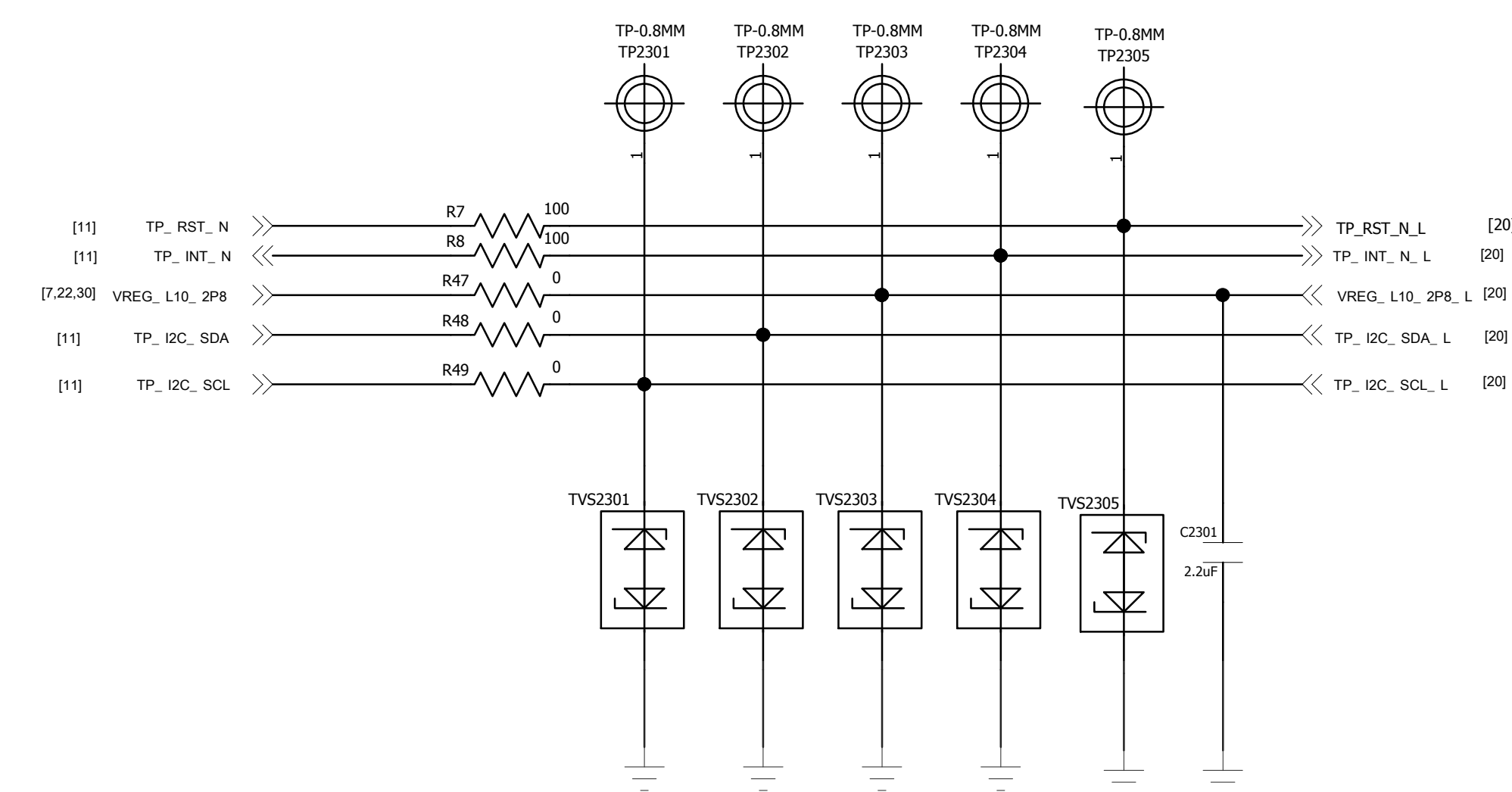
Note: Ferrite beads and their corresponding bypass capacitors on CDC_HPH_L, CDC_HPH_R and CDC_HPH_REF are needed to reduce noise generated by audio/FM concurrency



PROJECT:	22_EARPHONE	REV:	<REV>
DOCUMENT NO.:	22_EARPHONE	SIZED:	D
DEPARTMENT:	Hardware DEPT.		
COMPANY:			
DESIGNER:	<DESIGNER>	Last Saved Date:	2017/11/20
		SHEET:	19 OF 35

LCD interface and backlight

CTP



Power on sequence: Single Supply

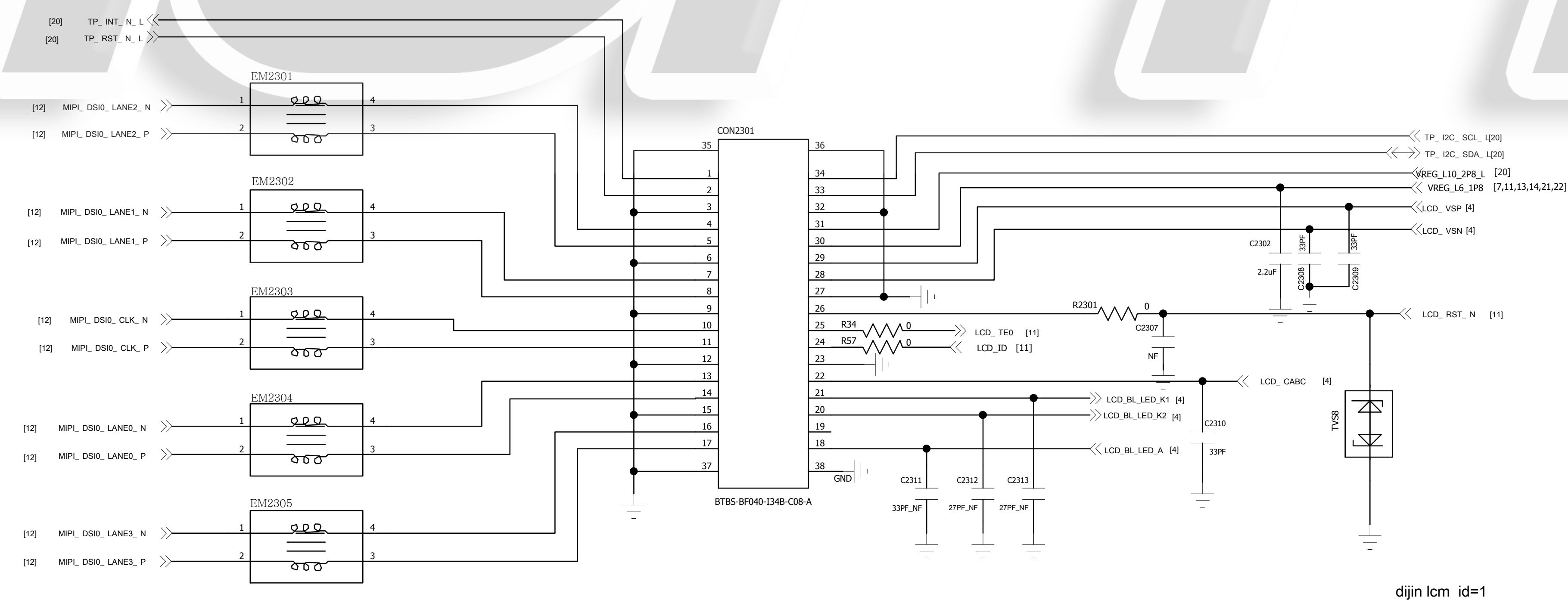
FocalTech: FT5336

I2C Address

Write	0x70	It can be changed by FACTORY.
-------	------	-------------------------------

Read	0x71
------	------

LCD



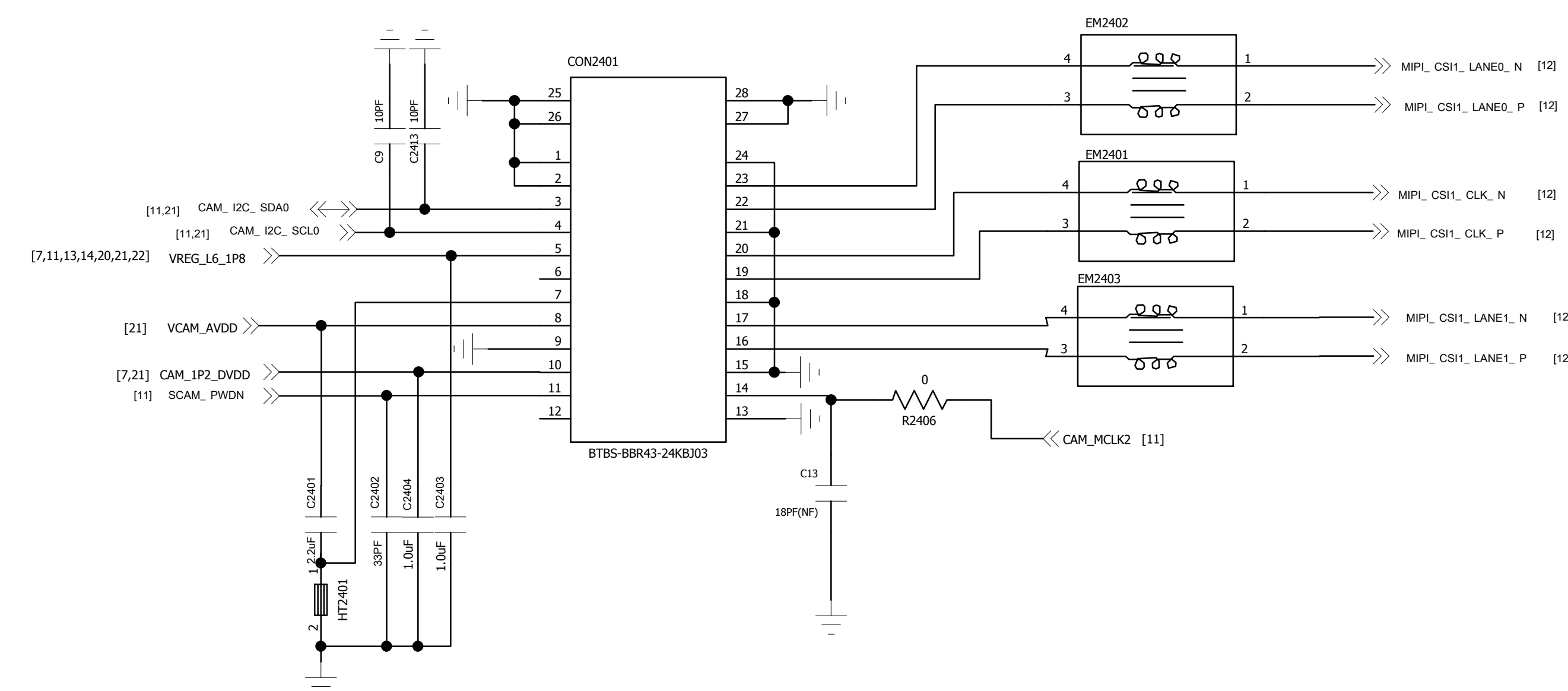
dijin lcm id=1

Note: If best EMI practices are followed for MIPI CSI/DSI signals, there is no need for common mode choke filters. You may choose to have placeholders for common mode depending upon your design constraints. Extreme care must be taken that no stubs are created by doing so.

PROJECT:			REV:	<REV>
DOCUMENT NO.:	23_LCD_INTERFACE/CTP		SIZED:	D
DEPARTMENT:	Hardware DEPT.			
COMPANY:				
DESIGNER:	<DESIGNER>	Last Saved Date:	2017/11/20	SHEET: 20 OF 35

Main/Slave Camera and Flash

FRONT_camera

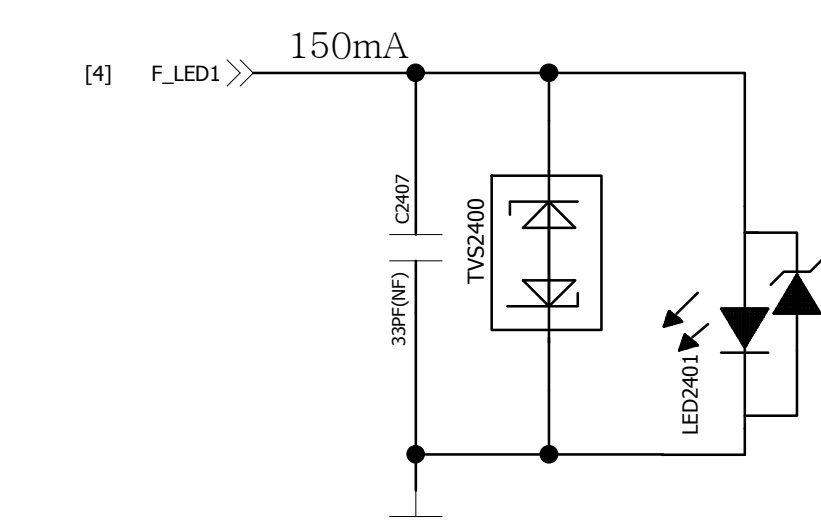
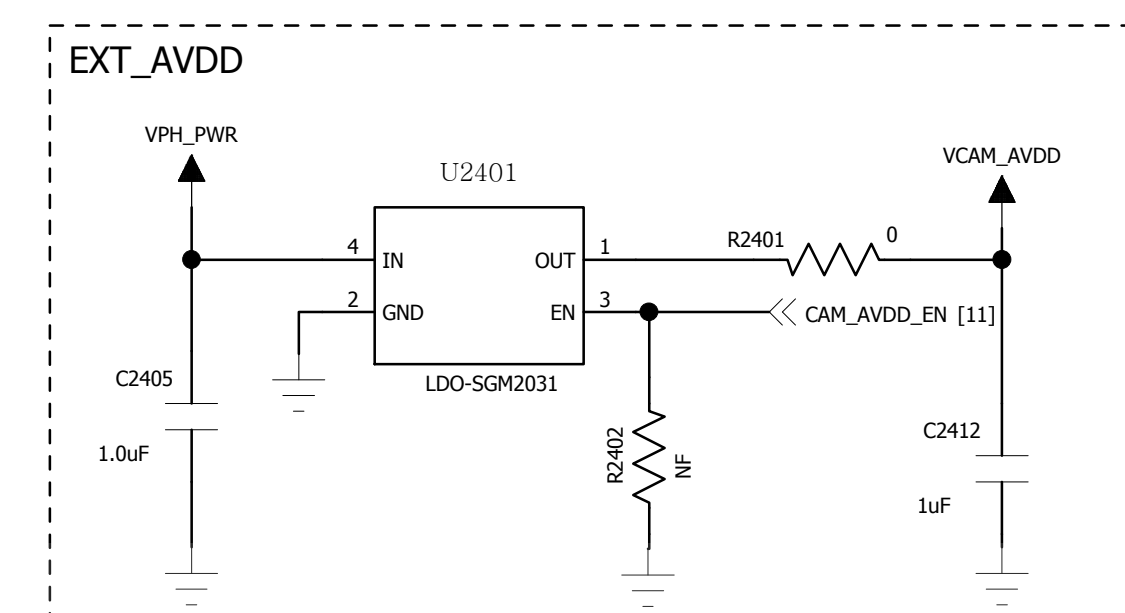


```
front Camera: 5M
Sensor: OV05675-GA4A
AVDD:2.8V
DOVDD:1.8V
DVDD:1.2V
```

I2C Address	
Write	0x6C
Read	0x6D

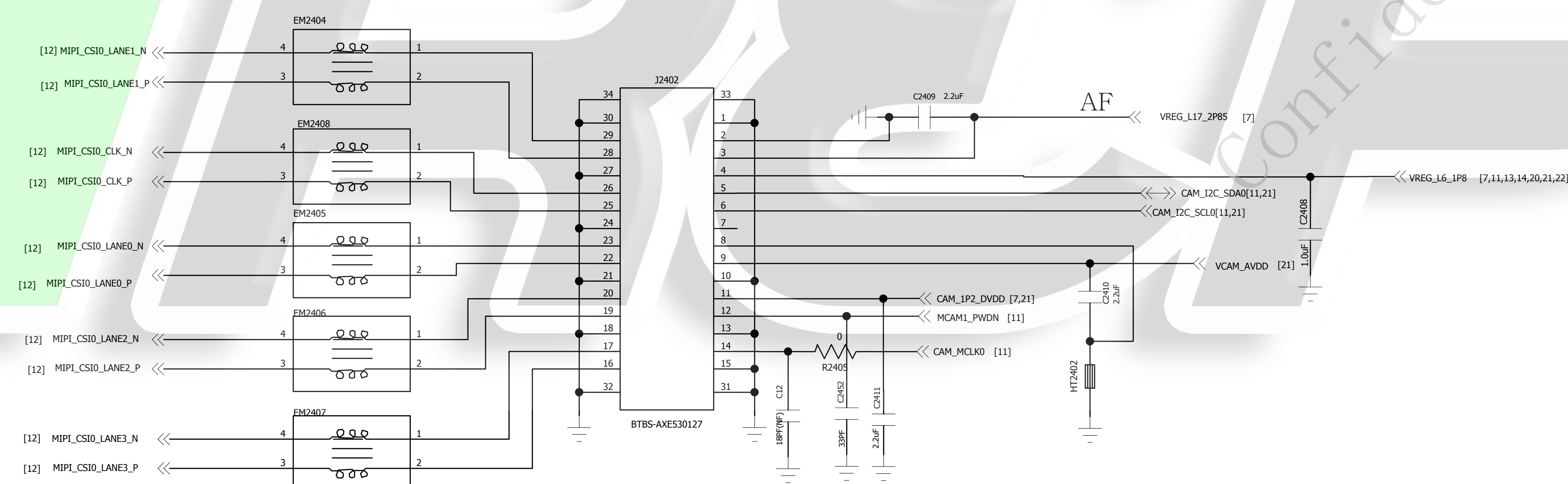
```
front Camera: 5M
Sensor: OV05675-GA4A
AVDD:2.8V
DOVDD:1.8V
DVDD:1.2V
```

I2C Address	
Write	0x6C
Read	0x6D



FLASH LED(front)

Main Camera



Rear Camera: 8M
Sensor: OV08856-GA4A
AVDD:2.8V
DOVDD:1.8V
DVDD:1.2V
AFVDD:2.8V
Driver ID :0X18

I2C Address	
Write	0x20
Read	0x21

Rear Camera: 8M
Sensor: OV08856-GA4A
AVDD:2.8V
DOVDD:1.8V
DVDD:1.2V
AFVDD:2.8V
Driver ID :0X18

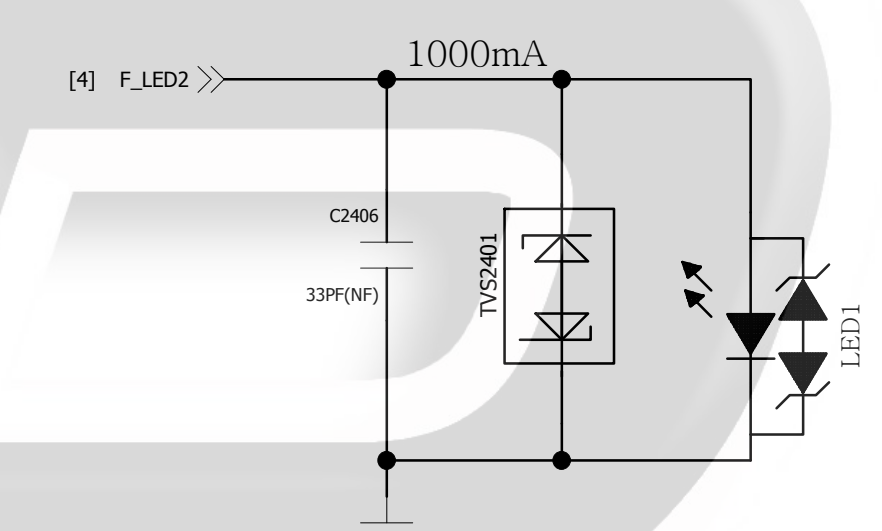
I2C Address	
Write	0x20
Read	0x21

```
Rear Camera: 13M
Sensor: OV13855-GA5A
AVDD:2.8V
DOVDD:1.8V
DVDD:1.2V
AFVDD:2.8V
Driver ID :0X18
```

I2C Address	
Write	0x6C
Read	0x6D

Rear Camera: 13M	
Sensor: S5K3L8XXM3-FGX9	
AVDD:2.8V	I2C
DOVDD:1.8V	Write
DVDD:1.2V	Read
AFVDD:2.8V	
Driver ID :0X18	

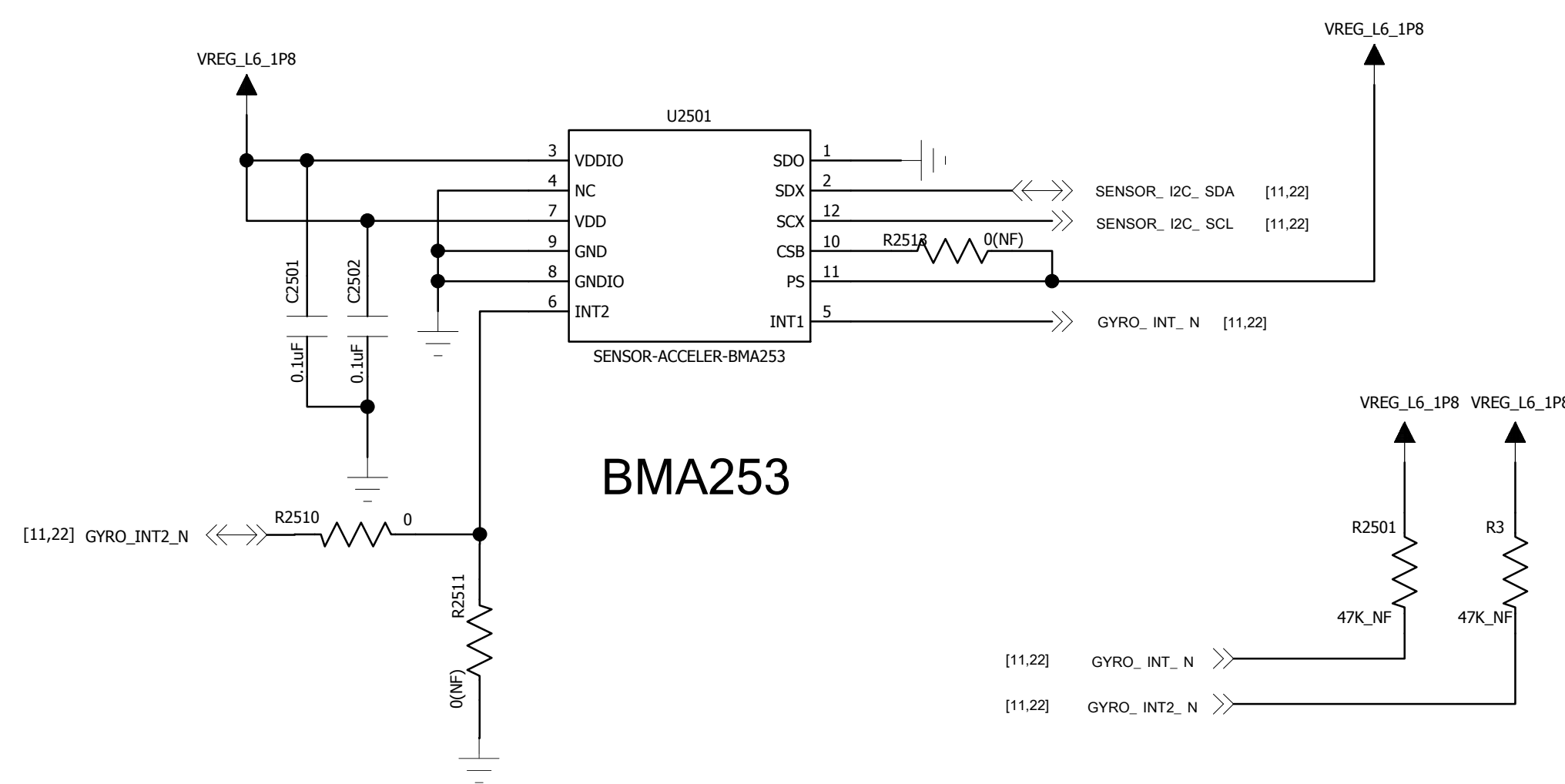
I2C Address	
Write	0x5A
Read	0x5B



FLASH LED(back)

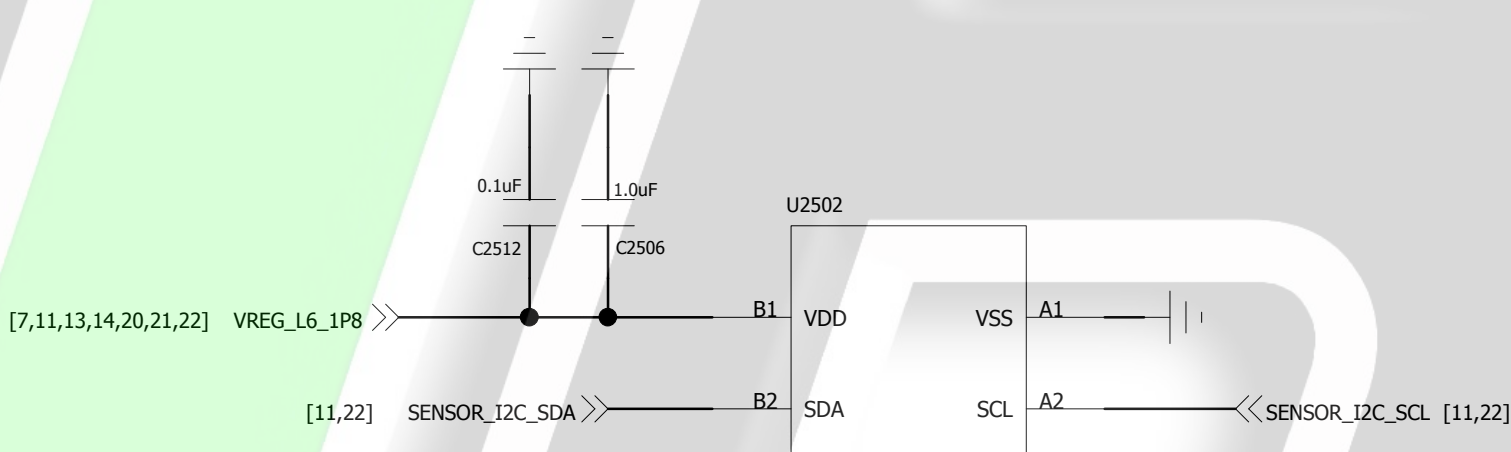
SENSORS

G-Sensor



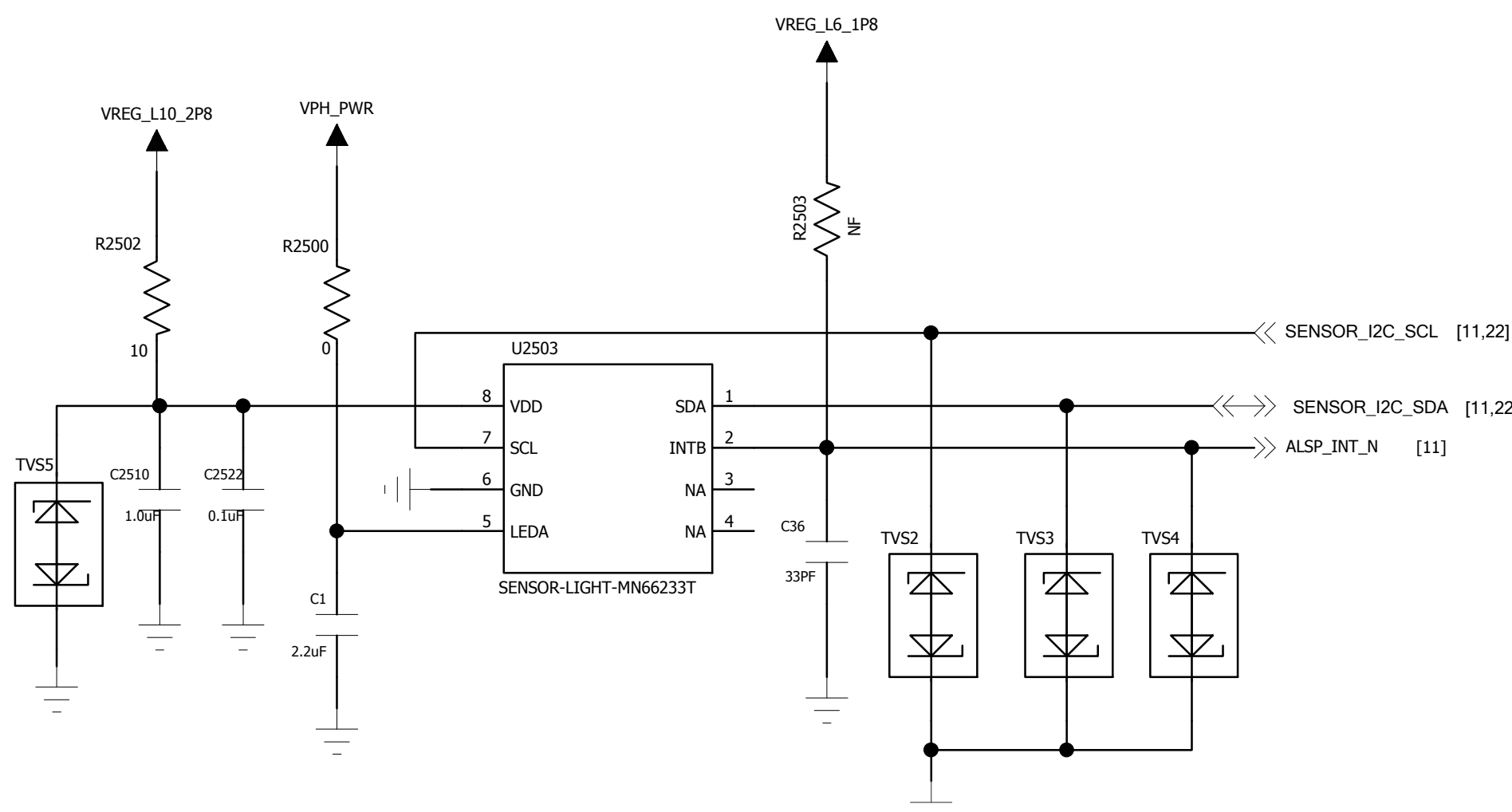
	BM253	KXJT3_1058
R2510	OR	NC
R2511	NC	OR

M-Sensor



AK09918C	
I2C Address	
Write	0x18
Read	0x19

ALS+PS+IR

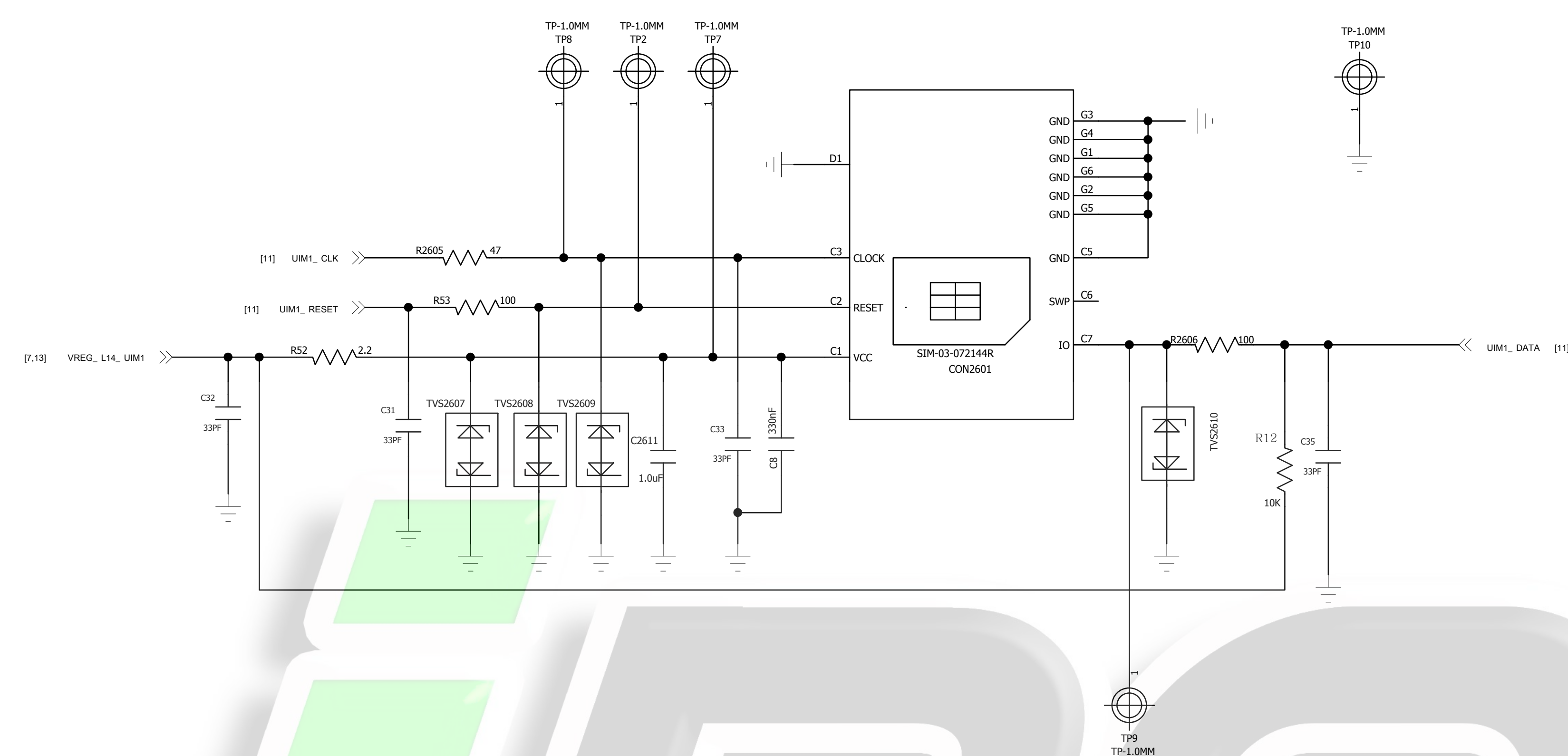


EMINENT:	
I2C Address	
Write	
Read	

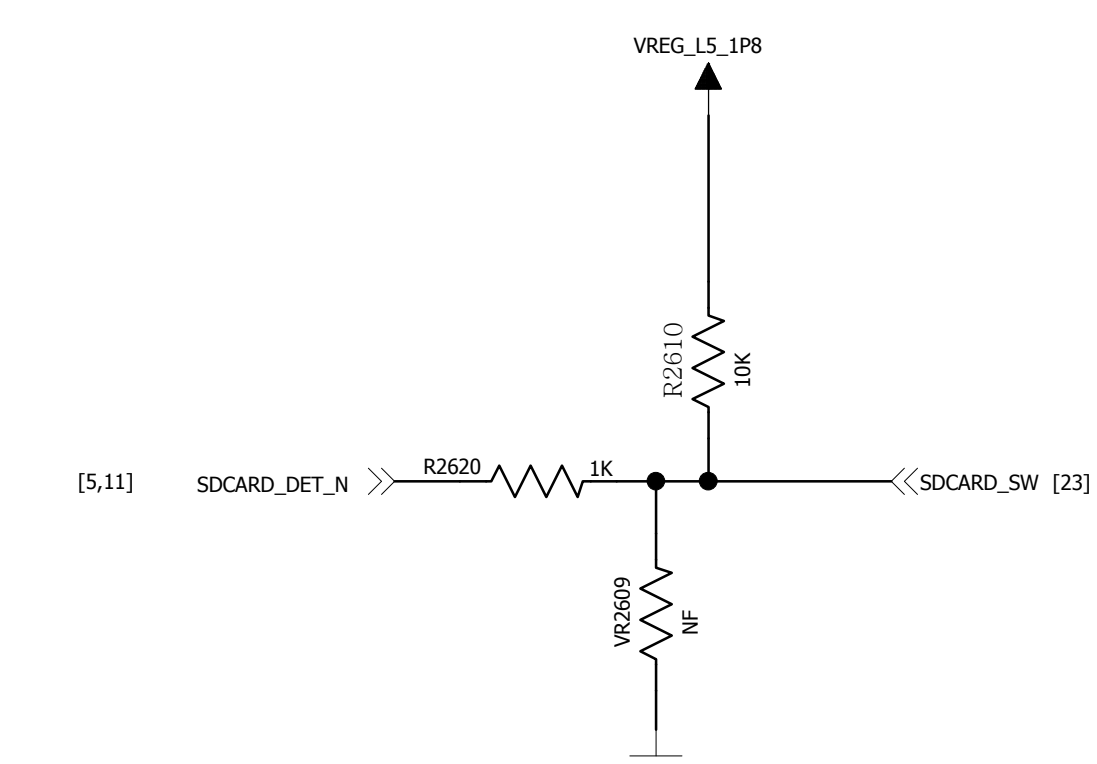
desense apk enable sensor when we enable the audio

SIM/TF card

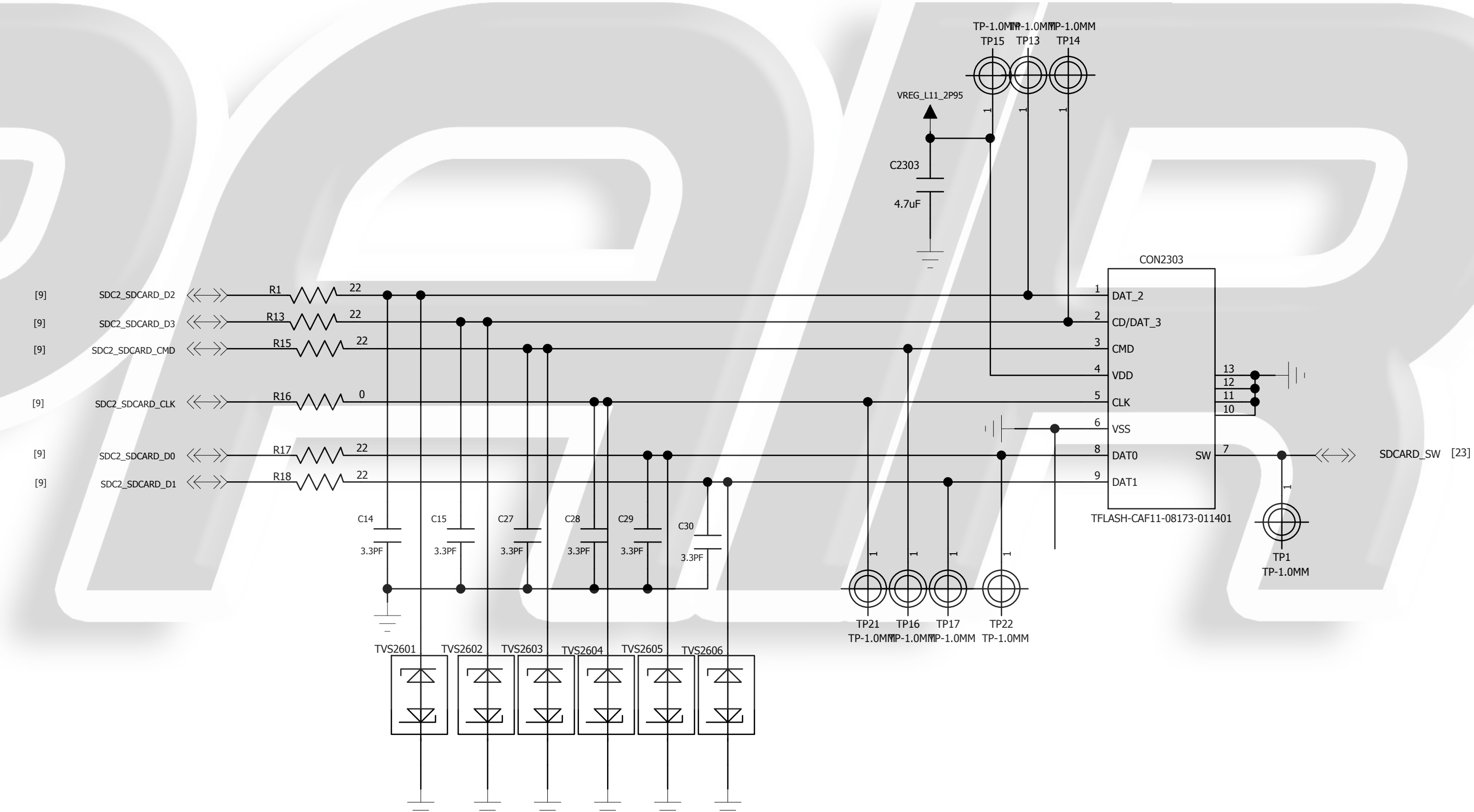
SIM1

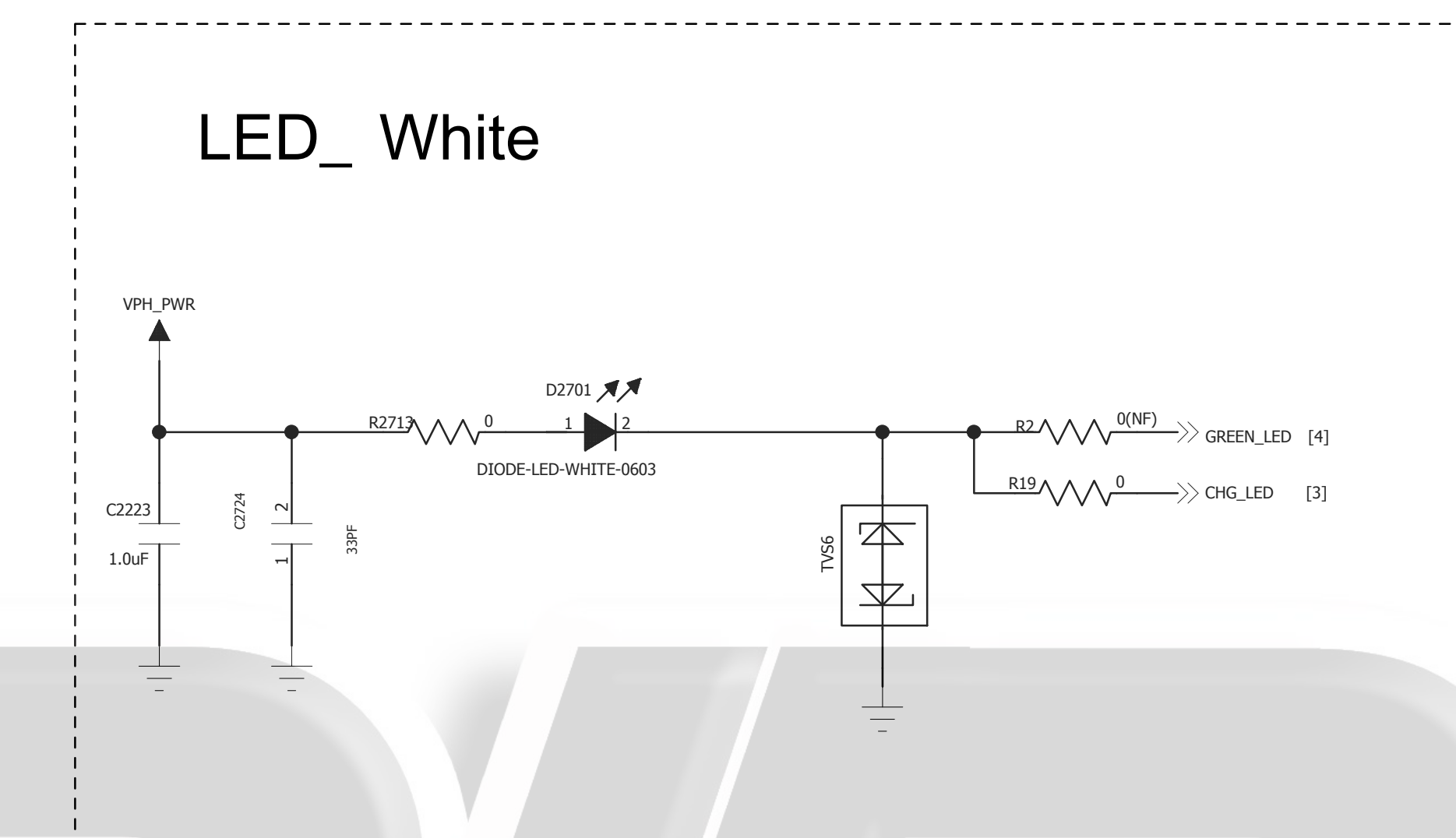


TF



This int should give to GPIO54/GPIO58/GPIO67





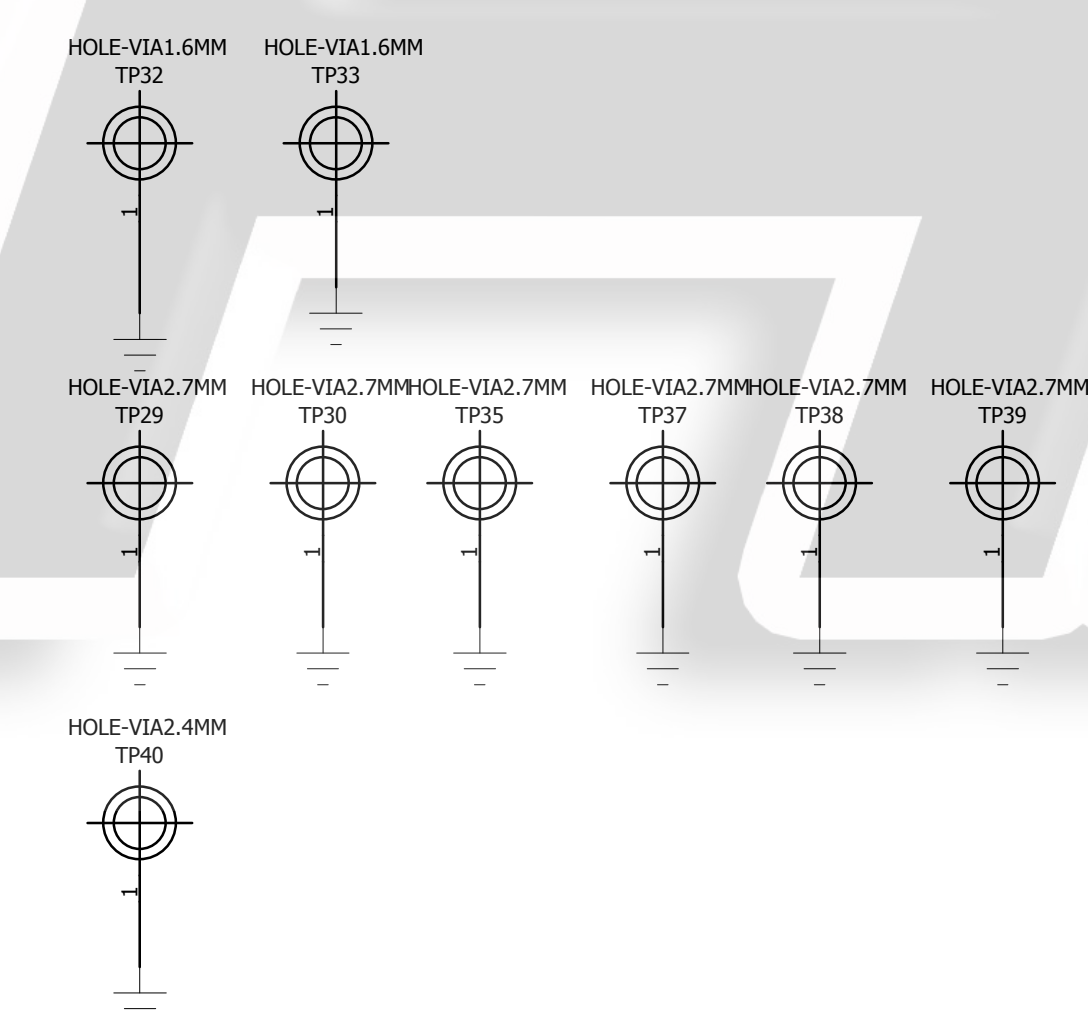
PROJECT:			REV:	<REV>
DOCUMENT NO.:	27_KEYPAD		SIZED:	D
DEPARTMENT:	Hardware DEPT.			
COMPANY:				
DESIGNER:	<DESIGNER>	Last Saved Date:	2017/11/20	SHEET: 24 OF 35

Figure 1 illustrates the pin assignment of the 40-pin QFN package. The package is shown with pins labeled J1 through J14. The pin assignments are as follows:

- P16: 焊盘区 (Pad Area), 1, M8BPA焊盘区
- P17: 焊盘区 (Pad Area), 1, L8BPA焊盘区
- P18: 焊盘区 (Pad Area), 1, AUDIO焊盘区
- P19: 焊盘区 (Pad Area), 1, APT焊盘区
- P20: 焊盘区 (Pad Area), 1, OVP焊盘区
- P7: 焊盘区 (Pad Area), 1, BREF焊盘区
- P8: 焊盘区 (Pad Area), 1, 引脚触点
- P9: 焊盘区 (Pad Area), 1, RF焊盘区
- P10: 焊盘区 (Pad Area), 1, NFC焊盘区
- P11: 焊盘区 (Pad Area), 1, POWER焊盘区
- P12: 焊盘区 (Pad Area), 1, WCN焊盘区
- P13: 焊盘区 (Pad Area), 1, LNA焊盘区

JTAG CONNECTOR

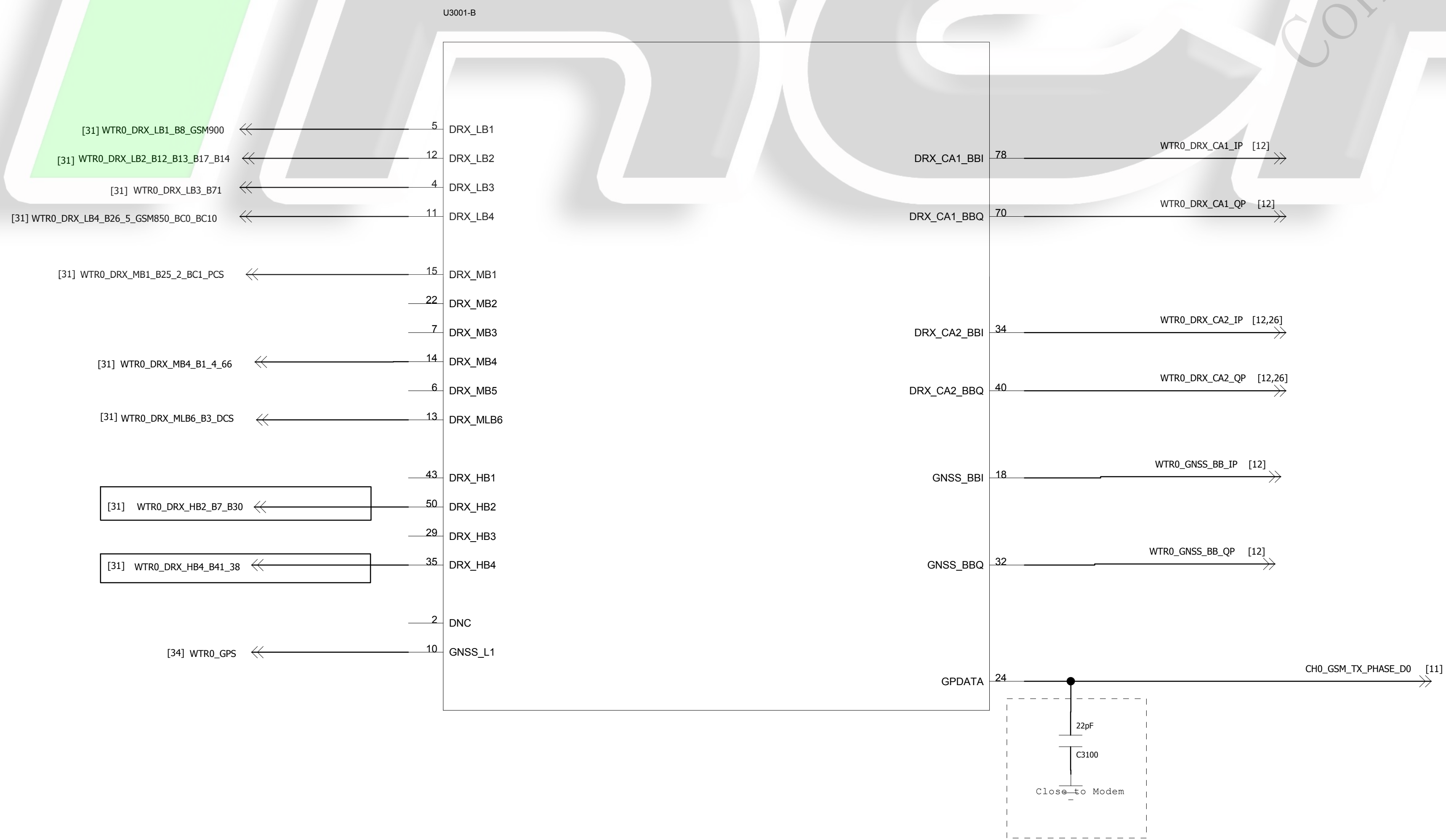
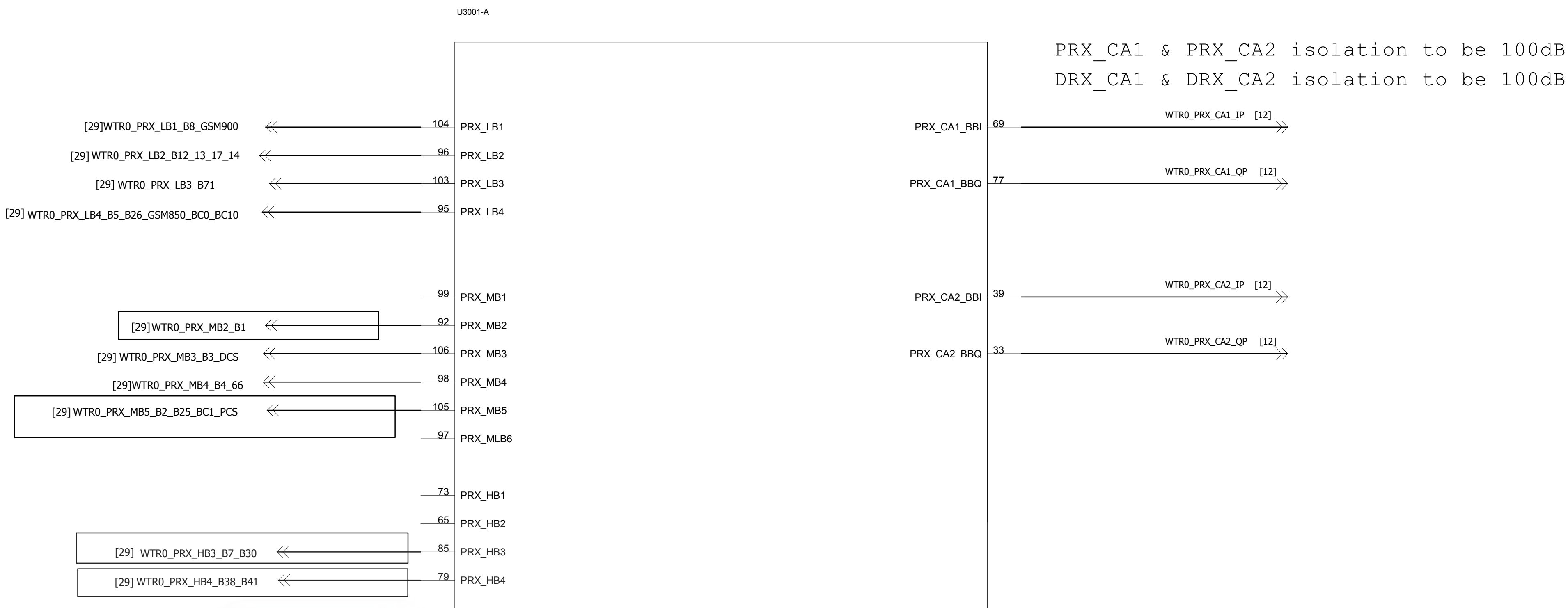
Test Point



PROJECT:		REV: <REV>	
DOCUMENT NO.:	29_TEST_POINT/GND/SHIELDS	SIZED:	D
DEPARTMENT: Hardware DEPT.			
COMPANY: 			
DESIGNER:	<DESIGNER>	Last Saved Date:	2017/11/20
		SHEET:	25 OF 35

WTR3925 PRX/DRX

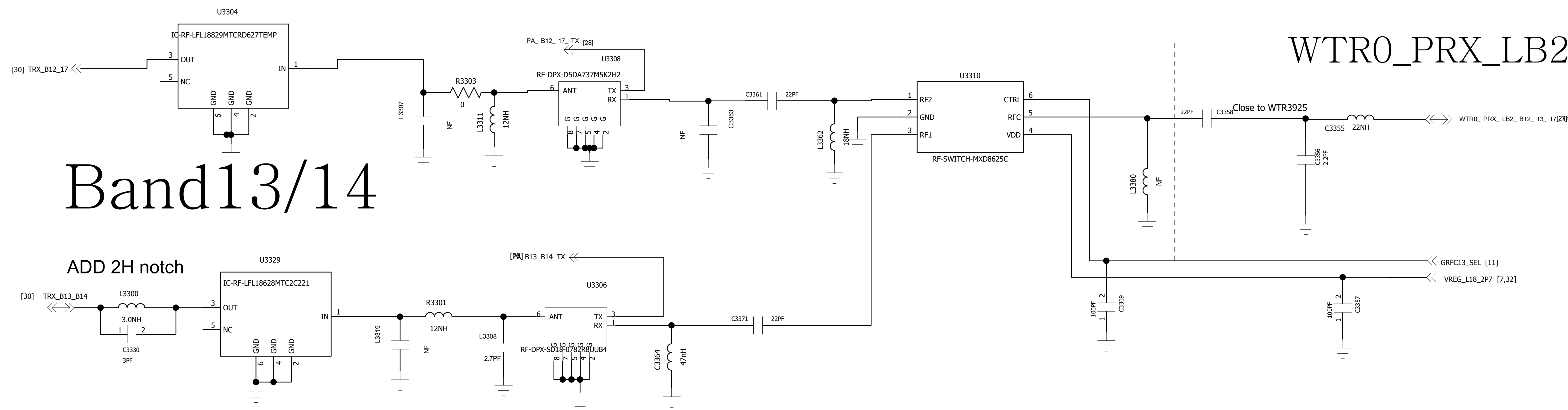
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Title		
Size	Name	Rev
Date:	Sheet	of

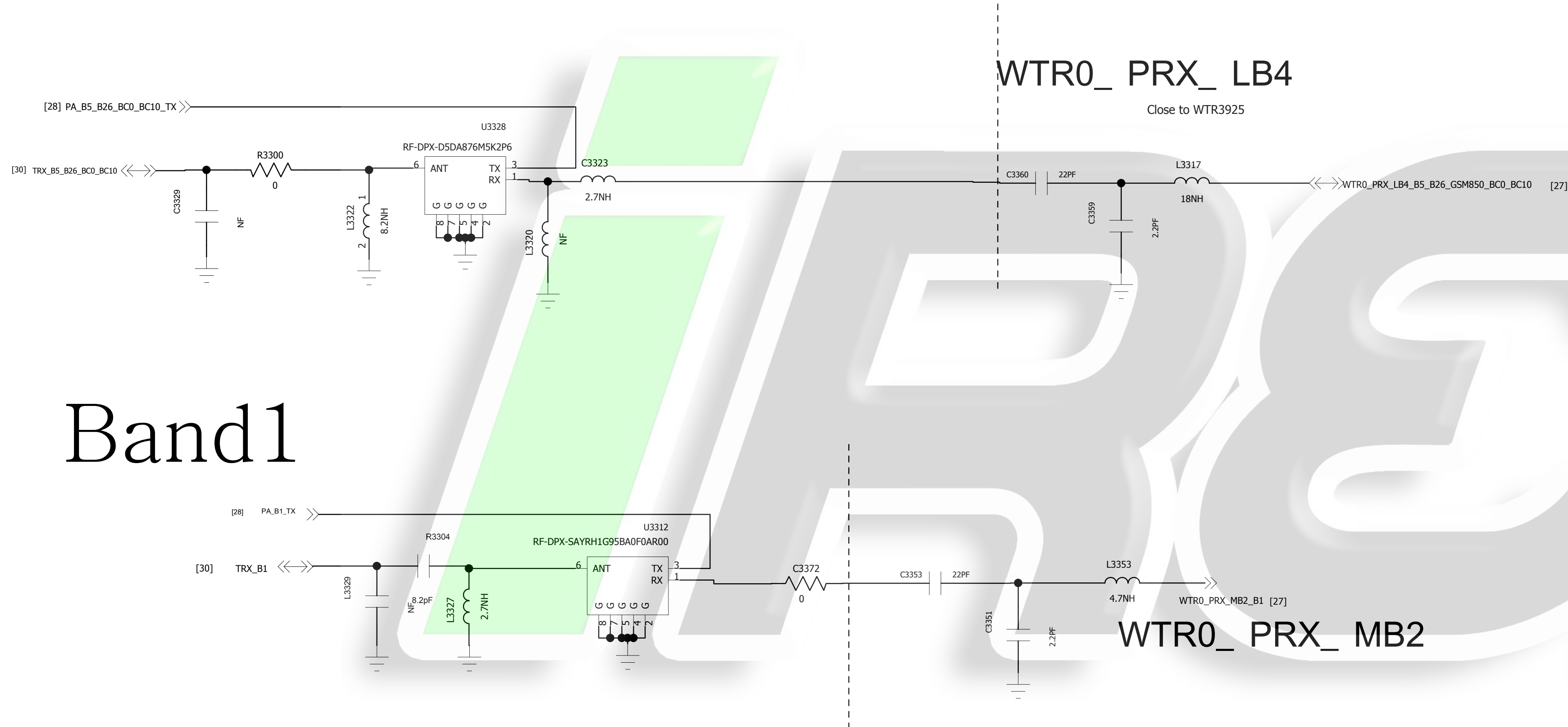
Title		
Sheet		
Size	Name	Rev
Date:	Sheet	of

Band12/17

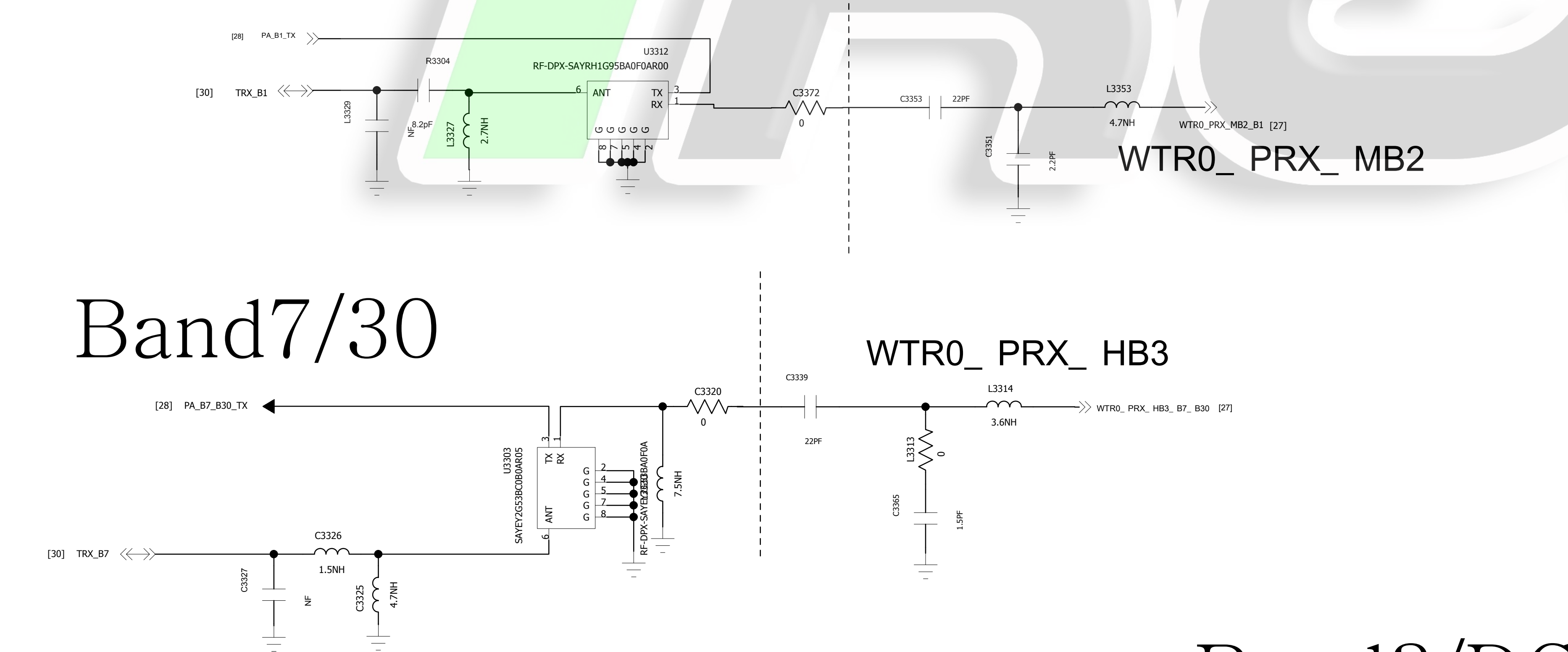


Band13/14

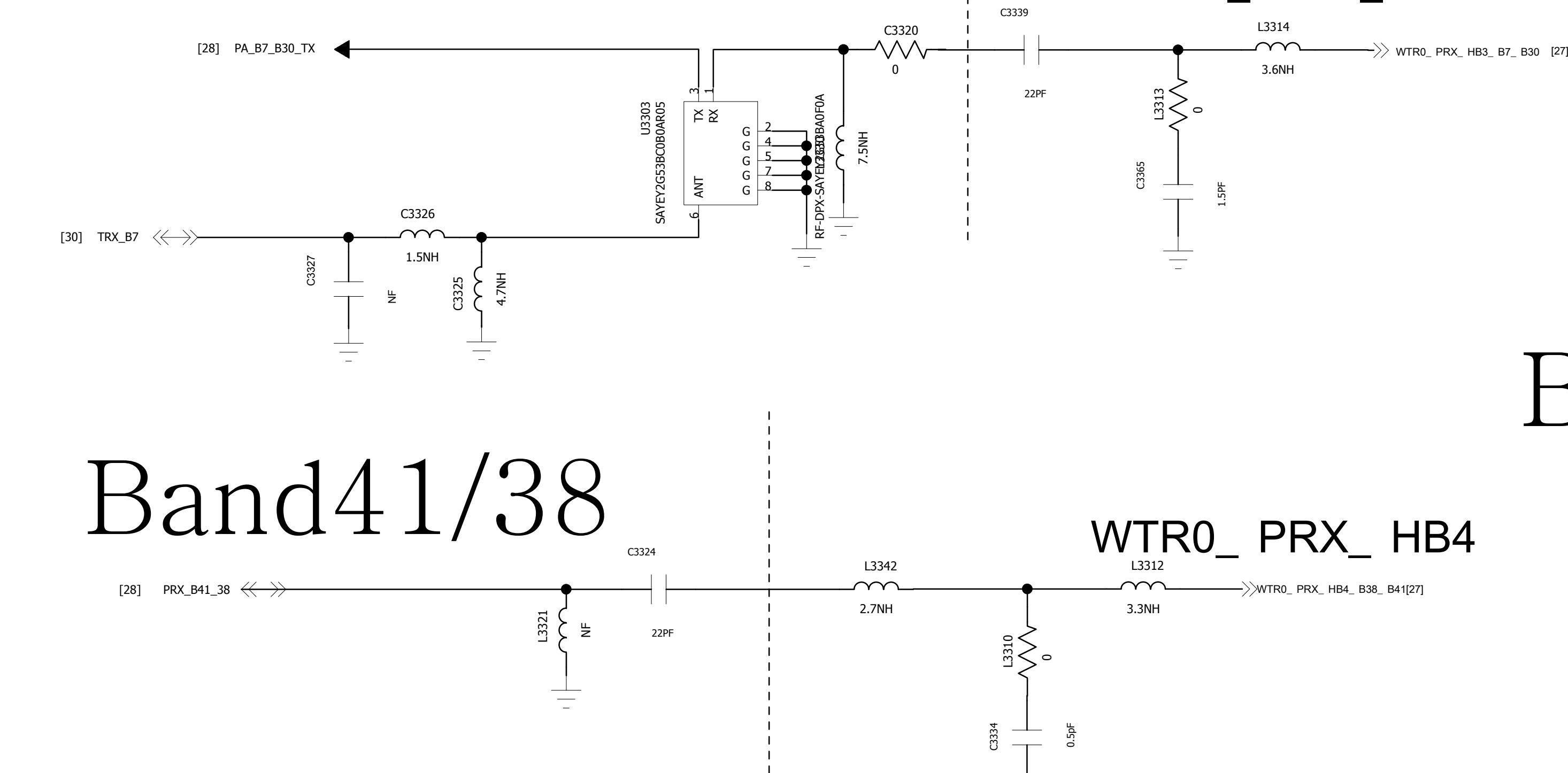
Band5/26/GSM850/BC0/BC10



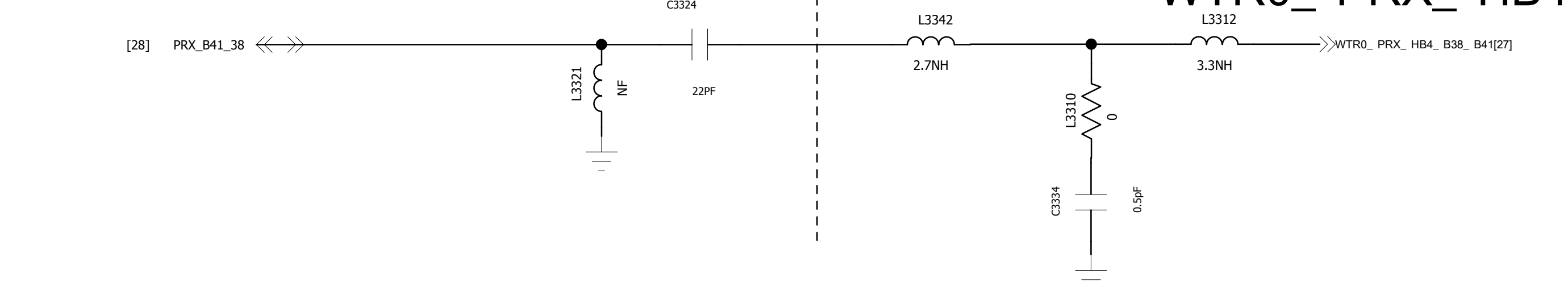
Band1



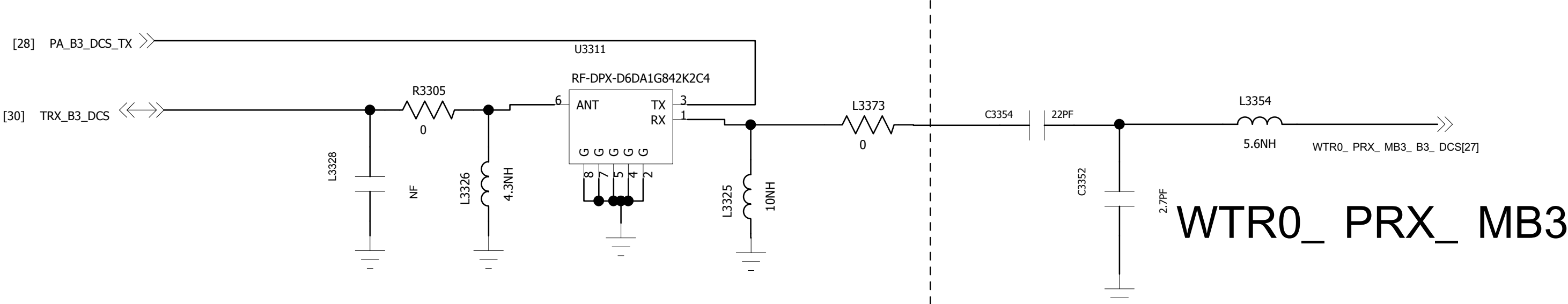
Band7/30



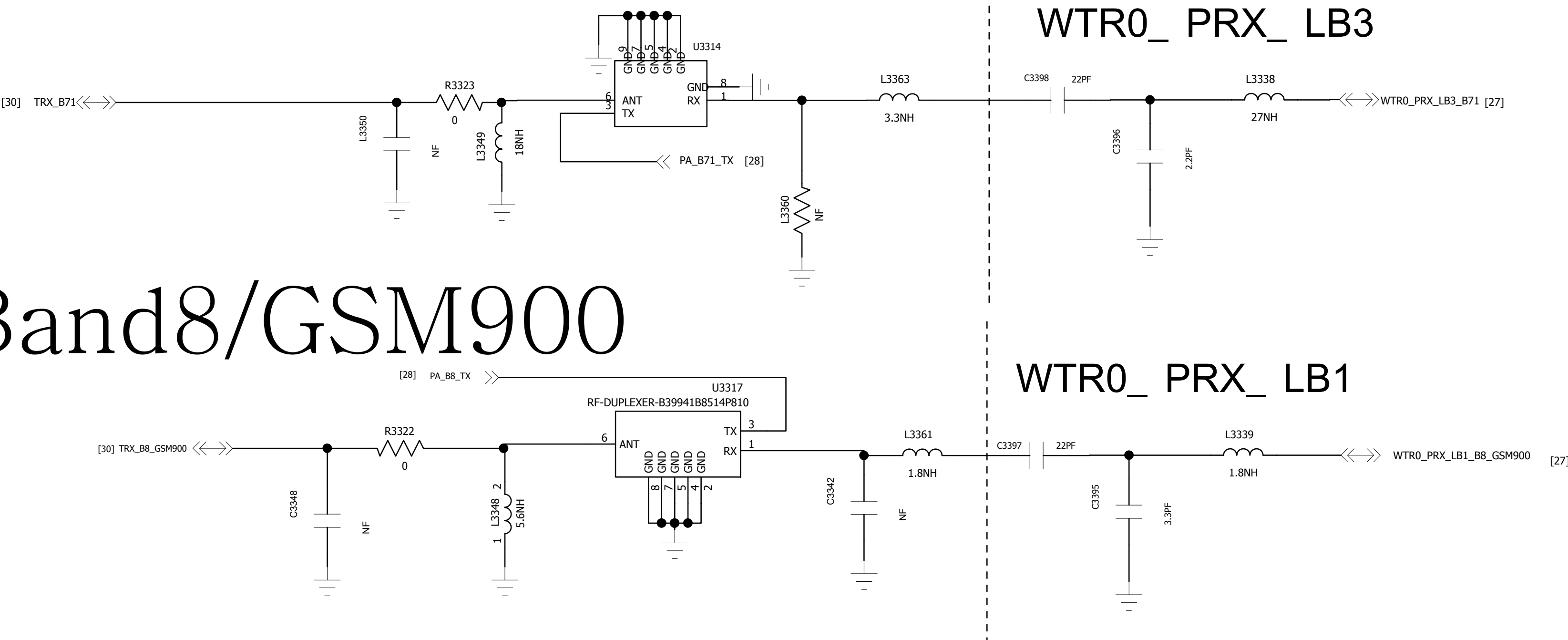
Band41/38



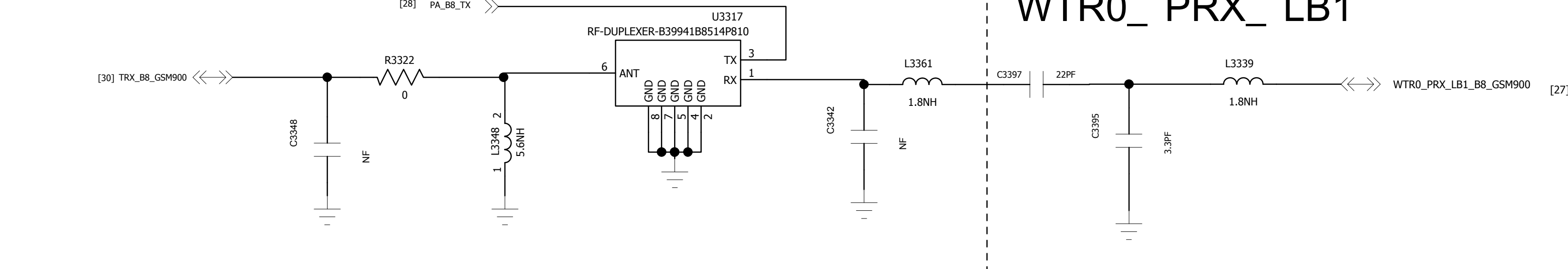
Band3/DCS



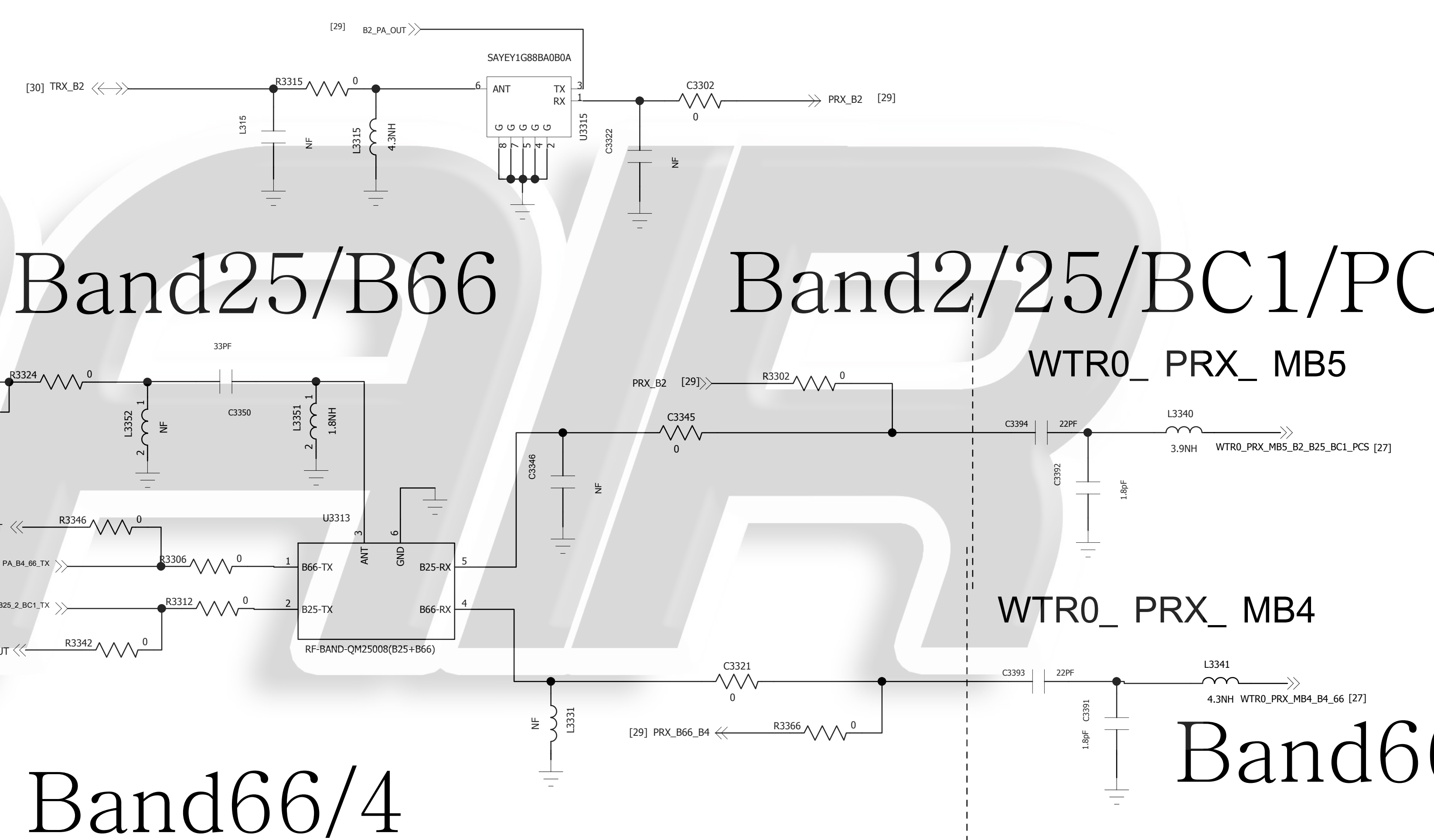
Band71



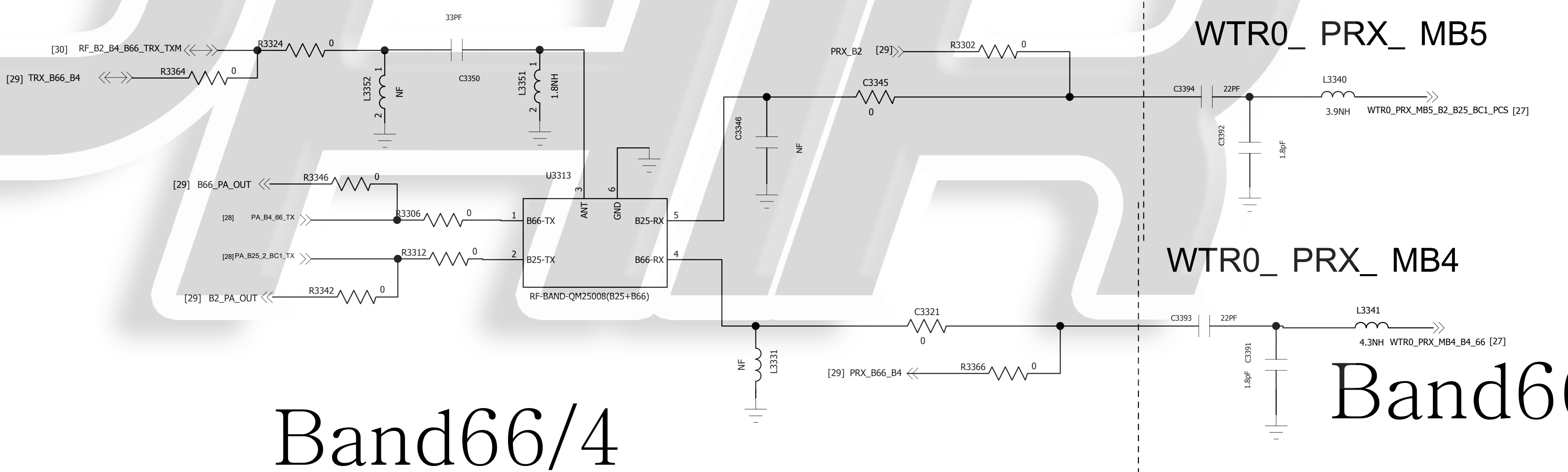
Band8/GSM900



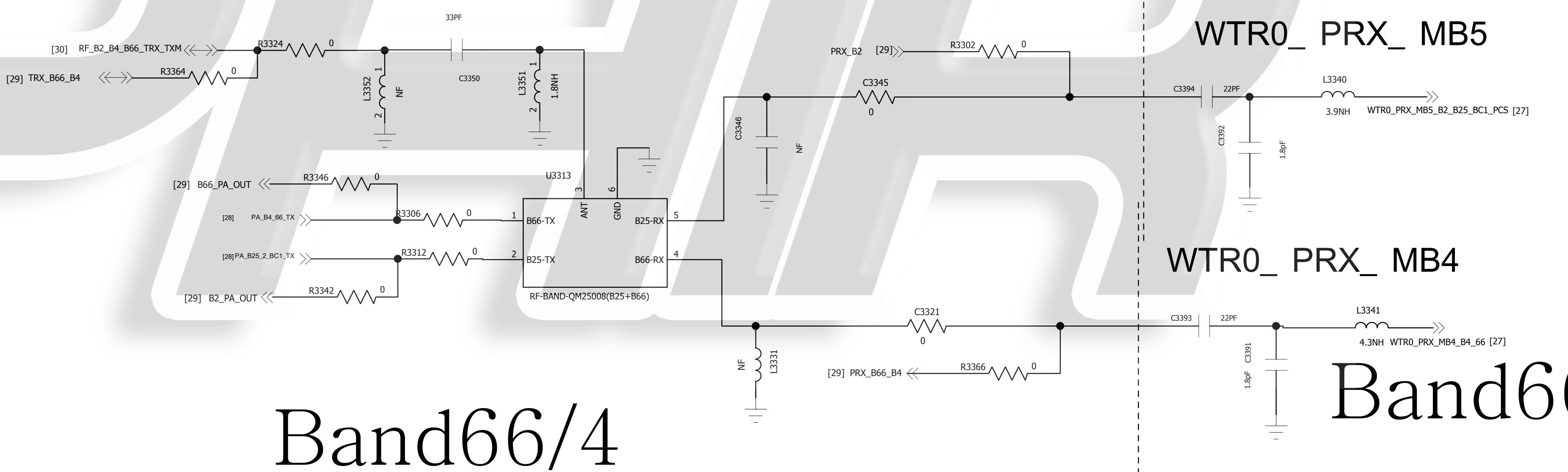
Band2



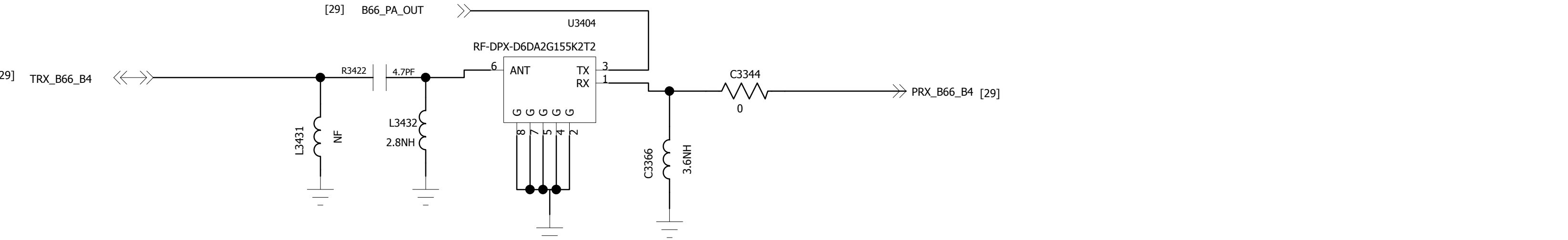
QPX Band25/B66



Band2/25/BC1/PCS



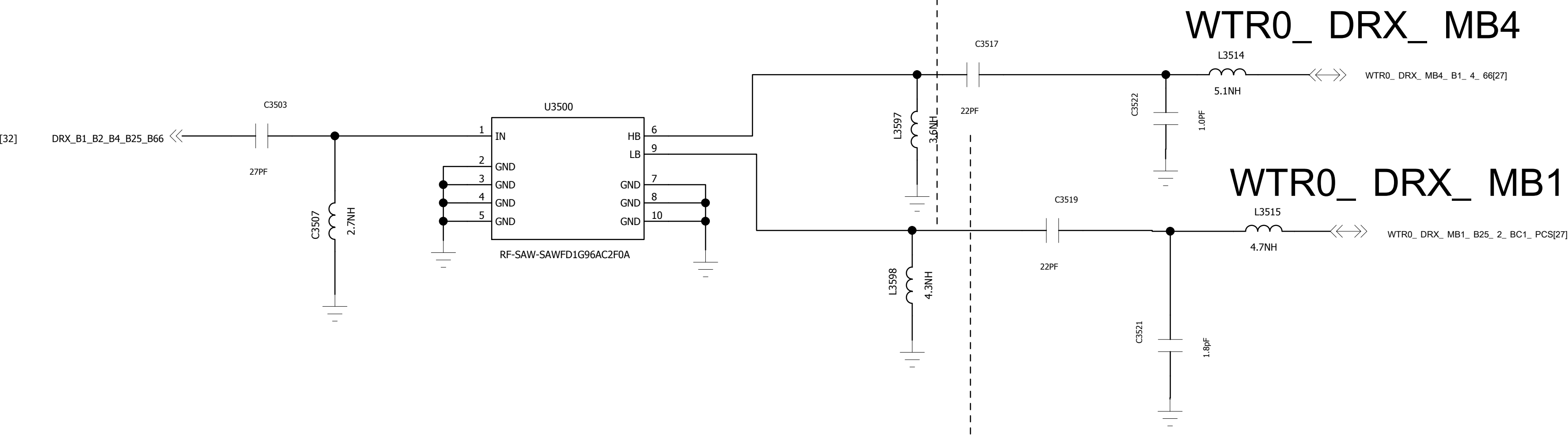
Band66/4



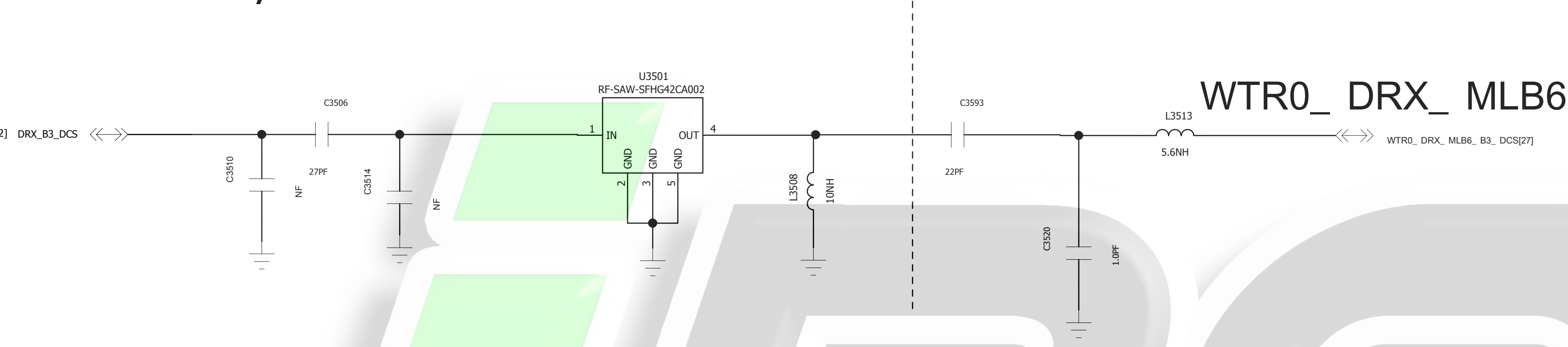
Band66/4

Title		
Sheet	Rev	
Size	Name	
Date:	Sheet	of

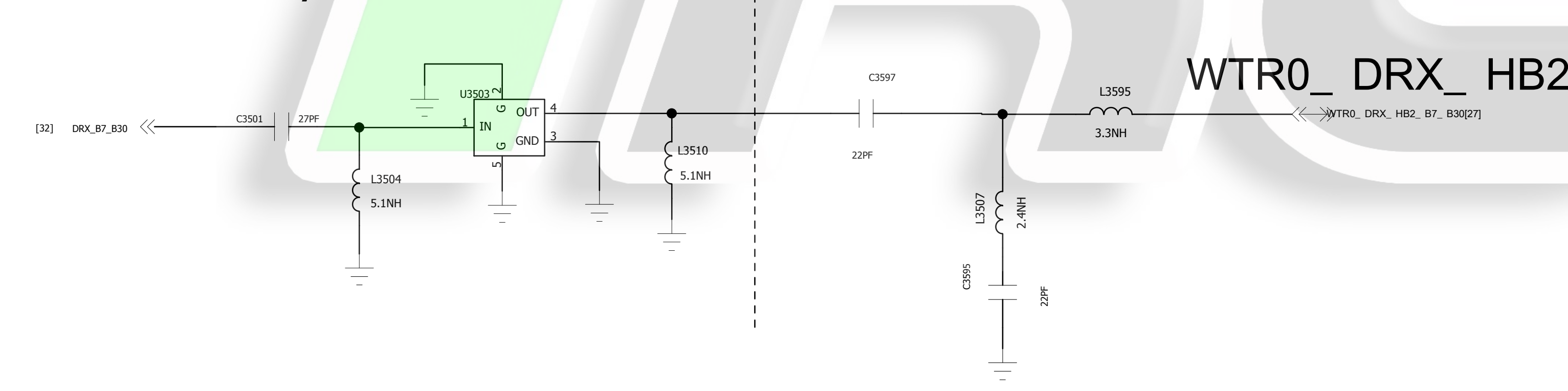
Band1/2/4/25/66/BC1/PCS



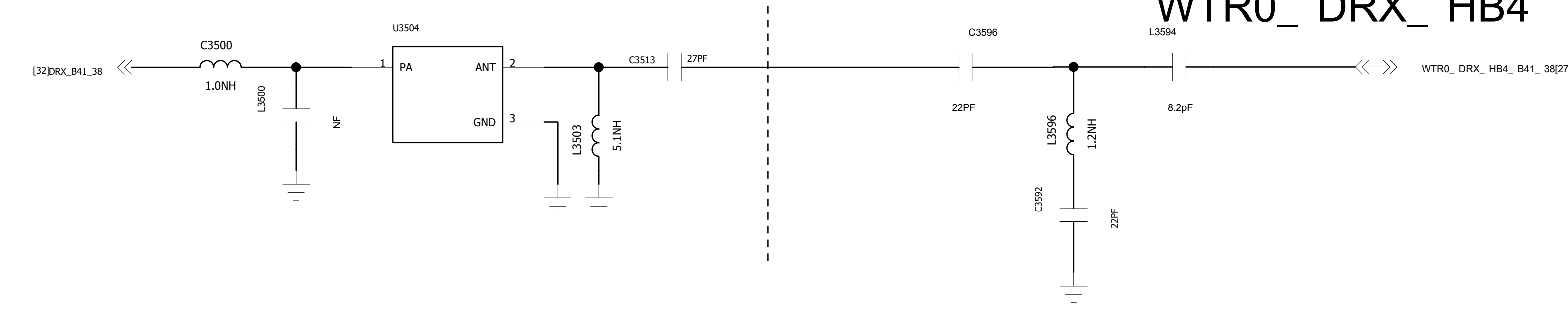
Band3/DCS



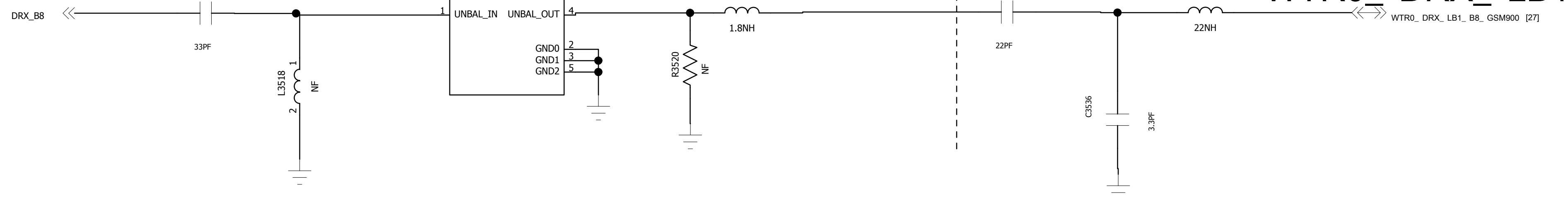
Band7/30



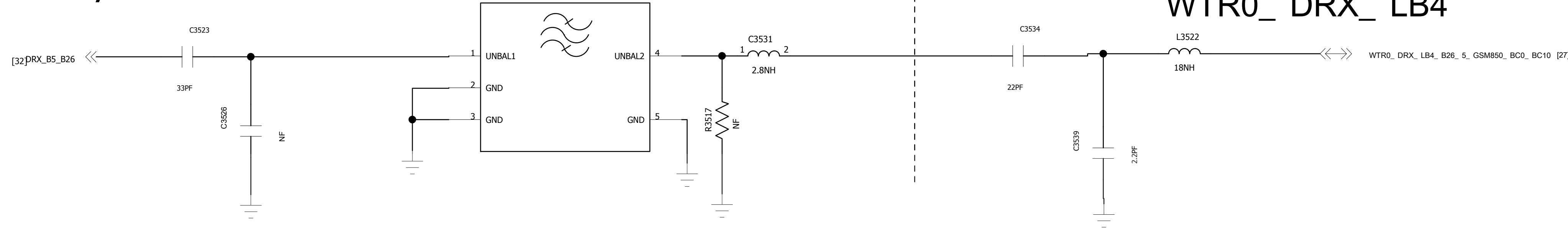
Band41_FULL/B38



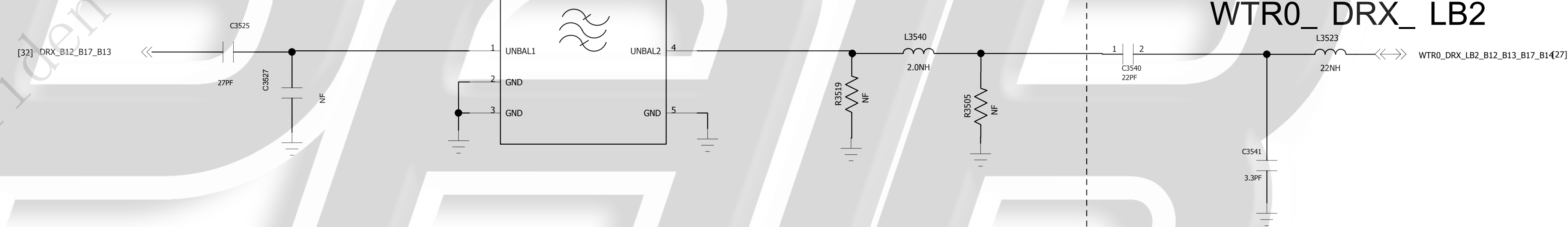
Band8



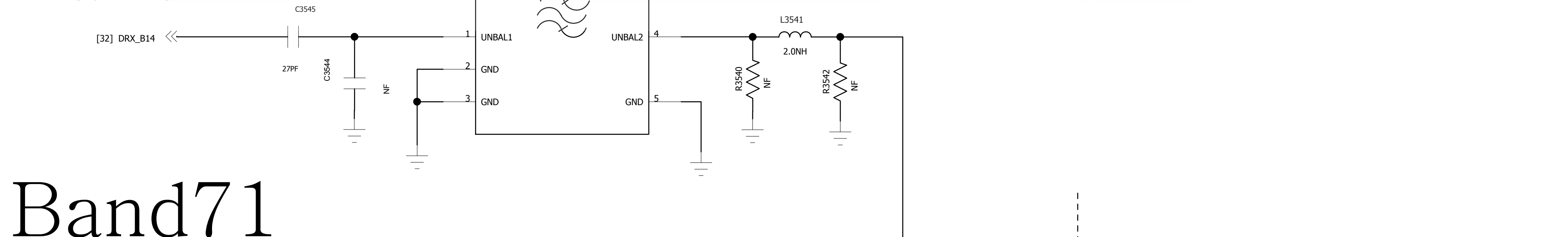
B5/B26



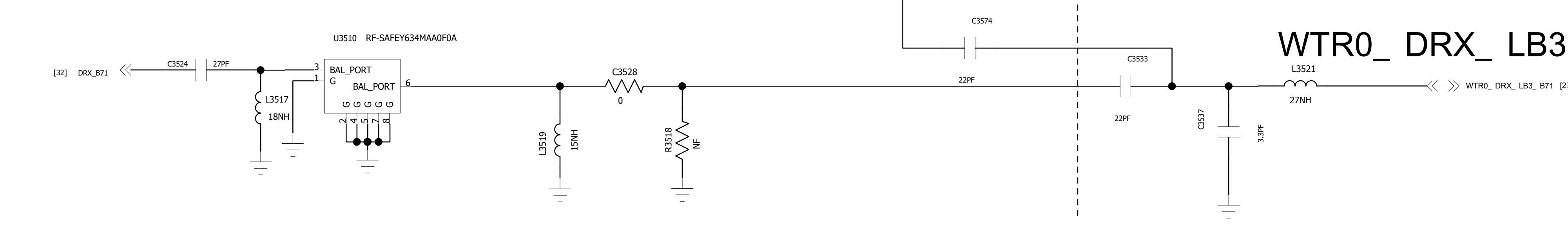
Band12+13

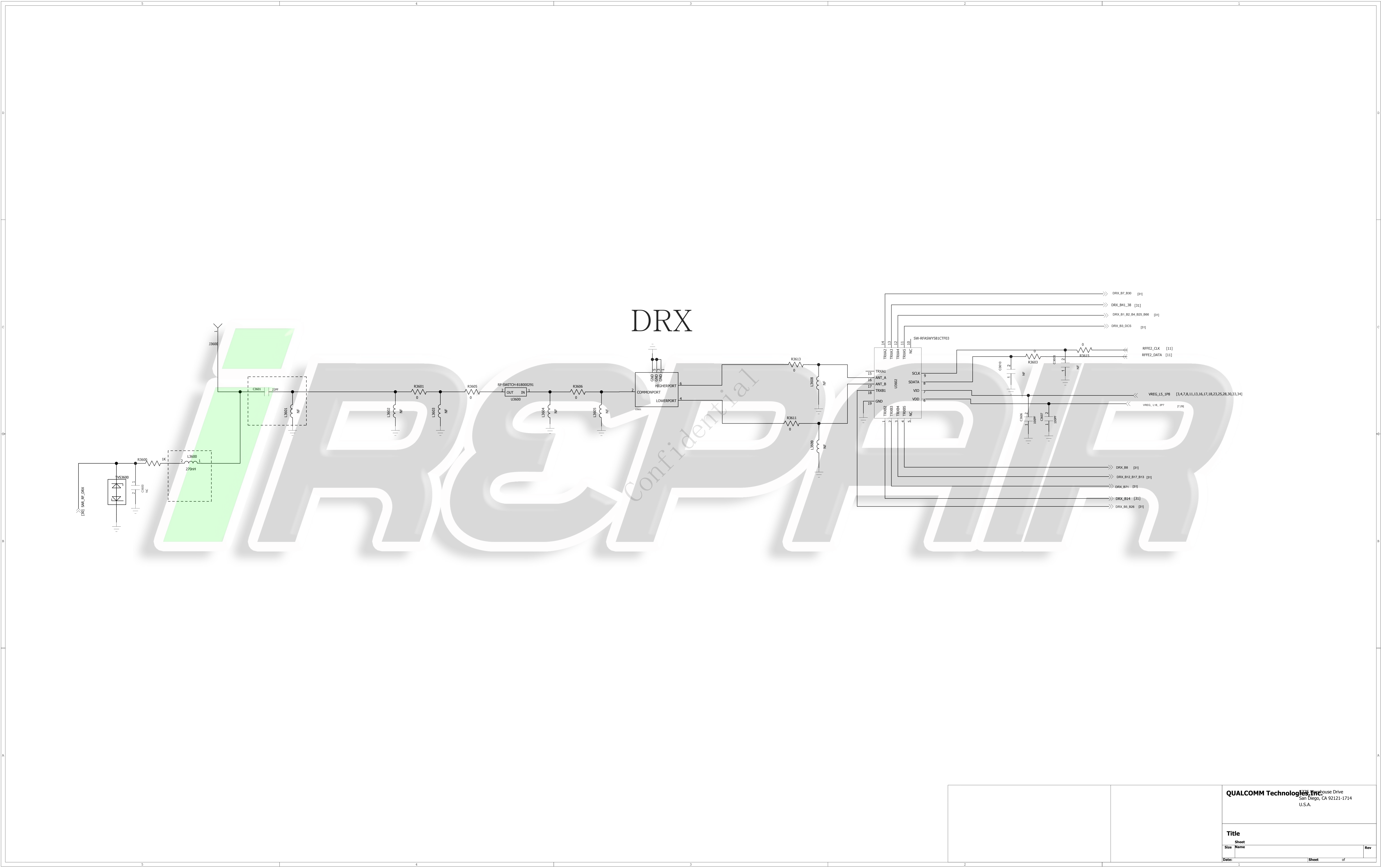


Band14



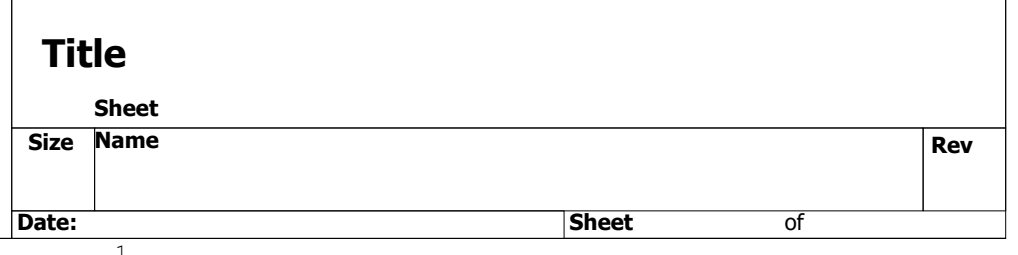
Band71



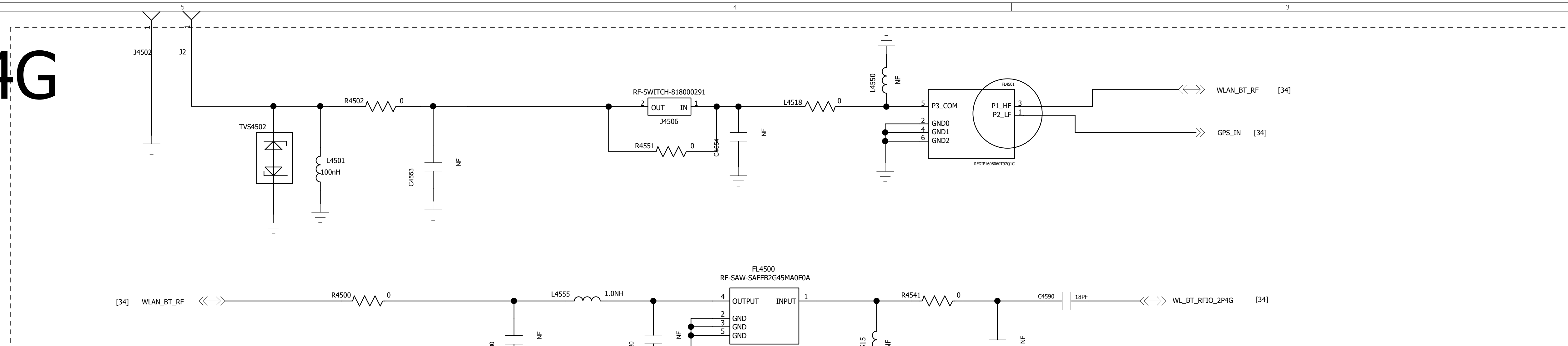


Refer to 80-NA681-91 rev.B or later revisions for more layout details.

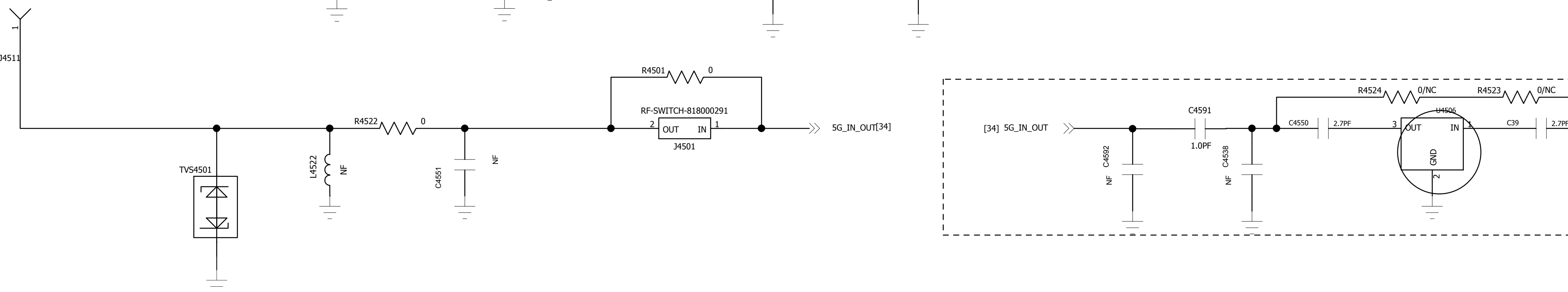
Note:for Sprint SKU,NC R3703;for other SKU,NC R3704 &TPS61280A



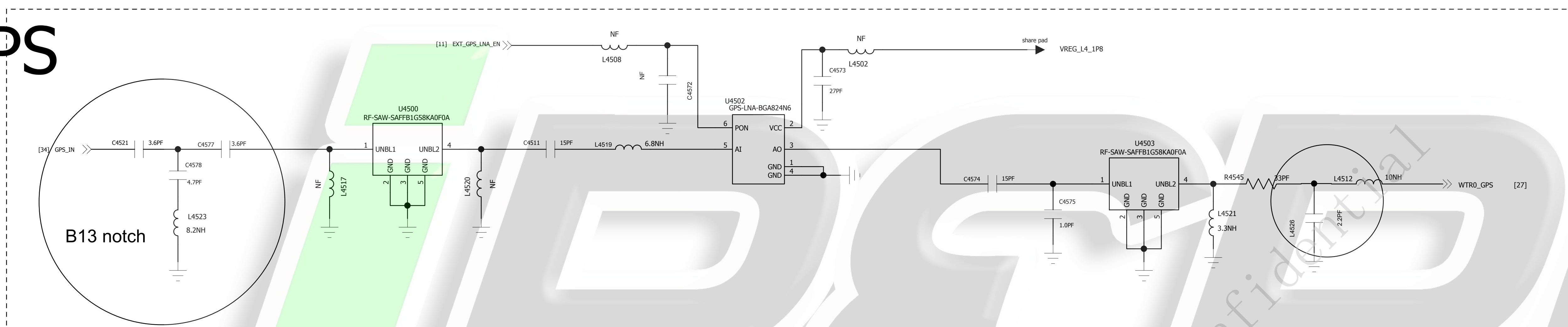
2.4G



5G

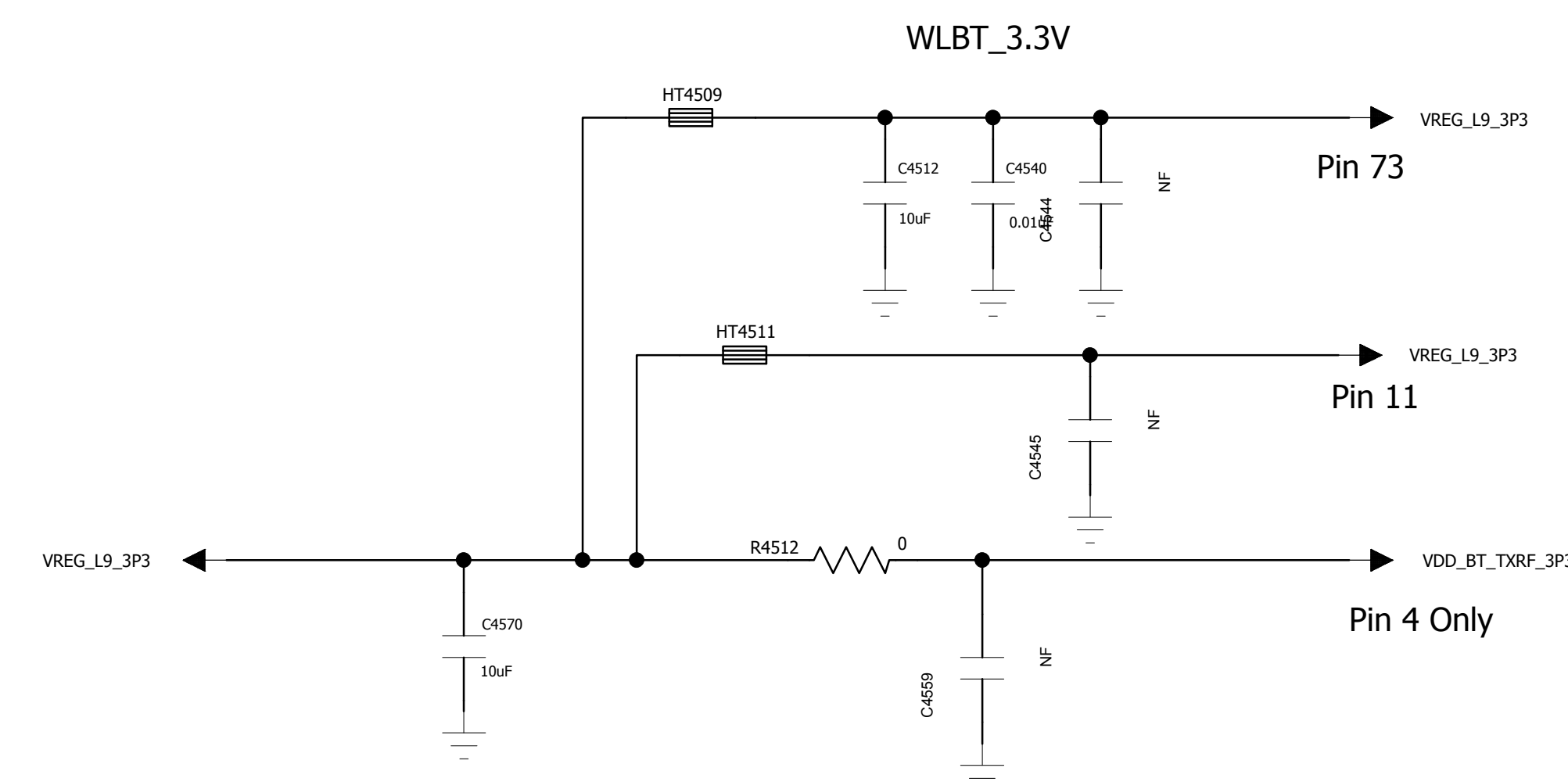
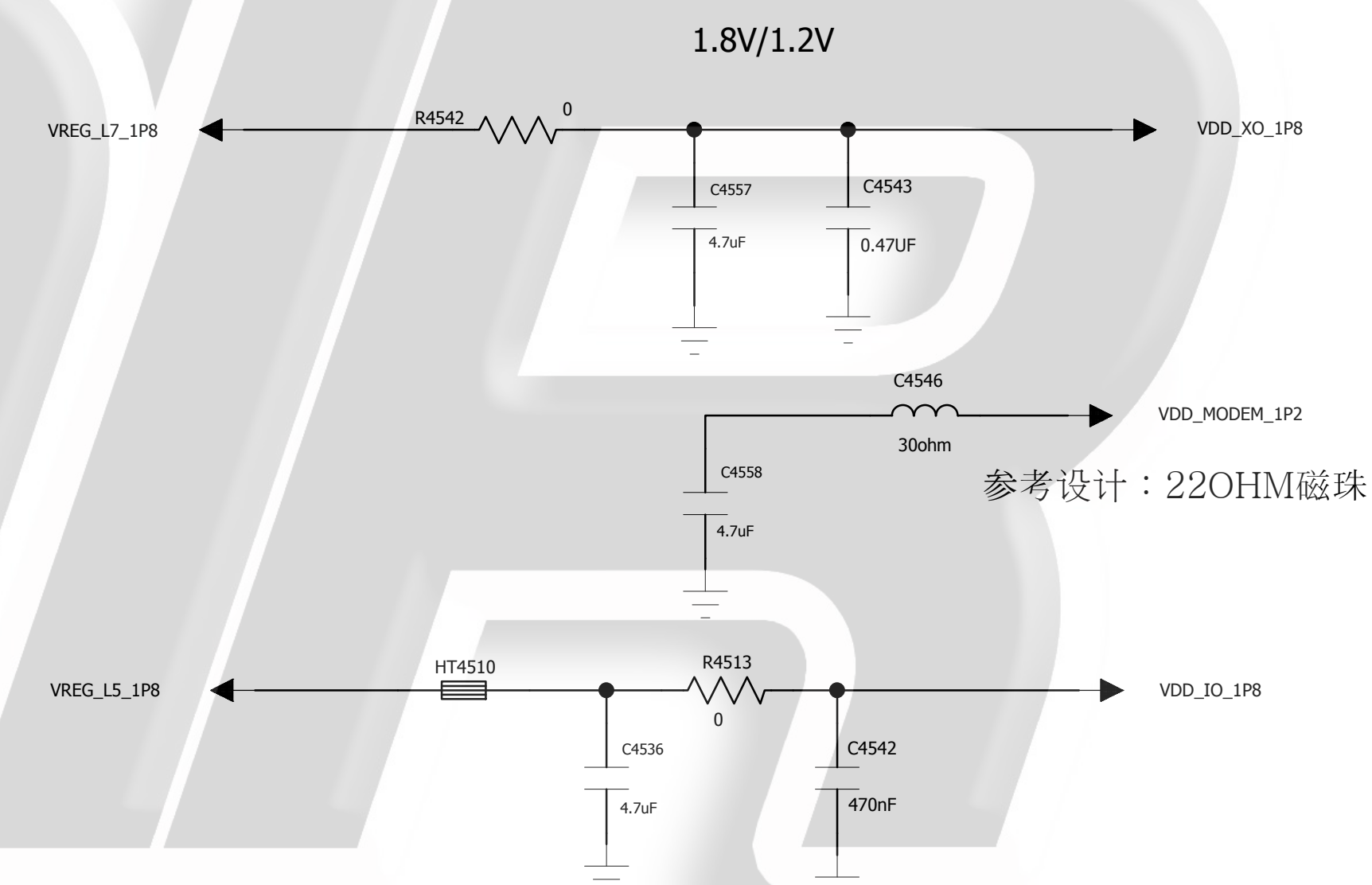
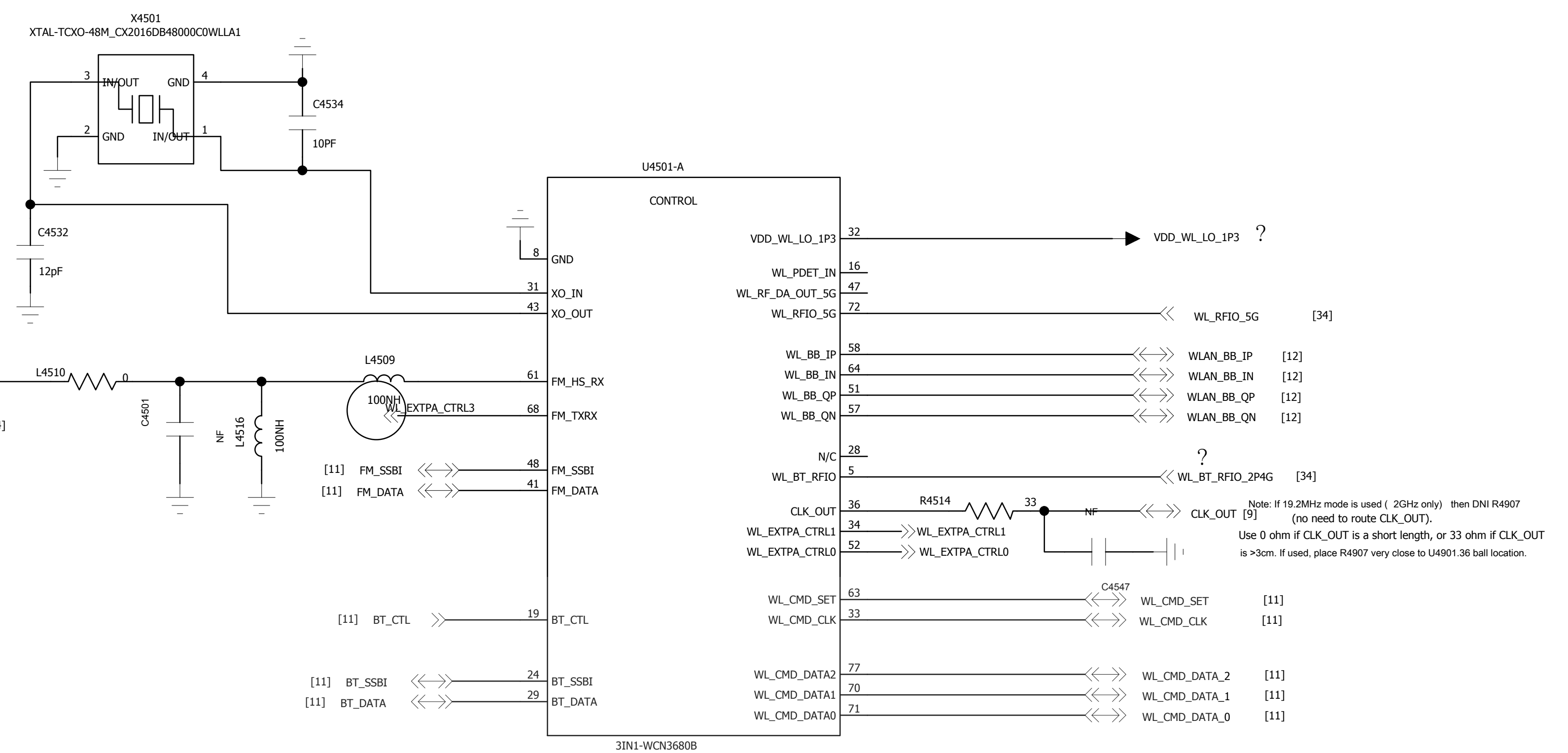


GPS



WIFI/BT

48 MHZ Crystal
Required for any design utilizing 5GHz



Note: If 2.4 GHz WLAN spurs are observed around 2.6 to 2.7 GHz, replace R4915 with ~20 nH inductor.

